Real Analysis

Lecture notes: Background and Chapters 4–7

Contents

Background: Results Used from Chapters 1–3 4

Chapter 4. Double Integrals 8

4.1 Product Measures 8

4.1.1 Setting up the problem . . . 8
4.1.2 The smallest σ-algebra . . . 8
4.1.3 Sections of sets in $\mathcal{S} \times \mathcal{T}$ . . . 8
4.1.4 The monotone class argument . . . 9
4.1.5 The product of Lebesgue measures; Borel sets . . . 11
4.1.6 Lebesgue measure on $\mathbb{R}^2$ . . . 12

4.2 Fubini's Theorem 13

4.2.1 Borel functions . . . 15

4.3 Convolution of Integrable Functions 17

4.3.1 Definition . . . 17
4.3.2 Convolution of L^p and L^q functions . . . 19
4.3.3 Averaging kernels . . . 20

Chapter 5. Fourier Transforms 22

5.1 Fourier Series 22

5.1.1 The circle $\mathbb{T}$ . . . 22
5.1.2 Fourier coefficients . . . 22
5.1.3 Approximate identities . . . 24
5.1.4 The Fejér kernel . . . 26
5.1.5 Consequences . . . 27

5.2 The Fourier Integral 30

5.2.1 Definition and continuity . . . 30
5.2.2 Differentiation . . . 31

5.2.3 Riemann–Lebesgue lemma . . . 32
5.2.4 Approximate identities on $\mathbb{R}$ . . . 32
5.2.5 Smooth functions are dense . . . 34
5.2.6 The inversion formula . . . 34
5.2.7 Fejér and Gauss kernels . . . 35
5.2.8 Transforms of the kernels . . . 36
5.2.9 The Fourier transform as a homomorphism . . . 37
5.2.10 Poisson summation . . . 37

Chapter 6. Normed Spaces **39**

6.1 Dual Spaces **39**
6.1.1 Bounded linear maps . . . 39
6.1.2 The duals of ℓ^p and c_0 . . . 42
6.1.3 The canonical embedding $X \to X^{**}$ . . . 43

6.2 Linear Maps between Banach Spaces **45**
6.2.1 The open mapping theorem . . . 46
6.2.2 The Fourier transform $L^1(\mathbb{T}) \to c_0(\mathbb{Z})$ is not onto . . . 47
6.2.3 The closed graph theorem . . . 48
6.2.4 The uniform boundedness principle . . . 48
6.2.5 A continuous function whose Fourier series diverges . . . 49

6.3 Hilbert Spaces **50**
6.3.1 Inner products . . . 50
6.3.2 The dual of a Hilbert space . . . 51
6.3.3 Orthonormal sets . . . 53
6.3.4 Fourier series in $L^2(\mathbb{T})$ . . . 54
6.3.5 The Plancherel transform . . . 55

Chapter 7. Complex Measures **58**

7.1 Signed Measures and Complex Measures **58**
7.1.1 Definitions . . . 58
7.1.2 The Hahn decomposition . . . 59
7.1.3 Jordan decomposition and total variation . . . 60
7.1.4 Total variation of a complex measure . . . 61

7.2 The Radon–Nikodym Theorem and the Lebesgue Decomposition **62**
7.2.1 Absolute continuity and singularity . . . 62

7.2.2 Polar decomposition 65
7.2.3 Integration with respect to a complex measure 66
7.2.4 The dual of $L^p(\mu)$ 66

7.3 Borel Measures on the Real Line **68**
7.3.1 Regular Borel measures and their distribution functions 68
7.3.2 Riemann–Stieltjes integrals 69
7.3.3 Continuity of α_λ 69
7.3.4 Regularity 70
7.3.5 Singular measures 72
7.3.6 The Radon–Nikodym derivative is α'_λ 73
7.3.7 Discrete, continuous singular, and absolutely continuous parts 73

Background: Results Used from Chapters 1–3

How to use this

Chapters 4–7 rely on standard facts about Lebesgue measure, the Lebesgue integral and L^p spaces that were developed in Chapters 1–3. They are collected here, in the order in which they arise, and are cited in the later chapters as *Fact B.1*, *Fact B.2*, ... Throughout, $(X, \mathcal{S}, \mu)$ is a measure space, m is Lebesgue measure on $\mathbb{R}$ with σ-algebra $\mathcal{M}$ of Lebesgue measurable sets, and $1 \le p \le \infty$.

Measures

Fact B.1 (translation invariance)

For $E \in \mathcal{M}$ and $x \in \mathbb{R}$, $E + x \in \mathcal{M}$ and $m(E + x) = m(E)$; consequently $\int f(y + x)\, dm(y) = \int f(y)\, dm(y)$ for integrable f.

Fact B.2 (completeness of Lebesgue measure)

$(\mathbb{R}, \mathcal{M}, m)$ is complete: every subset of a set of Lebesgue measure zero is Lebesgue measurable (and has measure zero).

Fact B.3 (regularity of Lebesgue measurable sets)

For every $S \in \mathcal{M}$ there are an F_σ-set A and a G_δ-set B with $A \subset S \subset B$ and $m(B \setminus A) = 0$.

Fact B.4 (counting measure)

On any set X with $\mathcal{S} = 2^X$, the counting measure $c(E) = \#E$ (the number of elements, possibly ∞) is a measure. On $X = \mathbb{N}$, $\int f\, dc = \sum_n f(n)$ for $f \ge 0$.

Measurable functions and the integral

Fact B.5 (approximation by simple functions)

If $f : X \to [0, \infty]$ is measurable, there is an increasing sequence of measurable simple functions $0 \le s_1 \le s_2 \le \cdots$ with $s_n \to f$ pointwise.

Fact B.6 (series of characteristic functions)

Every measurable $f : X \to [0, \infty]$ can be written as $f = \sum_{k=1}^{\infty} c_k \chi_{E_k}$ with constants $c_k \ge 0$ and measurable sets E_k.

Fact B.7 (measures defined by densities)

If $h : X \to [0, \infty]$ is measurable, then $\lambda(E) = \int_E h\, d\mu$ $(E \in \mathcal{S})$ defines a measure on $\mathcal{S}$.

Fact B.8 (functions with vanishing integrals)

(a) If $h \geq 0$ and $\int_X h\,d\mu = 0$ then $h = 0$ almost everywhere.
(b) If h is integrable and $\int_E h\,d\mu = 0$ for every $E \in \mathcal{S}$, then $h = 0$ almost everywhere.

Fact B.9 (triangle inequality for integrals)

For integrable f, $|\int_X f\,d\mu| \leq \int_X |f|\,d\mu$.

Fact B.10 (linearity of the integral)

If f, g are integrable and $a, b \in \mathbb{C}$, then $af + bg$ is integrable and $\int (af + bg)\,d\mu = a\int f\,d\mu + b\int g\,d\mu$.

Fact B.11 (differentiation under the integral sign)

Let $f(x,t)$ be integrable in x for each t in an open interval I, differentiable in t, and suppose $|\partial_t f(x,t)| \leq g(x)$ for an integrable g independent of t. Then $F(t) = \int f(x,t)\,d\mu(x)$ is differentiable on I with $F'(t) = \int \partial_t f(x,t)\,d\mu(x)$.

Fact B.12 (periodisation)

For $f \in L^1(\mathbb{R})$ the series $F(t) = \sum_{n\in\mathbb{Z}} f(t - 2n\pi)$ converges absolutely for almost every t, is 2π-periodic, and $\int_{-\pi}^{\pi} \sum_n |f(t-2n\pi)|\,dt = \int_{\mathbb{R}} |f|\,dm$ (term-by-term integration, Fact B.25).

Fact B.13 (Riemann integral via step functions)

For bounded $f : [a,b] \to \mathbb{R}$ and a partition $P = \{a = x_0 < x_1 < \cdots < x_n = b\}$ put $M_i = \sup_{[x_{i-1},x_i]} f$, $m_i = \inf_{[x_{i-1},x_i]} f$ and define the step functions

$$U_P = \sum_{i=1}^{n} M_i \chi_{(x_{i-1},x_i]}, \qquad L_P = \sum_{i=1}^{n} m_i \chi_{(x_{i-1},x_i]}.$$

f is Riemann integrable iff $\inf_P \int U_P = \sup_P \int L_P$, and then this common value is $\int_a^b f$. Improper Riemann integrals are defined as limits of such integrals; Lebesgue integrals need no such limiting procedure.

Differentiation

Fact B.14 (derivative of a monotone function)

If $\alpha : [a,b] \to \mathbb{R}$ is nondecreasing, then α' exists almost everywhere, $\alpha' \geq 0$ is integrable, and $\int_{[a,b]} \alpha'\,dm \leq \alpha(b) - \alpha(a)$.

Fact B.15 (absolute continuity of the integral)

If $f \in L^1(\mathbb{R})$, then for every $\epsilon > 0$ there is $\delta > 0$ such that $m(E) < \delta$ implies $\int_E |f|\,dm < \epsilon$.

Fact B.16 (fundamental theorem for absolutely continuous functions)

If $\alpha : [a, b] \to \mathbb{R}$ is absolutely continuous, then α' exists almost everywhere, is integrable, and $\alpha(x) - \alpha(a) = \int_{[a,x]} \alpha' \, dm$ for all $x \in [a, b]$.

L^p spaces

Fact B.17 (completeness of L^p)

For $1 \leq p \leq \infty$, $L^p(\mu)$ is a Banach space.

Fact B.18 (a.e. convergent subsequence)

If $\|f_n - f\|_p \to 0$ for some $1 \leq p < \infty$, then a subsequence $f_{n_k} \to f$ almost everywhere.

Fact B.19 (translates)

For a function f on $\mathbb{R}$ and $s \in \mathbb{R}$, the translate is $f_s(x) = f(x - s)$.

Fact B.20 (density of simple functions)

For $1 \leq p < \infty$, the measurable simple functions s with $\mu(\{s \neq 0\}) < \infty$ are dense in $L^p(\mu)$; moreover every $f \in L^p(\mu)$ is the L^p-limit of simple functions s_n with $|s_n| \leq |f|$.

Fact B.21 (density of continuous functions on an interval)

For $1 \leq p < \infty$, the continuous functions on $[a, b]$ are dense in $L^p[a, b]$.

Fact B.22 (density of $C_c(\mathbb{R})$)

For $1 \leq p < \infty$, the continuous functions with compact support are dense in $L^p(\mathbb{R})$.

Fact B.23 (continuity of translation)

For $f \in L^p(\mathbb{R})$ and $1 \leq p < \infty$, $\|f_s - f\|_p \to 0$ as $s \to 0$ (with f_s as in Fact B.19).

Further properties of measures, functions and sums

Fact B.24 (continuity of measures)

Let $E_1, E_2, \cdots \in \mathcal{S}$.
(a) Countable additivity: $\mu(\bigsqcup_n E_n) = \sum_n \mu(E_n)$ for disjoint E_n.
(b) Continuity from below: if $E_1 \subset E_2 \subset \cdots$, then $\mu(\bigcup_n E_n) = \lim_n \mu(E_n)$.
(c) Continuity from above: if $E_1 \supset E_2 \supset \cdots$ and $\mu(E_1) < \infty$, then $\mu(\bigcap_n E_n) = \lim_n \mu(E_n)$.

Fact B.25 (term-by-term integration)

For measurable $f_n : X \to [0,\infty]$, $\int_X \sum_{n=1}^{\infty} f_n \, d\mu = \sum_{n=1}^{\infty} \int_X f_n \, d\mu$ (a consequence of the monotone convergence theorem).

Fact B.26 (operations preserving measurability)

Let f, g be measurable (real or complex valued, with respect to a fixed σ-algebra).
(a) $f+g$ and fg are measurable.
(b) Pointwise sup, inf, lim sup, lim inf and limits of sequences of measurable functions are measurable; in particular a pointwise convergent series of measurable functions is measurable.
(c) $|f|$, $\operatorname{Re} f$, $\operatorname{Im} f$ are measurable.
(d) If $g : \mathbb{R} \to \mathbb{R}$ is continuous, then $g \circ f$ is measurable.

Fact B.27 (averages in a closed set)

Let $\mu(X) < \infty$, $g \in L^1(\mu)$ and S a closed subset of $\mathbb{C}$. If $\frac{1}{\mu(E)} \int_E g \, d\mu \in S$ for every $E \in \mathcal{S}$ with $\mu(E) > 0$, then $g(x) \in S$ for almost every x. (For σ-finite μ, apply this on each set of finite measure.)

Fact B.28 (sums over arbitrary index sets)

For $a_\alpha \geq 0$ ($\alpha \in A$, A any set), $\sum_{\alpha \in A} a_\alpha := \sup\{\sum_{\alpha \in F} a_\alpha : F \subset A \text{ finite}\}$. Hence if $\sum_{\alpha \in F} a_\alpha \leq M$ for every finite $F \subset A$, then $\sum_{\alpha \in A} a_\alpha \leq M$; and if the sum is finite, $a_\alpha \neq 0$ for at most countably many α.

Fact B.29 (inclusions between L^p spaces)

If $\mu(X) < \infty$ and $1 \leq p < q \leq \infty$, then $L^q(\mu) \subset L^p(\mu)$ and $\|f\|_p \leq \mu(X)^{1/p - 1/q} \|f\|_q$. In particular, on $\mathbb{T}$ with total mass 1, $L^2(\mathbb{T}) \subset L^1(\mathbb{T})$ and $\|f\|_1 \leq \|f\|_2$.

Chapter 4. Double Integrals

Overview

Results from Chapters 1–3 are cited as *Fact B.k* (see the *Background* notes). Interchanging the order of integration needs a measure on the plane. This chapter defines the **product** of two measures, the double integral, the conditions under which it equals the iterated integrals (Tonelli, Fubini), and the **convolution**.

4.1 Product Measures

Motivation. Iterated integration needs a measure on $X \times Y$ built from those on X and Y, defined on sets far more complicated than rectangles; hence the monotone class argument.

4.1.1 Setting up the problem

Let $(X, \mathcal{S}, \mu)$ and $(Y, \mathcal{T}, \nu)$ be measure spaces. A σ-algebra on $X \times Y$ should contain the **measurable rectangles**

$$\mathcal{R} = \{S \times T \subset X \times Y : \ S \in \mathcal{S}, \ T \in \mathcal{T}\}.$$

For $E \subset X \times Y$, $x \in X$, $y \in Y$ define the **sections**

$$E_x = \{y \in Y : (x, y) \in E\}, \qquad E^y = \{x \in X : (x, y) \in E\}.$$

If $E_x \in \mathcal{T}$ and $E^y \in \mathcal{S}$ for all x, y, let

$$\varphi : x \mapsto \nu(E_x) = \int_Y \chi_E(x, y)\, d\nu(y), \qquad \psi : y \mapsto \mu(E^y) = \int_X \chi_E(x, y)\, d\mu(x). \tag{1}$$

If φ, ψ are measurable and

$$\int_X \varphi \, d\mu = \int_Y \psi \, d\nu, \tag{2}$$

this common value should be the measure of E. We show that every E in the smallest σ-algebra $\mathcal{S} \times \mathcal{T}$ containing $\mathcal{R}$ has these properties.

4.1.2 The smallest σ-algebra

Let $\mathcal{U}$ be a collection of subsets of Z and $\Omega \neq \emptyset$ the family of σ-algebras containing $\mathcal{U}$ (the power set is one). Put

$$\mathcal{W} = \bigcap \{\mathcal{M} : \mathcal{M} \in \Omega\}.$$

$\emptyset \in \mathcal{W}$; if $A \in \mathcal{W}$ then $A^c \in \mathcal{M}$ for every $\mathcal{M} \in \Omega$, so $A^c \in \mathcal{W}$; likewise for countable unions. So $\mathcal{W}$ is a σ-algebra contained in every σ-algebra containing $\mathcal{U}$: the **smallest σ-algebra containing $\mathcal{U}$**.

4.1.3 Sections of sets in $\mathcal{S} \times \mathcal{T}$

Let

$$\mathcal{A} = \{E \in \mathcal{S} \times \mathcal{T} : \ E_x \in \mathcal{T} \ (x \in X)\}.$$

$\mathcal{R} \subset \mathcal{A}$, so it suffices that $\mathcal{A}$ is a σ-algebra: $\emptyset \in \mathcal{A}$, $(E^c)_x = (E_x)^c$ and $(\bigcup_n E_n)_x = \bigcup_n (E_n)_x$. Treating E^y likewise,

$$E \in \mathcal{S} \times \mathcal{T} \implies E_x \in \mathcal{T}, \ E^y \in \mathcal{S} \qquad (x \in X, \ y \in Y). \tag{3}$$

Exercise 4.1.1

In the argument above, show that $(E^c)_x = (E_x)^c$ and $(\bigcup_n E_n)_x = \bigcup_n (E_n)_x$.

Solution [Supplement]. $y \in (E^c)_x \iff (x,y) \in E^c \iff (x,y) \notin E \iff y \notin E_x \iff y \in (E_x)^c$. $y \in (\bigcup_n E_n)_x \iff (x,y) \in E_n$ for some $n \iff y \in (E_n)_x$ for some $n \iff y \in \bigcup_n (E_n)_x$. □

4.1.4 The monotone class argument

Assume from now on that

$$X = \bigsqcup_{n=1}^{\infty} X_n, \qquad Y = \bigsqcup_{m=1}^{\infty} Y_m, \qquad \mu(X_n),\ \nu(Y_m) < \infty.$$

To show (2) for all $E \in \mathcal{S} \times \mathcal{T}$, let

$$\mathcal{B} = \{E \in \mathcal{S} \times \mathcal{T} :\ \text{the functions } \varphi, \psi \text{ of (1) are measurable and (2) holds}\},$$
$$\mathcal{C} = \{E \in \mathcal{S} \times \mathcal{T} :\ E \cap (X_n \times Y_m) \in \mathcal{B}\ \ (m, n = 1, 2, \dots)\}.$$

Lemma 4.1.1

For the collections $\mathcal{B}$ and $\mathcal{C}$:

(a) $\mathcal{B}$ is closed under unions of increasing sequences.

(b) $\mathcal{B}$ is closed under countable disjoint unions.

(c) $\mathcal{C}$ is closed under unions of increasing sequences and intersections of decreasing sequences.

(d) If $E_1, \dots, E_N \in \mathcal{R}$ are pairwise disjoint, then $\bigcup_{n=1}^{N} E_n \in \mathcal{C}$.

Proof. (a) Let $E_i \in \mathcal{B}$ increase to E, with functions φ_i, ψ_i and φ, ψ. As $(E_i)_x \nearrow E_x$, Fact B.24(b) gives $\varphi_i \nearrow \varphi$, so φ is measurable; likewise ψ. By monotone convergence

$$\int_X \varphi \, d\mu = \lim_{i\to\infty} \int_X \varphi_i \, d\mu = \lim_{i\to\infty} \int_Y \psi_i \, d\nu = \int_Y \psi \, d\nu.$$

(b) For disjoint $E, F \in \mathcal{B}$ the functions in (1) add, so $\mathcal{B}$ is closed under finite disjoint unions; a countable disjoint union is an increasing union of finite ones, so use (a).

(c) If $E_i \in \mathcal{C}$ increase, then for each m, n

$$\langle E_i \cap (X_n \times Y_m) : i = 1, 2, \dots \rangle \tag{4}$$

increases in $\mathcal{B}$, so $\bigcup_i E_i \in \mathcal{C}$ by (a). For decreasing E_i, argue as in (a) with Fact B.24(c) (as $\nu(Y_m), \mu(X_n) < \infty$) and dominated convergence (bound $\nu(Y_m)\chi_{X_n}$).

(d) $\mathcal{R} \subset \mathcal{C}$: for $E = S \times T$, $\varphi = \nu(T \cap Y_m)\chi_{S \cap X_n}$ and $\psi = \mu(S \cap X_n)\chi_{T \cap Y_m}$ have equal integrals $\mu(S \cap X_n)\nu(T \cap Y_m)$. Then use (b). ■

Let $\mathcal{D}$ be the smallest collection containing $\mathcal{R}$ with (c) and (d) of Lemma 4.1.1. Then

$$\mathcal{D} \subset \mathcal{C} \subset \mathcal{S} \times \mathcal{T}, \tag{5}$$

and we show $\mathcal{D}$ is a σ-algebra. With $\mathcal{E}$ the finite disjoint unions of measurable rectangles, $\mathcal{D}$ is the smallest *monotone class* (closed under increasing unions and decreasing intersections) containing $\mathcal{E}$.

Exercise 4.1.2

Show that for any collection of sets there is a smallest collection containing it that is closed under unions of increasing sequences and intersections of decreasing sequences.

Solution [Supplement]**.** With $\Omega \neq \emptyset$ the monotone classes containing $\mathcal{U}$ (the power set is one), $\mathcal{W} = \bigcap \Omega$ contains $\mathcal{U}$, and a monotone sequence in $\mathcal{W}$ is one in every $\mathcal{M} \in \Omega$, so its limit is in $\mathcal{W}$. So $\mathcal{W}$ is the smallest monotone class containing $\mathcal{U}$. □

$\mathcal{E}$ is closed under differences and unions:

$$E, F \in \mathcal{E} \implies E \setminus F \in \mathcal{E},\ F \setminus E \in \mathcal{E},\ E \cup F \in \mathcal{E}. \tag{6}$$

Exercise 4.1.3

Prove (6).

Solution [Supplement]**.** (i) $(S \times T) \cap (S' \times T') = (S \cap S') \times (T \cap T')$, so for $E = \bigsqcup_i R_i$, $F = \bigsqcup_j R'_j$ in $\mathcal{E}$, $E \cap F = \bigsqcup_{i,j}(R_i \cap R'_j) \in \mathcal{E}$.

(ii) $(S \times T)^c = (S^c \times Y) \sqcup (S \times T^c)$, so $(\bigsqcup_i R_i)^c = \bigcap_i R_i^c \in \mathcal{E}$ by (i).

(iii) So $E \setminus F = E \cap F^c$, $F \setminus E \in \mathcal{E}$ and $E \cup F = (E \setminus F) \sqcup F \in \mathcal{E}$. □

For an arbitrary $E \subset X \times Y$ define

$$\Omega(E) = \{F \subset X \times Y :\ E \setminus F \in \mathcal{D},\ F \setminus E \in \mathcal{D},\ E \cup F \in \mathcal{D}\}.$$

$\Omega(E)$ also satisfies Lemma 4.1.1(c).

Exercise 4.1.4

Show that $\Omega(E)$ satisfies (c) of Lemma 4.1.1.

Solution [Supplement]**.** If $F_i \in \Omega(E)$ increase to F, then $E \setminus F_i \searrow E \setminus F$, $F_i \setminus E \nearrow F \setminus E$, $E \cup F_i \nearrow E \cup F$, all in the monotone class $\mathcal{D}$, so $F \in \Omega(E)$. Decreasing F_i likewise. □

For $E, F \in \mathcal{E}$, (6) gives $F \in \Omega(E)$, so $\mathcal{E} \subset \Omega(E)$ and $\mathcal{D} \subset \Omega(E)$:

$$E \in \mathcal{E} \implies \mathcal{D} \subset \Omega(E).$$

So for $E \in \mathcal{D}$, $F \in \mathcal{E}$: $E \in \Omega(F)$, i.e. $F \in \Omega(E)$ (symmetry). Hence $\mathcal{E} \subset \Omega(E)$ and $\mathcal{D} \subset \Omega(E)$:

$$E \in \mathcal{D} \implies \mathcal{D} \subset \Omega(E),$$

i.e. $\mathcal{D}$ is closed under differences and finite unions, hence complements ($X \times Y \in \mathcal{D}$), and countable unions $\bigcup_n E_n = \bigcup_n (E_1 \cup \cdots \cup E_n)$. So $\mathcal{D}$ is a σ-algebra, and by (5)

$$\mathcal{D} = \mathcal{C} = \mathcal{S} \times \mathcal{T}. \tag{7}$$

So for $E \in \mathcal{S} \times \mathcal{T} = \mathcal{C}$, the disjoint pieces $E \cap (X_n \times Y_m)$ lie in $\mathcal{B}$, and $E \in \mathcal{B}$ by Lemma 4.1.1(b): $\mathcal{S} \times \mathcal{T} \subset \mathcal{B}$.

Definition (σ-finite)

A measure space $(X, \mathcal{S}, \mu)$ is **σ-finite** if X is a countable union of sets of finite measure.

Theorem 4.1.2

Let $(X,\mathcal{S},\mu)$, $(Y,\mathcal{T},\nu)$ be measure spaces and let $\mathcal{S}\times\mathcal{T}$ be the smallest σ-algebra containing the measurable rectangles. If $E\in\mathcal{S}\times\mathcal{T}$, then $E_x\in\mathcal{T}$ and $E^y\in\mathcal{S}$ for all $x\in X$, $y\in Y$. If $(X,\mathcal{S},\mu)$ and $(Y,\mathcal{T},\nu)$ are σ-finite, then the functions φ,ψ of (1) are measurable and (2) holds.

(Proof: the discussion above.) Writing the common value of (2) as $(\mu\times\nu)(E)$,

$$\begin{aligned}(\mu\times\nu)(E) &= \int_X\Big[\int_Y \chi_E(x,y)\,d\nu(y)\Big]d\mu(x) = \int_X \nu(E_x)\,d\mu(x)\\ &= \int_Y\Big[\int_X \chi_E(x,y)\,d\mu(x)\Big]d\nu(y) = \int_Y \mu(E^y)\,d\nu(y).\end{aligned}\tag{8}$$

$\mu\times\nu$ is a measure: for disjoint E_n with union E, $E_x=\bigsqcup_n (E_n)_x$, and by Fact B.25

$$\sum_{n=1}^{\infty}(\mu\times\nu)(E_n)=\sum_{n=1}^{\infty}\int_X \nu((E_n)_x)\,d\mu(x)=\int_X\sum_{n=1}^{\infty}\nu((E_n)_x)\,d\mu(x)=\int_X\nu(E_x)\,d\mu(x)=(\mu\times\nu)(E).$$

$(X\times Y,\mathcal{S}\times\mathcal{T},\mu\times\nu)$ is the **product measure** of the σ-finite spaces.

Exercise 4.1.5

Show that $(\mu\times\nu)(S\times T)=\mu(S)\nu(T)$ for every $S\in\mathcal{S}$, $T\in\mathcal{T}$.

Solution [Supplement]. $\nu((S\times T)_x)=\nu(T)\chi_S(x)$, so by (8) $(\mu\times\nu)(S\times T)=\nu(T)\mu(S)$ $(0\cdot\infty=0)$. □

Exercise 4.1.6

If $(X,\mathcal{S},\mu)$ and $(Y,\mathcal{T},\nu)$ are σ-finite, show that $(X\times Y,\mathcal{S}\times\mathcal{T},\mu\times\nu)$ is σ-finite.

Solution [Supplement]. $X\times Y=\bigsqcup_{m,n}(X_n\times Y_m)$ with $(\mu\times\nu)(X_n\times Y_m)=\mu(X_n)\nu(Y_m)<\infty$ (Exercise 4.1.5). □

4.1.5 The product of Lebesgue measures; Borel sets

For Lebesgue measure $(\mathbb{R},\mathcal{M},m)$: an open $U\subset\mathbb{R}^2$ is the countable union of the

$$\{(a,b)\times(c,d)\subset U:\ a,b,c,d \text{ rational}\}$$

it contains, so open sets lie in $\mathcal{M}\times\mathcal{M}$. With $\mathcal{B}_n$ the smallest σ-algebra containing the open sets of $\mathbb{R}^n$,

$$\mathcal{B}_2\subset\mathcal{M}\times\mathcal{M}.$$

Members of $\mathcal{B}_n$ are the **Borel sets** (e.g. closed, G_δ, F_σ sets).

But $\mathcal{M}\times\mathcal{M}$ has a defect: for $\emptyset\neq E\in\mathcal{M}$ with $m(E)=0$ and $F\subset\mathbb{R}$, $F\notin\mathcal{M}$, (3) gives $E\times F\notin\mathcal{M}\times\mathcal{M}$ ($(E\times F)_x=F$ for $x\in E$), though $E\times F\subset E\times\mathbb{R}$, a null set.

Definition (complete measure)

A measure space $(X,\mathcal{S},\mu)$ is **complete** if

$$E\subset F,\ F\in\mathcal{S},\ \mu(F)=0\implies E\in\mathcal{S}.$$

So $(\mathbb{R}^2,\mathcal{M}\times\mathcal{M},m\times m)$ is not complete, while $(\mathbb{R},\mathcal{M},m)$ is (Fact B.2).

Proposition 4.1.3 (completion)

For a measure space $(X, \mathcal{S}, \mu)$ let

$$\mathcal{S}^* = \{E \subset X : \text{ there exist } A, B \in \mathcal{S} \text{ with } A \subset E \subset B,\ \mu(B \setminus A) = 0\},$$

and for $E \in \mathcal{S}^*$ put $\mu^*(E) = \mu(A)$. Then $(X, \mathcal{S}^*, \mu^*)$ is a complete measure space.

Proof. *Well-definedness.* If $A_i \subset E \subset B_i$, $\mu(B_i \setminus A_i) = 0$ $(i = 1, 2)$, then

$$A_2 \setminus A_1 \subset E \setminus A_1 \subset B_1 \setminus A_1,$$

so $\mu(A_2 \setminus A_1) = 0$, likewise $\mu(A_1 \setminus A_2) = 0$, and $\mu(A_1) = \mu(A_2)$.

$\mathcal{S}^*$ is a σ-algebra. $\emptyset \in \mathcal{S}^*$. If $A \subset E \subset B$, then $B^c \subset E^c \subset A^c$ with $A^c \setminus B^c = B \setminus A$. For $A_n \subset E_n \subset B_n$, $\mu(B_n \setminus A_n) = 0$, the unions $A \subset E \subset B$ satisfy

$$B \setminus A = \bigcup_n (B_n \setminus A) \subset \bigcup_n (B_n \setminus A_n),$$

so $\mu(B \setminus A) = 0$.

μ^* is a measure. Disjoint E_n have disjoint A_n, so

$$\mu^*(E) = \mu(A) = \sum_{n=1}^{\infty} \mu(A_n) = \sum_{n=1}^{\infty} \mu^*(E_n).$$

Completeness. If $E \subset F$, $\mu^*(F) = 0$, take $A \subset F \subset B$ with $\mu(B) = 0$; then $\emptyset \subset E \subset B$, so $E \in \mathcal{S}^*$. ■

Exercise 4.1.7

If $(X, \mathcal{S}, \mu)$ is complete, show that $\mathcal{S}^* = \mathcal{S}$.

Solution [Supplement]. $\mathcal{S} \subset \mathcal{S}^*$ $(A = B = E)$. Conversely, if $A \subset E \subset B$, $\mu(B \setminus A) = 0$, then $E \setminus A \subset B \setminus A$ is in $\mathcal{S}$ by completeness, and $E = A \cup (E \setminus A) \in \mathcal{S}$. □

4.1.6 Lebesgue measure on $\mathbb{R}^2$

Compare $\mathcal{B}_2^*$ and $(\mathcal{M} \times \mathcal{M})^*$; clearly $\mathcal{B}_2^* \subset (\mathcal{M} \times \mathcal{M})^*$.

For $S \in \mathcal{M}$, Fact B.3 gives an F_σ-set A and a G_δ-set B with $A \subset S \subset B$, $m(B \setminus A) = 0$; $A \times \mathbb{R}$, $B \times \mathbb{R} \in \mathcal{B}_2$ and

$$(m \times m)[(B \times \mathbb{R}) \setminus (A \times \mathbb{R})] = (m \times m)[(B \setminus A) \times \mathbb{R}] = 0,$$

so $S \times \mathbb{R} \in \mathcal{B}_2^*$; likewise $\mathbb{R} \times T \in \mathcal{B}_2^*$, hence $S \times T \in \mathcal{B}_2^*$. So $\mathcal{M} \times \mathcal{M} \subset \mathcal{B}_2^*$, and $(\mathcal{M} \times \mathcal{M})^* \subset \mathcal{B}_2^{**} = \mathcal{B}_2^*$ gives

$$(\mathcal{M} \times \mathcal{M})^* = \mathcal{B}_2^*.$$

Write $\mathcal{M}_2$, $m_2 = (m \times m)^*$: the complete space $(\mathbb{R}^2, \mathcal{M}_2, m_2)$ is **Lebesgue measure on** $\mathbb{R}^2$. (The $\mathbb{R}^1$ method also works directly on $\mathbb{R}^n$.)

Exercise 4.1.8

Use induction to define Lebesgue measure on $\mathbb{R}^n$.

Solution [Supplement]. Given complete $(\mathbb{R}^{n-1}, \mathcal{M}_{n-1}, m_{n-1})$ $(m_1 = m)$, σ-finite as $\mathbb{R}^{n-1} = \bigcup_k [-k,k]^{n-1}$, identify $\mathbb{R}^n = \mathbb{R}^{n-1} \times \mathbb{R}$, form $(\mathbb{R}^n, \mathcal{M}_{n-1} \times \mathcal{M}, m_{n-1} \times m)$ (Theorem 4.1.2), and set

$$\mathcal{M}_n = (\mathcal{M}_{n-1} \times \mathcal{M})^*, \qquad m_n = (m_{n-1} \times m)^*$$

(Proposition 4.1.3). As above, $\mathcal{B}_n \subset \mathcal{M}_{n-1} \times \mathcal{M}$ (rational boxes) and $\mathcal{M}_n = \mathcal{B}_n^*$ (each $S \in \mathcal{M}_{n-1}$ lies between an F_σ- and a G_δ-set differing by a null set). By Exercise 4.1.5, $m_n(\prod_i [a_i, b_i]) = \prod_i (b_i - a_i)$. □

4.2 Fubini's Theorem

Motivation. Tonelli swaps the order of integration for nonnegative functions, Fubini for integrable ones; counterexamples show a hypothesis is needed.

(8) swaps the order for characteristic functions; we extend to nonnegative and integrable functions. For f on $X \times Y$ let

$$f_x : y \mapsto f(x,y), \qquad f^y : x \mapsto f(x,y).$$

Proposition 4.2.1

If $f : X \times Y \to \mathbb{R}^*$ is $\mathcal{S} \times \mathcal{T}$-measurable, then for every $x \in X$ and $y \in Y$, f_x is $\mathcal{T}$-measurable and f^y is $\mathcal{S}$-measurable.

Proof. For $\alpha \in \mathbb{R}$, $E = \{f > \alpha\} \in \mathcal{S} \times \mathcal{T}$ has $E_x = \{f_x > \alpha\} \in \mathcal{T}$ by (3); likewise f^y. ■

"Measurable" on $X \times Y$ now means $\mathcal{S} \times \mathcal{T}$-measurable.

Exercise 4.2.1

If $f : \mathbb{R}^2 \to \mathbb{R}$ is measurable with respect to Lebesgue measure on $\mathbb{R}^2$, is $f_x : \mathbb{R} \to \mathbb{R}$ always Lebesgue measurable for every $x \in \mathbb{R}$?

Solution [Supplement]. No. For non-measurable $N \subset \mathbb{R}$, $f = \chi_{\{0\} \times N}$ is $\mathcal{M}_2$-measurable ($\{0\} \times N \subset \{0\} \times \mathbb{R}$, m_2-null, and completeness), but $f_0 = \chi_N$ is not. (Only a.e. x works: Theorem 4.2.6.) □

Theorem 4.2.2 (Tonelli)

Let $(X, \mathcal{S}, \mu)$ and $(Y, \mathcal{T}, \nu)$ be σ-finite. For every measurable $f : X \times Y \to [0, \infty]$:

(a) the functions $x \mapsto \int_Y f_x \, d\nu$ and $y \mapsto \int_X f^y \, d\mu$ are measurable;

(b)
$$\int_X \Big[\int_Y f_x \, d\nu \Big] d\mu = \int_{X \times Y} f \, d(\mu \times \nu) = \int_Y \Big[\int_X f^y \, d\mu \Big] d\nu.$$

Proof. By Fact B.5 take simple $s_n \nearrow f$ and put

$$\varphi(x) = \int_Y f_x \, d\nu, \qquad \varphi_n(x) = \int_Y (s_n)_x \, d\nu.$$

Characteristic functions satisfy (a), (b) by Theorem 4.1.2, hence so do simple functions. As $(s_n)_x \nearrow f_x$, monotone convergence gives $\varphi_n \nearrow \varphi$, so φ is measurable, and

$$\int_{X \times Y} f \, d(\mu \times \nu) = \lim_{n \to \infty} \int_{X \times Y} s_n \, d(\mu \times \nu) = \lim_{n \to \infty} \int_X \varphi_n \, d\mu = \int_X \varphi \, d\mu.$$

Likewise for the other order. ■

Exercise 4.2.2

Derive Fact B.25 from Tonelli's theorem.

Solution [Supplement]**.** Fact B.25: $\int_X \sum_n f_n\,d\mu = \sum_n \int_X f_n\,d\mu$ for measurable $f_n \ge 0$. Take $Y = \mathbb{N}$ with counting measure c (σ-finite) and $f(x,n) = f_n(x)$, measurable as $\{f > \alpha\} = \bigcup_n \{f_n > \alpha\} \times \{n\}$. As $\int_{\mathbb{N}} g\,dc = \sum_n g(n)$, Tonelli gives

$$\int_X \sum_n f_n(x)\,d\mu(x) = \int_{X\times\mathbb{N}} f\,d(\mu \times c) = \sum_n \int_X f_n\,d\mu.$$

□

For integrable functions we need a sufficient condition for integrability.

Theorem 4.2.3 (Fubini)

Let $(X, \mathcal{S}, \mu)$ and $(Y, \mathcal{T}, \nu)$ be σ-finite and let $f : X \times Y \to \mathbb{C}$ be measurable with

$$\int_X \Big[\int_Y |f(x,y)|\,d\nu(y)\Big] d\mu(x) < \infty.$$

Then:

(a) $f \in L^1(\mu \times \nu)$; moreover $f_x \in L^1(\nu)$ for almost every $x \in X$ and $f^y \in L^1(\mu)$ for almost every $y \in Y$;

(b) the functions $x \mapsto \int_Y f_x\,d\nu$ and $y \mapsto \int_X f^y\,d\mu$, defined almost everywhere, are integrable;

(c)
$$\int_X \Big[\int_Y f_x\,d\nu\Big] d\mu = \int_{X\times Y} f\,d(\mu\times\nu) = \int_Y \Big[\int_X f^y\,d\mu\Big] d\nu.$$

Proof. By Tonelli $\int_{X\times Y}|f| < \infty$, so $f \in L^1(\mu\times\nu)$. Take f real ($f = u + iv$) and put

$$\varphi(x) = \int_Y f_x\,d\nu, \qquad \varphi_1(x) = \int_Y (f_+)_x\,d\nu, \qquad \varphi_2(x) = \int_Y (f_-)_x\,d\nu.$$

By Tonelli,

$$\int_X \varphi_1\,d\mu = \int_{X\times Y} f_+\,d(\mu\times\nu) < \infty,$$

so $\varphi_1, \varphi_2 \in L^1(\mu)$, and

$$E = \{x \in X : \varphi_1(x) < \infty,\ \varphi_2(x) < \infty\}$$

has $\mu(E^c) = 0$. For $x \in E$, $f_x = (f_+)_x - (f_-)_x \in L^1(\nu)$ and $\varphi = \varphi_1 - \varphi_2 \in L^1(\mu)$. Tonelli for $f_\pm$ gives

$$\int_X \varphi\,d\mu = \int_X \varphi_1\,d\mu - \int_X \varphi_2\,d\mu = \int_{X\times Y} f_+\,d(\mu\times\nu) - \int_{X\times Y} f_-\,d(\mu\times\nu) = \int_{X\times Y} f\,d(\mu\times\nu).$$

The other half is identical. ■

Exercise 4.2.3

In Fubini's theorem let Y be the set of natural numbers with the measure of Fact B.4 (counting measure). What conclusion do you get? What if X is also taken to be $\mathbb{N}$ in the same way?

Solution [Supplement]**.** With $Y = \mathbb{N}$, counting measure c, $f_n(x) = f(x,n)$ and $\int_{\mathbb{N}} g\,dc = \sum_n g(n)$: if $\sum_n \int_X |f_n|\,d\mu < \infty$, then $\sum_n f_n(x)$ converges absolutely a.e., is integrable, and

$$\int_X \sum_{n=1}^{\infty} f_n\,d\mu = \sum_{n=1}^{\infty} \int_X f_n\,d\mu.$$

If also $X = \mathbb{N}$, for $a_{mn} = f(m,n)$ with $\sum_m \sum_n |a_{mn}| < \infty$,

$$\sum_{m=1}^{\infty}\sum_{n=1}^{\infty} a_{mn} = \sum_{n=1}^{\infty}\sum_{m=1}^{\infty} a_{mn},$$

both absolutely convergent; for $a_{mn} \geq 0$ (Tonelli) no condition is needed. □

Exercise 4.2.4

On $\mathbb{N} \times \mathbb{N}$ consider

$$f(m,n) = \begin{cases} 2 - 2^{-m}, & m = n, \\ -2 + 2^{-m}, & m = n+1, \\ 0, & \text{otherwise.} \end{cases}$$

Compute the two iterated sums and see how this relates to the hypotheses of Tonelli's and Fubini's theorems. Do the same for

$$f(x,y) = \frac{x^2 - y^2}{(x^2+y^2)^2}, \qquad x, y \in [1, \infty).$$

Solution [Supplement]. (i) For fixed n, only $m = n, n+1$ contribute: $(2 - 2^{-n}) + (-2 + 2^{-n-1}) = -2^{-n-1}$, so $\sum_n \sum_m f(m,n) = -\frac{1}{2}$. For fixed m: $m = 1$ gives $2 - \frac{1}{2} = \frac{3}{2}$; for $m \geq 2$, $n = m, m-1$ give 0. So $\sum_m \sum_n f(m,n) = \frac{3}{2}$. No contradiction: f changes sign (no Tonelli), and $\sum\sum|f| \geq \sum_n (2 - 2^{-n}) = \infty$ (no Fubini).

(ii) Since $\dfrac{\partial}{\partial y}\dfrac{y}{x^2+y^2} = \dfrac{x^2-y^2}{(x^2+y^2)^2}$,

$$\int_1^{\infty} f(x,y)\,dy = \Big[\frac{y}{x^2+y^2}\Big]_{y=1}^{\infty} = -\frac{1}{1+x^2}, \qquad \int_1^{\infty}\int_1^{\infty} f\,dy\,dx = -\int_1^{\infty}\frac{dx}{1+x^2} = -\frac{\pi}{4}.$$

As $f(y,x) = -f(x,y)$, the other order gives $+\frac{\pi}{4}$; f changes sign and $\iint|f| = \infty$, so neither theorem applies. □

4.2.1 Borel functions

A **Borel function** on $\mathbb{R}^n$ is $\mathcal{B}_n$-measurable; "measurable" means $\mathcal{M}_n$-measurable. Borel functions are measurable. The next result extends Fact B.26(d).

Proposition 4.2.4

If $f : X \to \mathbb{R}$ is measurable with respect to $(X, \mathcal{S}, \mu)$ and $g : \mathbb{R} \to \mathbb{R}$ is a Borel function, then $g \circ f : X \to \mathbb{R}$ is measurable.

Proof. As $g^{-1}(\alpha, \infty) \in \mathcal{B}_1$, show $f^{-1}(E) \in \mathcal{S}$ for $E \in \mathcal{B}_1$. Let

$$\mathcal{A} = \{E \in \mathcal{B}_1 : \ f^{-1}(E) \in \mathcal{S}\}.$$

$\mathcal{A}$ is a σ-algebra (Exercise 4.2.5) containing the open sets ($f^{-1}(a,b) = \{f > a\} \cap \{f < b\}$), so $\mathcal{A} = \mathcal{B}_1$. ■

Exercise 4.2.5

In the proof of Proposition 4.2.4, show that $\mathcal{A}$ is a σ-algebra.

Solution [Supplement]. $f^{-1}(\emptyset)=\emptyset$, $f^{-1}(E^c)=X\setminus f^{-1}(E)$, $f^{-1}(\bigcup_n E_n)=\bigcup_n f^{-1}(E_n)$, and $\mathcal{B}_1$ is a σ-algebra. □

Tonelli–Fubini need $\mathcal{M}\times\mathcal{M}$-measurability, tedious to check directly; Borel functions on $\mathbb{R}^2$ have it ($\mathcal{B}_2\subset\mathcal{M}\times\mathcal{M}$), so the next result extends the theorems to all measurable functions.

Proposition 4.2.5

For every measurable function f on $\mathbb{R}^n$ there is a Borel function g with $f=g$ almost everywhere.

Proof. Take $f\ge 0$ and, by Fact B.6, $f=\sum_k c_k\chi_{E_k}$ with $c_k\ge 0$, $E_k\in\mathcal{M}_n$. Choose Borel $A_k\subset E_k\subset B_k$ with $m_n(B_k\setminus A_k)=0$ and put

$$g(x)=\sum_{k=1}^{\infty} c_k\chi_{A_k}(x),\qquad x\in\mathbb{R}^n.$$

g is Borel (Fact B.26), and

$$\{x: f(x)\neq g(x)\}\subset\bigcup_{k=1}^{\infty}(E_k\setminus A_k)\subset\bigcup_{k=1}^{\infty}(B_k\setminus A_k),$$

a null set. ■

Theorem 4.2.6 (Tonelli–Fubini on $\mathbb{R}^2$)

Let f be a measurable function on $\mathbb{R}^2$.

(a) For almost every x and almost every y, f_x and f^y are measurable.

(b) If $0\le f\le\infty$, then

$$\int_{\mathbb{R}}\int_{\mathbb{R}} f(x,y)\,dx\,dy=\int_{\mathbb{R}\times\mathbb{R}} f\,dm_2=\int_{\mathbb{R}}\int_{\mathbb{R}} f(x,y)\,dy\,dx. \tag{9}$$

(c) If $f:\mathbb{R}^2\to\mathbb{C}$ and $\int_{\mathbb{R}}\int_{\mathbb{R}}|f(x,y)|\,dx\,dy<\infty$, then $f\in L^1(m_2)$; moreover $f_x, f^y\in L^1(m_1)$ for almost every x,y, and (9) holds.

Proof. By Proposition 4.2.5, $f=g+h$ with g Borel and $h=0$ a.e. g is $\mathcal{M}\times\mathcal{M}$-measurable, so it satisfies the theorem by Proposition 4.2.1 and Theorems 4.2.2, 4.2.3 ($\int g\,dm_2=\int g\,d(m\times m)$). It remains to treat h. Let

$$E=\{(x,y)\in\mathbb{R}^2: h(x,y)\neq 0\}.$$

Then $m_2(E)=0$, so $E\subset F\in\mathcal{M}\times\mathcal{M}$ with $(m\times m)(F)=0$, and by (8) $\int_{\mathbb{R}} m(F_x)\,dx=0$, so

$$A=\{x\in\mathbb{R}: m(F_x)>0\}$$

has $m(A)=0$. For $x\notin A$, $m(E_x)\le m(F_x)=0$ (completeness), and $h_x=0$ off E_x, so h_x is measurable and 0 a.e. Likewise h^y, proving (a);

(b) and (c) for h follow as all integrals of h, h_x, h^y (a.e. x,y) vanish. ■

Exercise 4.2.6

Complete the proof of Theorem 4.2.6.

Solution [Supplement]. Write $f=g+h$. (b): for $f\ge 0$ take $g\ge 0$ (replace g by g_+, h by $f-g_+$). For $x\notin A$, $f_x=g_x$ a.e., so $\int\int f\,dy\,dx=\int\int g\,dy\,dx$; and $\int f\,dm_2=\int g\,dm_2=\int g\,d(m\times m)$. Tonelli

for g and the symmetric argument with h^y give (9). (c): by (b) $\int|f|\,dm_2 = \int\int|f|\,dx\,dy < \infty$, so $f \in L^1(m_2)$, $g \in L^1(m \times m)$ and Fubini applies to g; as $f_x = g_x$ a.e. for a.e. x (likewise f^y), integrability of sections and (9) transfer to f. □

Example 1 (the Dirichlet integral)

Fubini for $f(x,t) = e^{-xt}\sin x$ on $[0,A]\times[0,\infty)$: as

$$\frac{1}{x} = \int_0^\infty e^{-xt}\,dt, \qquad x > 0,$$

we have

$$\int_0^1\int_0^\infty |e^{-xt}\sin x|\,dt\,dx \le \int_0^1 x\int_0^\infty e^{-xt}\,dt\,dx \le \int_0^1 dx = 1,$$
$$\int_1^A\int_0^\infty |e^{-xt}\sin x|\,dt\,dx \le \int_1^A\int_0^\infty e^{-xt}\,dt\,dx = \int_1^A \frac{1}{x}\,dx < \infty,$$

(with $|\sin x| \le x$ on $[0,1]$), Fubini applies:

$$\int_0^A \frac{\sin x}{x}\,dx = \int_0^A\int_0^\infty e^{-xt}\sin x\,dt\,dx = \int_0^\infty\int_0^A e^{-xt}\sin x\,dx\,dt$$
$$= \int_0^\infty \Big[\frac{e^{-At}}{1+t^2}(-t\sin A - \cos A) + \frac{1}{1+t^2}\Big]dt.$$

As

$$\int_0^\infty \Big|\frac{e^{-tA}}{1+t^2}(-t\sin A - \cos A)\Big|dt \le \int_0^\infty 2e^{-tA}\,dt = \frac{2}{A}$$

($|t\sin A + \cos A| \le 1+t$, $\frac{1+t}{1+t^2} \le 2$),

$$\lim_{A\to\infty}\int_0^A \frac{\sin x}{x}\,dx = \int_0^\infty \frac{dt}{1+t^2} = \frac{\pi}{2}.$$

4.3 Convolution of Integrable Functions

Motivation. Convolution averages f with weights g; it smooths, becomes multiplication under the Fourier transform, and approximates L^p functions; Fubini makes it well defined.

$L^1(X,\mu)$ carries a product when X has an operation and μ is invariant (Fact B.1); we treat only $\mathbb{R}$.

4.3.1 Definition

For measurable $f, g : \mathbb{R} \to \mathbb{C}$ and $x \in \mathbb{R}$ consider

$$\int_{\mathbb{R}} f(y)g(x-y)\,dy.$$

By Proposition 4.2.5 take f, g Borel. $(x,y) \mapsto y$ and $(x,y) \mapsto x-y$ are continuous, so by Proposition 4.2.4 and Fact B.26(a)

$$(x,y) \mapsto f(y)g(x-y) \tag{10}$$

is Borel (for measurable f, g, still measurable). By translation invariance (Fact B.1),

$$\int_{\mathbb{R}}\Big[\int_{\mathbb{R}}|f(y)g(x-y)|\,dx\Big]dy = \int_{\mathbb{R}}|f(y)|\Big[\int_{\mathbb{R}}|g(x-y)|\,dx\Big]dy = \int_{\mathbb{R}}|f(y)|\,dy\int_{\mathbb{R}}|g(x)|\,dx = \|f\|_1\|g\|_1,$$

so by Fubini

$$x \mapsto \int_{\mathbb{R}} f(y)g(x-y)\,dy \tag{11}$$

is defined a.e., with

$$\int_{\mathbb{R}} \Big| \int_{\mathbb{R}} f(y)g(x-y)\,dy \Big| dx \le \int_{\mathbb{R}} \Big[\int_{\mathbb{R}} |f(y)g(x-y)|\,dx \Big] dy = \|f\|_1 \|g\|_1.$$

Definition (convolution)

For $f, g \in L^1(\mathbb{R})$, (11) is the **convolution** $f * g \in L^1(\mathbb{R})$, and

$$\|f * g\|_1 \le \|f\|_1 \|g\|_1, \qquad f, g \in L^1(\mathbb{R}). \tag{12}$$

Proposition 4.3.1

For $f, g, h \in L^1(\mathbb{R})$ and constants $\alpha, \beta \in \mathbb{C}$:

(a) $[(f * g) * h](x) = [f * (g * h)](x)$ for almost every x;

(b) $(f * g)(x) = (g * f)(x)$ for every x (for which either side is defined);

(c) $[(\alpha f + \beta g) * h](x) = [\alpha(f * h) + \beta(g * h)](x)$ for almost every x.

Proof. (a) Fix x. Then

$$[(f * g) * h](x) = \int_{\mathbb{R}} (f * g)(y)h(x-y)\,dy = \int_{\mathbb{R}} \Big[\int_{\mathbb{R}} f(z)g(y-z)\,dz \Big] h(x-y)\,dy,$$

and

$$\begin{aligned} [f * (g * h)](x) &= \int_{\mathbb{R}} f(z)(g * h)(x-z)\,dz = \int_{\mathbb{R}} f(z) \Big[\int_{\mathbb{R}} g(y)h(x-z-y)\,dy \Big] dz \\ &= \int_{\mathbb{R}} f(z) \Big[\int_{\mathbb{R}} g(y-z)h(x-y)\,dy \Big] dz \end{aligned}$$

$(y \mapsto y - z)$. Now

$$\int_{\mathbb{R}} \int_{\mathbb{R}} |f(z)g(y-z)h(x-y)|\,dy\,dz = [(|f| * |g|) * |h|](x),$$

is finite for a.e. x ($|f| * |g|$, $|h| \in L^1$); for such x apply Fubini to $(y, z) \mapsto f(z)g(y-z)h(x-y)$.

(b) Directly,

$$(f * g)(x) = \int_{\mathbb{R}} f(y)g(x-y)\,dy = \int_{\mathbb{R}} f(-y)g(x+y)\,dy = \int_{\mathbb{R}} f(x-y)g(y)\,dy = (g * f)(x),$$

by reflection and translation invariance.

(c) For a.e. x, $y \mapsto f(y)h(x-y)$ and $g(y)h(x-y)$ are in L^1; use linearity (Fact B.10). ■

Definition (Banach algebra)

A Banach space X with a multiplication $(x, y) \mapsto xy$ that is associative, distributive and satisfies

$$\|xy\| \le \|x\|\|y\|, \qquad x, y \in X,$$

is a **Banach algebra**. By (12) and Proposition 4.3.1(a),(c), $L^1(\mathbb{R})$ is one under convolution.

Proposition 4.3.2

If g is differentiable and $f, g, g' \in L^1(\mathbb{R})$, then $f * g$ is differentiable and

$$(f * g)' = f * g'.$$

Exercise 4.3.1

Prove Proposition 4.3.2.

Solution [Supplement]. *Step 1: g is bounded and $g(b) - g(a) = \int_a^b g'$.* An everywhere differentiable function with integrable derivative satisfies the fundamental theorem of calculus (Ch. 5): $g(x) = g(0) + \int_0^x g'$. As $g' \in L^1$, $g(\pm\infty)$ exist, and they are 0 as $g \in L^1$; so g is bounded and $(f * g)(x)$ is defined for *every* x.

Step 2: an integral formula. For $x \in \mathbb{R}$, $h \neq 0$,

$$(f * g)(x + h) - (f * g)(x) = \int_{\mathbb{R}} f(y) \int_0^h g'(x - y + t)\, dt\, dy = \int_0^h (f * g')(x + t)\, dt,$$

by Fubini, which applies because $\int_{\mathbb{R}} \int_0^h |f(y)g'(x - y + t)|\, dt\, dy \leq |h|\, \|f\|_1 \|g'\|_1 < \infty$.

Step 3. So $\frac{(f*g)(x+h)-(f*g)(x)}{h}$ is the average of $f * g'$ between x and $x + h$, which tends to $(f * g')(x)$ wherever $f * g'$ is continuous (everywhere if g' is bounded, by Exercise 4.3.2). In general Step 2 makes $f * g$ absolutely continuous with $(f * g)' = f * g'$ a.e. (Lebesgue differentiation, Ch. 5).

Alternative under bounded g'. If $|g'| \leq M$, the difference quotient $f(y)\frac{g(x+h-y)-g(x-y)}{h}$ is bounded by $M|f(y)|$ and tends to $f(y)g'(x - y)$; dominated convergence gives $(f * g)'(x) = (f * g')(x)$ for every x. □

4.3.2 Convolution of L^p and L^q functions

Let $p, q > 1$ be conjugate exponents. For a function g and $s \in \mathbb{R}$ write $g_s(t) = g(t - s)$. If $f \in L^p(\mathbb{R})$ and $g \in L^q(\mathbb{R})$, Hölder's inequality gives

$$\int_{\mathbb{R}} |f(y)g(x - y)|\, dy = \int_{\mathbb{R}} |f(y)g_{-x}(-y)|\, dy \leq \|f\|_p \|g_{-x}\|_q = \|f\|_p \|g\|_q.$$

Hence $(f * g)(x)$ is defined for every $x \in \mathbb{R}$ and $\|f * g\|_\infty \leq \|f\|_p \|g\|_q$.

Exercise 4.3.2

If $f \in L^1(\mathbb{R})$ and $g \in L^\infty(\mathbb{R})$, show that $f * g$ is uniformly continuous.

Solution [Supplement]. Using $f * g = g * f$, for $x, y \in \mathbb{R}$,

$$\begin{aligned} |(f * g)(x) - (f * g)(y)| &= \left| \int_{\mathbb{R}} g(z)(f(x - z) - f(y - z))dz \right| \\ &\leq \|g\|_\infty \int_{\mathbb{R}} |f(x - z) - f(y - z)|\, dz = \|g\|_\infty \|f_{x-y} - f\|_1, \end{aligned}$$

after the substitution $w = y - z$. By the continuity of translation in L^1 (Fact B.23), $\|f_s - f\|_1 \to 0$ as $s \to 0$. Hence for $\epsilon > 0$ there is $\delta > 0$ with $|(f * g)(x) - (f * g)(y)| < \epsilon$ whenever $|x - y| < \delta$. □

Proposition 4.3.3

Let $p, q \in (1, \infty)$ be conjugate exponents. Then $f * g \in C_0(\mathbb{R})$ for every $f \in L^p(\mathbb{R})$ and $g \in L^q(\mathbb{R})$.

Proof. By Hölder's inequality again, for $x, y \in \mathbb{R}$

$$|(f * g)(x) - (f * g)(y)| \le \int_{\mathbb{R}} |f(z)(g_{-x}(-z) - g_{-y}(-z))|\, dz \le \|f\|_p \|g_{-x} - g_{-y}\|_q,$$

so by Fact B.23 (continuity of translation in L^q, $q < \infty$), $f * g$ is continuous. Next, by Fact B.22 there are sequences $\langle f_n \rangle, \langle g_n \rangle$ in $C_c(\mathbb{R})$ with

$$\lim_{n\to\infty} \|f_n - f\|_p = 0, \qquad \lim_{n\to\infty} \|g_n - g\|_q = 0.$$

If the supports of f_n and g_n lie in $[-A, A]$ and $[-B, B]$, then the support of $f_n * g_n$ lies in $[-A - B, A + B]$, so $\langle f_n * g_n \rangle$ is a sequence in $C_c(\mathbb{R})$. Then

$$\begin{aligned} \|f_n * g_n - f * g\|_\infty &\le \|f_n * g_n - f_n * g\|_\infty + \|f_n * g - f * g\|_\infty \\ &\le \|f_n\|_p \|g_n - g\|_q + \|f_n - f\|_p \|g\|_q \to 0, \end{aligned}$$

so $\langle f_n * g_n \rangle$ converges uniformly to $f * g$. A uniform limit of functions in $C_c(\mathbb{R})$ belongs to $C_0(\mathbb{R})$. ■

4.3.3 Averaging kernels

It is usually hard to guess what $f * g$ looks like, but sometimes one can. Consider

$$k_r = \frac{1}{2r} \chi_{[-r,r]}, \qquad r > 0. \tag{13}$$

By definition

$$(k_r * f)(x) = \int_{-\infty}^{\infty} k_r(x - y) f(y)\, dy = \frac{1}{2r} \int_{x-r}^{x+r} f(y)\, dy,$$

the average of f over the neighborhood $[x - r, x + r]$ of x. So $k_r * f$ should be better behaved than f and approach f as $r \to 0$.

$k_r * f$ is uniformly continuous for $f \in L^1(\mathbb{R})$. By Fact B.15 (absolute continuity of the integral), for any $\epsilon > 0$ there is $\delta \in (0, 2r)$ with

$$E \in \mathcal{M},\ m(E) \le \delta \implies \int_E |f(t)|\, dt < 2r\epsilon.$$

If $|x - y| < \frac{\delta}{2} < r$, let $E = [x - r, x + r] \triangle [y - r, y + r]$, where $A \triangle B = (A \cup B) \setminus (A \cap B)$; then $m(E) = 2|x - y| \le \delta$. Therefore

$$|(k_r * f)(x) - (k_r * f)(y)| = \frac{1}{2r} \Big| \int_{x-r}^{x+r} f(t)\, dt - \int_{y-r}^{y+r} f(t)\, dt \Big| \le \frac{1}{2r} \int_E |f(t)|\, dt < \epsilon,$$

so $f * k_r$ is uniformly continuous on $\mathbb{R}$.

Uniform approximation. If f is uniformly continuous, for each $\epsilon > 0$ there is $\delta > 0$ with $|x - y| < \delta \Rightarrow |f(x) - f(y)| < \epsilon$. If $r < \delta$, then

$$|f(x) - (f * k_r)(x)| \le \frac{1}{2r} \int_{x-r}^{x+r} |f(x) - f(t)|\, dt < \epsilon,$$

so $f * k_r \to f$ uniformly as $r \to 0$.

Exercise 4.3.3

For $f = \chi_{[-1,1]}$ sketch $x \mapsto (f * k_r)(x)$ for $r = 1, \frac{1}{2}, \frac{1}{3}, \frac{1}{5}, \frac{1}{10}$. Do the same for $f(x) = \max\{1 - |x|, 0\}$.

Solution [Supplement]. *(i)* $f = \chi_{[-1,1]}$. $(f * k_r)(x) = \frac{1}{2r}\, m([x-r, x+r] \cap [-1,1])$, an even function. For $r \le 1$:

$$(f * k_r)(x) = \begin{cases} 1, & |x| \le 1 - r, \\ \dfrac{1 + r - |x|}{2r}, & 1 - r \le |x| \le 1 + r, \\ 0, & |x| \ge 1 + r. \end{cases}$$

A trapezoid: height 1 on $[-(1-r), 1-r]$, linear ramps of width $2r$ through $\frac{1}{2}$ at ± 1. For $r = 1$ it is the triangle $(2 - |x|)/2$ on $[-2, 2]$; as $r \to 0$ it tends to $\chi_{[-1,1]}$ pointwise (to $\frac{1}{2}$ at ± 1), not uniformly, f being discontinuous.

(ii) $f(x) = \max\{1 - |x|, 0\}$ *(tent).* For $r \le \frac{1}{2}$ and $x \ge 0$ (the function is even), a direct integration gives

$$(f * k_r)(x) = \begin{cases} 1 - \dfrac{x^2 + r^2}{2r}, & 0 \le x \le r, \\ 1 - x, & r \le x \le 1 - r, \\ \dfrac{(1 + r - x)^2}{4r}, & 1 - r \le x \le 1 + r, \\ 0, & x \ge 1 + r. \end{cases}$$

The tent's corners at $0, \pm 1$ become parabolic arcs of width $2r$: the result is C^1 with maximum $1 - \frac{r}{2}$, and $\|f * k_r - f\|_\infty = \frac{r}{2} \to 0$. For $r = 1$: $(f * k_1)(x) = \frac{1}{2} - \frac{x^2}{4}$ for $|x| \le 1$ and $\frac{(2-|x|)^2}{4}$ for $1 \le |x| \le 2$. □

Exercise 4.3.4

Let $f_0 = \chi_{[-1,1]}$ and $f_{n+1} = f_n * f_0$ $(n = 0, 1, 2, \dots)$. Sketch how the graphs of this sequence change.

Solution [Supplement]. $f_1(x) = m([x-1, x+1] \cap [-1,1]) = \max\{2 - |x|, 0\}$, a triangle of height 2 on $[-2, 2]$. Inductively, since $(f_n * f_0)(x) = \int_{x-1}^{x+1} f_n$:

- f_n is even, nonnegative, supported in $[-(n+1), n+1]$, increasing on $(-(n+1), 0]$;
- f_n is a piecewise polynomial of degree n with breakpoints at the integers of the same parity as $n + 1$, and $f_n \in C^{n-1}$ (each convolution with f_0 gains one derivative: $f'_{n+1}(x) = f_n(x + 1) - f_n(x - 1)$);
- $\int f_n = \left(\int f_0\right)^{n+1} = 2^{n+1}$ by Fubini.

So the graphs go step → triangle → quadratic bell → $\cdots$, smoother and wider: $f_n / 2^{n+1}$ is the density of a sum of $n + 1$ independent uniforms on $[-1, 1]$ (variance $(n+1)/3$), so by the CLT it approaches $\frac{2^{n+1}}{\sqrt{2\pi(n+1)/3}} e^{-3x^2/(2(n+1))}$. □

Chapter 5. Fourier Transforms

Overview

Results from Chapters 1–3 are cited as *Fact B.k* (see the *Background* notes). This chapter studies Fourier series of periodic functions and Fourier transforms of integrable functions, with Fubini's theorem and convolution as the key tools.

5.1 Fourier Series

Motivation. Is a periodic function the sum of its harmonics e^{inx}? Fejér's kernel gives a clean yes for Cesàro means and shows the coefficients determine the function.

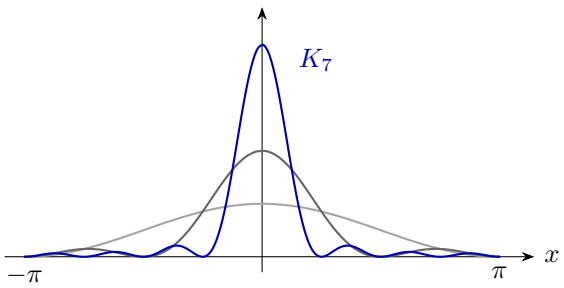


Figure. *Fejér kernels K_N ($N = 1, 3, 7$): nonnegative, integral 1, concentrating at 0, so $K_N * f \to f$.*

5.1.1 The circle $\mathbb{T}$

Let $\mathbb{T} = \mathbb{R}/\{2\pi n : n \in \mathbb{Z}\}$ with the distance from $\mathbb{R}$; 2π-periodic functions are functions on $\mathbb{T}$, and $C(\mathbb{T})$ consists of the continuous f on $[-\pi, \pi]$ with $f(-\pi) = f(\pi)$. As $\mathbb{T} = (-\pi, \pi]$, use Lebesgue measure normalized to mass 1:

$$\int_{\mathbb{T}} f = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x)\, dx.$$

The convolution of §4.3 is defined on $L^1(\mathbb{T})$ in the same way, making $L^1(\mathbb{T})$ a Banach algebra: for $f, g : \mathbb{T} \to \mathbb{C}$, $t \in \mathbb{T}$,

$$(f * g)(t) = \int_{\mathbb{T}} f(s) g(t - s)\, ds.$$

If $f, g \in L^1(\mathbb{T})$, then as in §4.3, $(f * g)(t)$ is defined for almost every t, $f * g \in L^1(\mathbb{T})$,

$$\|f * g\|_1 \le \|f\|_1 \|g\|_1, \qquad f, g \in L^1(\mathbb{T}),$$

and the identities of Proposition 4.3.1 hold.

5.1.2 Fourier coefficients

Definition (Fourier coefficients, Fourier series)

For $f \in L^1(\mathbb{T})$ and $n = 0, \pm 1, \pm 2, \dots$ define

$$\widehat{f}(n) = \int_{\mathbb{T}} f(t) e^{-int}\, dt,$$

so that $|\widehat{f}(n)| \le \|f\|_1$. With

$$u_n(t) = e^{int}, \qquad t \in \mathbb{T}, \tag{1}$$

one checks directly that

$$u_n * f = \widehat{f}(n) u_n, \qquad n \in \mathbb{Z},\ f \in L^1(\mathbb{T}); \tag{2}$$

in particular $(u_n * f)(0) = \widehat{f}(n)$. Also $\widehat{u_n}(m) = 1$ if $m = n$ and 0 if $m \neq n$. The series

$$\sum_{n=-\infty}^{\infty} \widehat{f}(n) u_n \tag{3}$$

is the **Fourier series** of $f \in L^1(\mathbb{T})$, with **Fourier coefficients** $\widehat{f}(n)$; without conditions on them, (3) need not define a function.

Indeed (2): $(u_n * f)(t) = \int_{\mathbb{T}} f(s) e^{in(t-s)}\, ds = e^{int}\widehat{f}(n)$.

Exercise 5.1.1

If $\widehat{f} \in \ell^1(\mathbb{Z})$, show that the series (3) converges uniformly to a continuous function.

Solution [Supplement]. $|\widehat{f}(n)u_n(t)| = |\widehat{f}(n)|$ and $\sum_n |\widehat{f}(n)| < \infty$, so by the M-test the partial sums (any $\sum_{-M}^{N}$) are uniformly Cauchy, with continuous uniform limit. □

Proposition 5.1.1

For $f, g \in L^1(\mathbb{T})$, constants $a, b \in \mathbb{C}$ and $n \in \mathbb{Z}$,

$$\widehat{af + bg}(n) = a\widehat{f}(n) + b\widehat{g}(n), \qquad \widehat{f * g}(n) = \widehat{f}(n)\,\widehat{g}(n). \tag{4}$$

Moreover for each $t \in \mathbb{T}$

$$\widehat{f_t}(n) = \widehat{f}(n) u_n(-t), \qquad n \in \mathbb{Z}, \tag{5}$$

where $f_t(s) = f(s - t)$.

Proof. The first identity is linearity (Fact B.10); for the second, by Fubini,

$$\begin{aligned}\widehat{f * g}(n) &= \int_{\mathbb{T}} (f * g)(t) e^{-int}\, dt = \int_{\mathbb{T}} e^{-int} \int_{\mathbb{T}} f(t - s) g(s)\, ds\, dt \\ &= \int_{\mathbb{T}} g(s) e^{-ins} \int_{\mathbb{T}} f(t - s) e^{-in(t-s)}\, dt\, ds = \int_{\mathbb{T}} g(s) e^{-ins}\, ds \int_{\mathbb{T}} f(t) e^{-int}\, dt = \widehat{g}(n)\widehat{f}(n).\end{aligned}$$

For (5),

$$\widehat{f_t}(n) = \int_{\mathbb{T}} f(s - t) e^{-ins}\, ds = \int_{\mathbb{T}} f(s) e^{-in(s+t)}\, ds = \widehat{f}(n) u_n(-t).$$

■

Proposition 5.1.2

If f is a 2π-periodic differentiable function with $f' \in L^1(\mathbb{T})$, then

$$\widehat{f'}(n) = in\widehat{f}(n), \qquad n \in \mathbb{Z}.$$

Proof. Integrating by parts (the boundary terms cancel by periodicity),

$$\widehat{f'}(n) = \int_{\mathbb{T}} f'(t) e^{-int}\, dt = in \int_{\mathbb{T}} f(t) e^{-int}\, dt = in\widehat{f}(n).$$

■

The next theorem is a variant of Fact B.21, proved as in Fact B.22.

Theorem 5.1.3

For each $p \in [1, \infty)$, the space $C(\mathbb{T})$ of continuous functions is dense in $L^p(\mathbb{T})$.

For a function f on $\mathbb{T}$ and $t \in \mathbb{T}$ define the translate as in Fact B.19:

$$f_t(s) = f(s - t), \qquad s \in \mathbb{T}.$$

The next theorem is proved like Fact B.23, with Theorem 5.1.3 in place of Fact B.22.

Theorem 5.1.4

For each $f \in L^p(\mathbb{T})$ $(1 \le p < \infty)$, the map

$$t \mapsto f_t : \mathbb{T} \to L^p(\mathbb{T})$$

is uniformly continuous.

Exercise 5.1.2

Prove Theorems 5.1.3 and 5.1.4.

Solution [Supplement]. *Theorem 5.1.3.* Simple functions are dense in $L^p(\mathbb{T})$ (Fact B.21), so approximate χ_E, $E \subset (-\pi, \pi)$ measurable. By regularity take compact K and open U with $K \subset E \subset U \subset (-\pi, \pi)$, $m(U \setminus K) < \epsilon$. Let

$$\varphi(x) = \frac{d(x, U^c)}{d(x, U^c) + d(x, K)},$$

continuous, $0 \le \varphi \le 1$, $\varphi = 1$ on K, $\varphi = 0$ off U; it vanishes near $\pm\pi$, so $\varphi \in C(\mathbb{T})$. As $|\chi_E - \varphi| \le \chi_{U \setminus K}$,

$$\|\chi_E - \varphi\|_p^p \le \frac{m(U \setminus K)}{2\pi} < \epsilon.$$

Taking linear combinations proves density.

Theorem 5.1.4. For $f \in L^p(\mathbb{T})$ take $g \in C(\mathbb{T})$ with $\|f - g\|_p < \epsilon$, uniformly continuous: $|g(u) - g(v)| < \epsilon$ for $d(u, v) < \delta$. By translation invariance $\|f_t - g_t\|_p = \|f - g\|_p$, and for $d(t, s) < \delta$, $\|g_t - g_s\|_p \le \|g_t - g_s\|_\infty \le \epsilon$ (mass 1), so

$$\|f_t - f_s\|_p \le \|f_t - g_t\|_p + \|g_t - g_s\|_p + \|g_s - f_s\|_p < 3\epsilon.$$

□

5.1.3 Approximate identities

The central theorem on Fourier series: by (7), $\langle h_n \rangle$ acts like an identity in the Banach algebra $L^1(\mathbb{T})$.

Theorem 5.1.5

If a sequence $\langle h_n \rangle$ satisfies

$$h_n \geq 0, \qquad \int_{\mathbb{T}} h_n = 1, \qquad \lim_{n\to\infty} \int_{\mathbb{T}\setminus[-\delta,\delta]} h_n = 0 \quad (0 < \delta < \pi), \tag{6}$$

(the third condition meaning: for *every* $\delta \in (0,\pi)$ the limit is 0), then for every $f \in L^1(\mathbb{T})$

$$\lim_{n\to\infty} \|f - h_n * f\|_1 = 0. \tag{7}$$

Proof. Let $\epsilon > 0$. By Theorem 5.1.4 there is $\delta > 0$ with

$$|s| < \delta \implies \|f - f_s\|_1 < \epsilon,$$

and then N with

$$n \geq N \implies \int_{\mathbb{T}\setminus[-\delta,\delta]} h_n(s)\,ds < \frac{\epsilon}{2\|f\|_1}.$$

For every $t \in \mathbb{T}$,

$$f(t) - (h_n * f)(t) = f(t)\int_{\mathbb{T}} h_n(s)\,ds - \int_{\mathbb{T}} h_n(s) f(t-s)\,ds = \int_{\mathbb{T}} h_n(s)\big[f(t) - f_s(t)\big]ds.$$

Hence for $n \geq N$, using Tonelli,

$$\begin{aligned}
\|f - h_n * f\|_1 &\leq \int_{\mathbb{T}}\int_{\mathbb{T}} h_n(s)|f(t) - f_s(t)|\,ds\,dt = \int_{\mathbb{T}} h_n(s)\int_{\mathbb{T}} |f(t) - f_s(t)|\,dt\,ds \\
&= \int_{\mathbb{T}} h_n(s)\|f - f_s\|_1\,ds \\
&= \int_{[-\delta,\delta]} h_n(s)\|f - f_s\|_1\,ds + \int_{\mathbb{T}\setminus[-\delta,\delta]} h_n(s)\|f - f_s\|_1\,ds \\
&\leq \epsilon\int_{[-\delta,\delta]} h_n(s)\,ds + 2\|f\|_1\int_{\mathbb{T}\setminus[-\delta,\delta]} h_n(s)\,ds \leq \epsilon + \epsilon = 2\epsilon.
\end{aligned}$$

■

The same method gives the next theorem, more simply (no Tonelli–Fubini).

Theorem 5.1.6

If $\langle h_n \rangle$ satisfies (6), then $\lim_{n\to\infty}\|f - h_n * f\|_\infty = 0$ for every $f \in C(\mathbb{T})$.

Proof. Since f is uniformly continuous, choose $\delta > 0$ with

$$|s| < \delta \implies \|f - f_s\|_\infty < \epsilon,$$

and N with

$$n \geq N \implies \int_{\mathbb{T}\setminus[-\delta,\delta]} h_n(s)\,ds < \frac{\epsilon}{2\|f\|_\infty}.$$

Then for every $t \in \mathbb{T}$ and $n \geq N$,

$$\begin{aligned}
|f(t) - (h_n * f)(t)| &\leq \int_{\mathbb{T}} h_n(s)|f(t) - f_s(t)|\,ds = \Big(\int_{[-\delta,\delta]} + \int_{\mathbb{T}\setminus[-\delta,\delta]}\Big) h_n(s)|f(t) - f_s(t)|\,ds \\
&\leq \epsilon\int_{[-\delta,\delta]} h_n(s)\,ds + 2\|f\|_\infty\int_{\mathbb{T}\setminus[-\delta,\delta]} h_n(s)\,ds \leq \epsilon + \epsilon = 2\epsilon.
\end{aligned}$$

■

Exercise 5.1.3

In the proof of Theorem 5.1.6, show that there is $\delta > 0$ with $|s| < \delta \Rightarrow \|f - f_s\|_\infty < \epsilon$.

Solution [Supplement]. f is uniformly continuous on the compact $\mathbb{T}$: $|f(u) - f(v)| < \epsilon/2$ for $d(u,v) < \delta$. For $|s| < \delta$, $d(t, t-s) < \delta$, so $\|f - f_s\|_\infty \le \epsilon/2 < \epsilon$. □

Exercise 5.1.4

Use Theorem 5.1.6 to prove Theorem 5.1.5.

Solution [Supplement]. Let $f \in L^1(\mathbb{T})$, $\epsilon > 0$, and choose $g \in C(\mathbb{T})$ with $\|f - g\|_1 < \epsilon$ (Theorem 5.1.3). Since $\mathbb{T}$ has total mass 1, $\|\cdot\|_1 \le \|\cdot\|_\infty$, and $\|h_n\|_1 = 1$, so

$$\begin{aligned}\|f - h_n * f\|_1 &\le \|f - g\|_1 + \|g - h_n * g\|_1 + \|h_n * (g - f)\|_1 \\ &\le \epsilon + \|g - h_n * g\|_\infty + \|h_n\|_1 \|g - f\|_1 < 2\epsilon + \|g - h_n * g\|_\infty.\end{aligned}$$

By Theorem 5.1.6 the last term tends to 0, so $\limsup_n \|f - h_n * f\|_1 \le 2\epsilon$. □

5.1.4 The Fejér kernel

Sequences satisfying (6) are easy to find, e.g. $\langle n\pi\chi_{[-1/n,1/n]} : n = 1, 2, \dots \rangle$. A more important example: for $n = 1, 2, \dots$ put

$$K_n = \sum_{k=-n}^{n} \Big(1 - \frac{|k|}{n+1}\Big) u_k. \tag{8}$$

$\langle K_n \rangle$ is the **Fejér kernel**.

Exercise 5.1.5

Show that $\lim_{n\to\infty} \widehat{K_n}(k) = 1$ for every $k \in \mathbb{Z}$.

Solution [Supplement]. Since $\widehat{u_j}(k) = \delta_{jk}$, $\widehat{K_n}(k) = 1 - \frac{|k|}{n+1}$ for $|k| \le n$ and 0 for $|k| > n$. For fixed k and $n \ge |k|$ this is $1 - |k|/(n+1) \to 1$. □

Since $\int_{\mathbb{T}} u_n = 0$ for $n \in \mathbb{Z} \setminus \{0\}$, the second condition $\int_{\mathbb{T}} K_n = 1$ is immediate. Expanding the left side shows

$$\Big(-\frac{1}{4}u_{-1} + \frac{1}{2} - \frac{1}{4}u_1\Big) K_n(t) = \frac{1}{n+1}\Big(-\frac{1}{4}u_{-(n+1)} + \frac{1}{2} - \frac{1}{4}u_{n+1}\Big),$$

from which

$$K_n(t) = \frac{1}{n+1} \frac{\sin^2\big(\frac{n+1}{2}t\big)}{\sin^2\big(\frac{1}{2}t\big)}, \qquad n = 0, 1, 2, \dots \tag{9}$$

Exercise 5.1.6

Prove (9).

Solution [Supplement]. For $0 \le j, l \le n$, the number of pairs with $j - l = k$ is $n + 1 - |k|$ ($|k| \le n$). Hence

$$\Big|\sum_{j=0}^{n} e^{ijt}\Big|^2 = \sum_{j,l=0}^{n} e^{i(j-l)t} = \sum_{k=-n}^{n} (n + 1 - |k|) e^{ikt} = (n+1) K_n(t).$$

For $t \neq 0$,

$$\sum_{j=0}^{n} e^{ijt} = \frac{e^{i(n+1)t} - 1}{e^{it} - 1} = \frac{e^{i(n+1)t/2}}{e^{it/2}} \cdot \frac{\sin\frac{(n+1)t}{2}}{\sin\frac{t}{2}},$$

whose squared modulus is $\sin^2 \frac{(n+1)t}{2} / \sin^2 \frac{t}{2}$: this is (9). (Equivalently, $-\frac{1}{4}u_{-1} + \frac{1}{2} - \frac{1}{4}u_1 = \sin^2 \frac{t}{2}$, and similarly on the right.) □

Since $t \mapsto \sin^2(\frac{1}{2}t)$ is increasing on $[0, \pi]$ and K_n is even, for every $\delta > 0$

$$0 \le K_n(t) \le \frac{1}{n+1} \frac{1}{\sin^2(\frac{1}{2}\delta)}, \qquad t \in \mathbb{T} \setminus [-\delta, \delta],$$

so $\langle K_n \rangle$ satisfies all of (6).

Exercise 5.1.7

Sketch the graph of K_n for $n = 1, 2, 3, 5, 10$.

Solution [Supplement]. $K_1(t) = 1 + \cos t$, $K_2(t) = 1 + \frac{4}{3}\cos t + \frac{2}{3}\cos 2t$; in general K_n is even, nonnegative, with maximum $K_n(0) = n + 1$, zeros at $t = \frac{2\pi j}{n+1}$, humps of height $\le \frac{1}{(n+1)\sin^2(t/2)}$ between them, and average 1. For $n = 1, 2, 3, 5, 10$ the peak height is $2, 3, 4, 6, 11$ and its width $\frac{4\pi}{n+1}$ shrinks: the mass concentrates at 0. □

5.1.5 Consequences

Each K_n is a finite linear combination of the u_n: a **trigonometric polynomial** (space $T(\mathbb{T})$). $P = \sum_{n=-N}^{N} a_n u_n$ has $\widehat{P}(n) = a_n$.

By (2), $K_n * f$ is a trigonometric polynomial for every $f \in L^1(\mathbb{T})$, and by Theorem 5.1.5, $K_n * f \to f$ in L^1. Hence:

Theorem 5.1.7

$T(\mathbb{T})$ is dense in $L^1(\mathbb{T})$.

If $\widehat{f}(n) = 0$ for all $n \in \mathbb{Z}$, then $K_n * f = 0$ by (2), so $\|f\|_1 = \lim \|f - K_n * f\|_1 = 0$:

Theorem 5.1.8 (uniqueness)

If $f \in L^1(\mathbb{T})$ and $\widehat{f} = 0$, then $f = 0$ almost everywhere.

For $a \in \ell^1(\mathbb{Z})$, $\sum_k |a_k u_k(t)| = \sum_k |a_k| < \infty$, so $g = \sum_k a_k u_k$ is defined, and by Fubini for counting measure (Exercise 4.2.3)

$$\widehat{g}(n) = \int_{\mathbb{T}} \Big(\sum_{k \in \mathbb{Z}} a_k u_{k-n} \Big) = \sum_{k \in \mathbb{Z}} a_k \int_{\mathbb{T}} u_{k-n} = a_n.$$

In particular, if $f \in L^1(\mathbb{T})$ and $\widehat{f} \in \ell^1(\mathbb{Z})$, then $g = \sum_k \widehat{f}(k) u_k$ and f have the same Fourier coefficients, so Theorem 5.1.8 gives:

Corollary 5.1.9

If $f \in L^1(\mathbb{T})$ and $\widehat{f} \in \ell^1(\mathbb{Z})$, then

$$f(t) = \sum_{n \in \mathbb{Z}} \widehat{f}(n) e^{int}$$

for almost every $t \in \mathbb{T}$.

On the other hand, given $\epsilon > 0$ take $P \in T(\mathbb{T})$ with $\|f - P\|_1 < \epsilon$ and let $N = \max\{|n| : \widehat{P}(n) \neq 0\}$. For $|n| > N$,

$$|\widehat{f}(n)| = |\widehat{f}(n) - \widehat{P}(n)| \leq \|f - P\|_1 < \epsilon.$$

This proves:

Theorem 5.1.10 (Riemann–Lebesgue)

$\lim_{n\to\pm\infty} \widehat{f}(n) = 0$ for every $f \in L^1(\mathbb{T})$.

Exercise 5.1.8

If a 2π-periodic function $f : \mathbb{R} \to \mathbb{C}$ is twice differentiable and $f'' \in L^1(\mathbb{T})$, show that $\widehat{f} \in \ell^1(\mathbb{Z})$.

Solution [Supplement]**.** f' is differentiable, hence continuous and so bounded on $\mathbb{T}$; thus $f' \in L^1(\mathbb{T})$. Applying Proposition 5.1.2 to f and to f',

$$\widehat{f''}(n) = in\,\widehat{f'}(n) = (in)^2\widehat{f}(n) = -n^2\widehat{f}(n).$$

Hence $|\widehat{f}(n)| = |\widehat{f''}(n)|/n^2 \leq \|f''\|_1/n^2$ for $n \neq 0$, and $\sum_n |\widehat{f}(n)| \leq |\widehat{f}(0)| + \|f''\|_1 \sum_{n\neq 0} n^{-2} < \infty$. □

Let $c_0(\mathbb{Z})$ be the vector space of functions $a : \mathbb{Z} \to \mathbb{C}$ with $\lim_{n\to\pm\infty} a(n) = 0$, with the norm of $\ell^\infty(\mathbb{Z})$.

Exercise 5.1.9

Show that the normed space $c_0(\mathbb{Z})$ is a Banach space.

Solution [Supplement]**.** As $\ell^\infty(\mathbb{Z})$ is complete, show $c_0(\mathbb{Z})$ is closed. If $\|a_k - a\|_\infty < \epsilon$ and $|a_k(n)| < \epsilon$ for $|n| > N$, then $|a(n)| < 2\epsilon$ for $|n| > N$. □

Summarizing:

Theorem 5.1.11

The map $f \mapsto \widehat{f}$ is a norm-decreasing, injective linear map from the Banach algebra $L^1(\mathbb{T})$ into $c_0(\mathbb{Z})$.

Proof. Linearity is (4); $\|\widehat{f}\|_\infty \leq \|f\|_1$ is the estimate $|\widehat{f}(n)| \leq \|f\|_1$; $\widehat{f} \in c_0(\mathbb{Z})$ is Theorem 5.1.10; injectivity is Theorem 5.1.8. ■

With $(ab)(n) = a(n)b(n)$, $c_0(\mathbb{Z})$ is a Banach algebra, and the last identity of (4) says $f \mapsto \widehat{f}$ preserves multiplication: a **homomorphism** of Banach algebras.

Remark ($L^1(\mathbb{T})$ has no identity)

If the Banach algebra $L^1(\mathbb{T})$ had an identity e, then for every $n \in \mathbb{Z}$

$$\widehat{e}(n) = \widehat{e}(n)\widehat{u_n}(n) = \widehat{e * u_n}(n) = \widehat{u_n}(n) = 1,$$

contradicting $\widehat{e} \in c_0(\mathbb{Z})$. So $L^1(\mathbb{T})$ has no identity; Theorem 5.1.5 supplies an *approximate identity*.

Exercise 5.1.10

If $\langle h_n \rangle$ satisfies (6), show that $\|a\widehat{h_n} - a\|_\infty \to 0$ for every $a \in c_0(\mathbb{Z})$.

Solution [Supplement]. First, $|\widehat{h_n}(m)| \le \|h_n\|_1 = 1$. Next, for fixed m and $0 < \delta < \pi$,

$$|1 - \widehat{h_n}(m)| = \Big| \int_{\mathbb{T}} h_n(s)(1 - e^{-ims})\, ds \Big| \le \int_{[-\delta,\delta]} h_n(s)|ms|\, ds + 2\int_{\mathbb{T}\setminus[-\delta,\delta]} h_n \le |m|\delta + 2\int_{\mathbb{T}\setminus[-\delta,\delta]} h_n,$$

so $\limsup_n |1 - \widehat{h_n}(m)| \le |m|\delta$ for every δ, i.e. $\widehat{h_n}(m) \to 1$. Now let $\epsilon > 0$ and choose M with $|a(m)| < \epsilon$ for $|m| > M$. Then

$$\|a\widehat{h_n} - a\|_\infty \le \max_{|m|\le M} |a(m)||\widehat{h_n}(m) - 1| + \sup_{|m|>M} |a(m)| \cdot 2,$$

where the first term tends to 0 (finitely many m) and the second is at most 2ϵ. □

Finally, consider some consequences of Theorem 5.1.6. With the Fejér kernel, $K_n * f \in T(\mathbb{T})$ and $K_n * f \to f$ uniformly for $f \in C(\mathbb{T})$:

Corollary 5.1.12

$T(\mathbb{T})$ is dense in $C(\mathbb{T})$ (in the uniform norm).

If $f \in L^p(\mathbb{T})$ $(1 \le p < \infty)$, Theorem 5.1.3 gives $g \in C(\mathbb{T})$ with $\|f - g\|_p < \epsilon$, and Corollary 5.1.12 gives $P \in T(\mathbb{T})$ with $\|g - P\|_\infty < \epsilon$. Then

$$\|f - P\|_p \le \|f - g\|_p + \|g - P\|_p < \epsilon + \|g - P\|_\infty < 2\epsilon.$$

Corollary 5.1.13

$T(\mathbb{T})$ is dense in $L^p(\mathbb{T})$ for $1 \le p < \infty$.

Corollary 5.1.12 implies the Weierstrass polynomial approximation theorem on bounded closed intervals.

Theorem 5.1.14 (Weierstrass)

The space of polynomial functions on $[0,1]$ is dense in $C[0,1]$.

Proof. Extend a continuous $f : [0,1] \to \mathbb{C}$ to a continuous periodic $g : \mathbb{T} \to \mathbb{C}$. By Corollary 5.1.12 choose $\{a_k \in \mathbb{C} : k = 0, \pm 1, \dots, \pm n\}$ with

$$\Big| f(t) - \sum_{k=-n}^{n} a_k u_k(t) \Big| < \epsilon, \qquad t \in [0,1].$$

Since the Taylor series of the exponential converges uniformly on $[0,1]$, for each k there is $M(k) \in \mathbb{N}$ with

$$\Big| \sum_{\ell=0}^{M(k)} \frac{(ikt)^\ell}{\ell!} - u_k(t) \Big| \le \frac{\epsilon}{(2n+1)|a_k| + 1}, \qquad t \in [0,1].$$

Put

$$P(t) = \sum_{k=-n}^{n} a_k \sum_{\ell=0}^{M(k)} \frac{(ikt)^\ell}{\ell!}, \qquad t \in [0,1].$$

Then $|f(t) - P(t)| < 2\epsilon$ (Exercise 5.1.11). ■

Exercise 5.1.11

Complete the proof of Theorem 5.1.14.

Solution [Supplement]. *Extension.* Define g on $[-\pi,\pi]$: $g=f$ on $[0,1]$, linear from $f(1)$ to 0 on $[1,2]$, 0 on $[2,\pi]\cup[-\pi,-1]$, linear from 0 to $f(0)$ on $[-1,0]$. Then $g\in C(\mathbb{T})$.

Estimate. P is a polynomial in t. For $t\in[0,1]$,

$$\begin{aligned}|f(t)-P(t)| &\le \Big|f(t)-\sum_{k=-n}^{n} a_k u_k(t)\Big| + \sum_{k=-n}^{n}|a_k|\Big|u_k(t)-\sum_{\ell=0}^{M(k)}\frac{(ikt)^\ell}{\ell!}\Big| \\ &< \epsilon + \sum_{k=-n}^{n}\frac{|a_k|\epsilon}{(2n+1)|a_k|+1} < \epsilon + (2n+1)\cdot\frac{\epsilon}{2n+1} = 2\epsilon.\end{aligned}$$

(If f is real-valued, $\operatorname{Re}P$ is a real polynomial with $|f-\operatorname{Re}P|\le|f-P|$.) □

5.2 The Fourier Integral

Motivation. On $\mathbb{R}$ sums over frequencies become integrals: continuity and decay of $\hat f$, inversion via Gauss and Fejér kernels, differentiation as multiplication.

5.2.1 Definition and continuity

Definition (Fourier transform)

For $f\in L^1(\mathbb{R})$ and $\alpha\in\mathbb{R}$ define

$$\widehat f(\alpha)=\int_{\mathbb{R}} f(x)e^{-i\alpha x}\,dx.$$

$\widehat f$ is the **Fourier transform** of f.

Clearly $|\widehat f(\alpha)|\le\|f\|_1$ for all $\alpha\in\mathbb{R}$. If $\alpha_n\to\alpha$, then

$$|\widehat f(\alpha_n)-\widehat f(\alpha)|\le\int_{\mathbb{R}}|f(x)||e^{-i\alpha_n x}-e^{-i\alpha x}|\,dx,$$

and $|e^{-i\alpha_n x}-e^{-i\alpha x}|\le 2$, $e^{-i\alpha_n x}\to e^{-i\alpha x}$, so dominated convergence gives $\widehat f(\alpha_n)\to\widehat f(\alpha)$: *$\widehat f$ is bounded and continuous.* Writing

$$u_\alpha(x)=e^{i\alpha x},\qquad x\in\mathbb{R},$$

one checks immediately that

$$u_\alpha * f=\widehat f(\alpha)u_\alpha,\qquad \alpha\in\mathbb{R}.$$

Example 1

For $f(x)=e^{-|x|/a}$ $(a>0)$,

$$\widehat f(\alpha)=\int_{\mathbb{R}}e^{-|x|/a}e^{-i\alpha x}\,dx=\int_{-\infty}^{0}e^{(1/a-i\alpha)x}\,dx+\int_{0}^{\infty}e^{(-1/a-i\alpha)x}\,dx=\frac{2a}{1+a^2\alpha^2}.$$

For $g = \chi_{[a,b]}$,

$$\widehat{g}(\alpha) = \int_{-\infty}^{\infty} \chi_{[a,b]} e^{-i\alpha x}\, dx = \frac{e^{-i\alpha a} - e^{-i\alpha b}}{i\alpha},$$

and in particular

$$\widehat{\chi_{[-a,a]}}(\alpha) = \frac{2\sin \alpha a}{\alpha}, \qquad a > 0.$$

Exercise 5.2.1

In Example 1, describe how the graphs of f and $\widehat{f}$ change as $a \to \infty$. Do the same for $\widehat{\chi_{[-a,a]}}$.

Solution [Supplement]. As $a \to \infty$, $e^{-|x|/a}$ spreads out to 1, while $\widehat{f}(\alpha) = \frac{2a}{1+a^2\alpha^2}$ (peak $2a$, half-width $1/a$, $\int \widehat{f} = 2\pi f(0) = 2\pi$) concentrates at 0 like $2\pi\delta_0$. Likewise $\widehat{\chi_{[-a,a]}}(\alpha) = \frac{2\sin a\alpha}{\alpha}$ has peak $2a$, zeros at $k\pi/a$ ($k \neq 0$) and oscillations decaying like $2/|\alpha|$: wide functions have narrow transforms. (It is not in $L^1(\mathbb{R})$.) □

The basic properties of Proposition 5.1.1 hold for Fourier integrals as well.

Proposition 5.2.1

For $f, g \in L^1(\mathbb{R})$ and constants $a, b \in \mathbb{C}$,

$$\widehat{af + bg}(\alpha) = a\widehat{f}(\alpha) + b\widehat{g}(\alpha), \qquad \widehat{f * g}(\alpha) = \widehat{f}(\alpha)\,\widehat{g}(\alpha), \qquad \alpha \in \mathbb{R}.$$

Moreover for each $\alpha \in \mathbb{R}$

$$\widehat{fu_\alpha} = (\widehat{f})_\alpha, \qquad \widehat{f_x}(\alpha) = \widehat{f}(\alpha)u_\alpha(-x), \quad x \in \mathbb{R}.$$

Proof. For example, for every $\beta \in \mathbb{R}$

$$\widehat{fu_\alpha}(\beta) = \int_{\mathbb{R}} f(x)u_\alpha(x)e^{-i\beta x}\, dx = \int_{\mathbb{R}} f(x)e^{-i(\beta-\alpha)x}\, dx = \widehat{f}(\beta - \alpha),$$

so $\widehat{fu_\alpha} = (\widehat{f})_\alpha$. The rest is Exercise 5.2.2. ■

Exercise 5.2.2

Complete the proof of Proposition 5.2.1.

Solution [Supplement]. Linearity is clear. $(x, y) \mapsto f(x - y)g(y)e^{-i\alpha x}$ is integrable on $\mathbb{R}^2$ (with integral of the modulus $\|f\|_1\|g\|_1$), so by Fubini

$$\widehat{f * g}(\alpha) = \int_{\mathbb{R}} g(y)e^{-i\alpha y} \int_{\mathbb{R}} f(x - y)e^{-i\alpha(x-y)}\, dx\, dy = \widehat{g}(\alpha)\widehat{f}(\alpha).$$

For translation, $\widehat{f_x}(\alpha) = \int_{\mathbb{R}} f(y - x)e^{-i\alpha y}\, dy = \int_{\mathbb{R}} f(y)e^{-i\alpha(y+x)}\, dy = \widehat{f}(\alpha)e^{-i\alpha x} = \widehat{f}(\alpha)u_\alpha(-x)$. □

5.2.2 Differentiation

Proposition 5.2.2

Let $f \in L^1(\mathbb{R})$.

(a) Let $g(x) = -ixf(x)$. If $g \in L^1(\mathbb{R})$, then $\widehat{f}$ is differentiable and

$$\widehat{f}\,'(\alpha) = \widehat{g}(\alpha), \qquad \alpha \in \mathbb{R}.$$

(b) If f is differentiable and $f' \in L^1(\mathbb{R})$, then

$$\widehat{f'}(\alpha) = i\alpha \widehat{f}(\alpha), \qquad \alpha \in \mathbb{R}.$$

Proof. (a) The difference quotient is

$$\frac{\widehat{f}(\alpha+h) - \widehat{f}(\alpha)}{h} = \int_{\mathbb{R}} f(x) e^{-i\alpha x} \frac{e^{-ihx} - 1}{h}\, dx,$$

and

$$\left| f(x) e^{-i\alpha x} \frac{e^{-ihx} - 1}{h} \right| \le |xf(x)|.$$

Since $x \mapsto |xf(x)|$ is integrable and $\lim_{h\to 0} \frac{e^{-ihx}-1}{h} = -ix$, the Lebesgue convergence theorem gives, for every real sequence $h_n \to 0$,

$$\lim_{n\to\infty} \frac{\widehat{f}(\alpha+h_n) - \widehat{f}(\alpha)}{h_n} = \widehat{g}(\alpha).$$

(b) By parts as in Proposition 5.1.2: $\widehat{f'}(\alpha) = i\alpha \int_{\mathbb{R}} f(x) e^{-i\alpha x}\, dx$, the boundary terms vanishing as $f(\pm\infty)$ exist ($f' \in L^1$) and are 0 ($f \in L^1$). ∎

5.2.3 Riemann–Lebesgue lemma

As for Fourier series, $\lim_{\alpha\to\pm\infty} \widehat{f}(\alpha) = 0$ for $f \in L^1(\mathbb{R})$. Using $e^{\pi i} = -1$,

$$\widehat{f}(\alpha) = -\int_{\mathbb{R}} f(x) e^{-i\alpha(x+\pi/\alpha)}\, dx = -\int_{\mathbb{R}} f\Big(x - \frac{\pi}{\alpha}\Big) e^{-i\alpha x}\, dx,$$

and hence

$$2\widehat{f}(\alpha) = \int_{\mathbb{R}} [f(x) - f_{\pi/\alpha}(x)] e^{-i\alpha x}\, dx.$$

By Fact B.23,

$$0 \le \lim_{\alpha\to\pm\infty} |\widehat{f}(\alpha)| \le \frac{1}{2} \lim_{\alpha\to\pm\infty} \|f - f_{\pi/\alpha}\|_1 = 0.$$

Since $\widehat{f}$ is already known to be continuous:

Theorem 5.2.3 (Riemann–Lebesgue)

$\widehat{f} \in C_0(\mathbb{R})$ for every $f \in L^1(\mathbb{R})$.

5.2.4 Approximate identities on $\mathbb{R}$

Using Fact B.23 instead of Theorem 5.1.4 in the proof of Theorem 5.1.5 yields:

Theorem 5.2.4

If a family $\{h_\lambda : \lambda \in (0,\infty)\}$ satisfies

$$h_\lambda \geq 0, \qquad \int_{\mathbb{R}} h_\lambda = 1, \qquad \lim_{\lambda\to\infty} \int_{\mathbb{R}\setminus[-\delta,\delta]} h_\lambda = 0 \quad (\delta > 0), \tag{10}$$

then:

(a) $\lim_{\lambda\to\infty} \|f - h_\lambda * f\|_1 = 0$ for every $f \in L^1(\mathbb{R})$;

(b) $\lim_{\lambda\to\infty} \|f - h_\lambda * f\|_\infty = 0$ for every bounded uniformly continuous $f : \mathbb{R} \to \mathbb{C}$.

Exercise 5.2.3

Prove Theorem 5.2.4.

Solution [Supplement]. (a) Let $\epsilon > 0$. By Fact B.23 choose $\delta > 0$ with $|s| < \delta \Rightarrow \|f - f_s\|_1 < \epsilon$, and Λ with $\int_{\mathbb{R}\setminus[-\delta,\delta]} h_\lambda < \epsilon/(2\|f\|_1)$ for $\lambda \geq \Lambda$. Since $f(t) - (h_\lambda * f)(t) = \int_{\mathbb{R}} h_\lambda(s)[f(t) - f_s(t)]\,ds$, Tonelli gives for $\lambda \geq \Lambda$

$$\|f - h_\lambda * f\|_1 \leq \int_{\mathbb{R}} h_\lambda(s)\|f - f_s\|_1\,ds \leq \epsilon \int_{[-\delta,\delta]} h_\lambda + 2\|f\|_1 \int_{\mathbb{R}\setminus[-\delta,\delta]} h_\lambda < 2\epsilon.$$

(b) Choose δ with $|s| < \delta \Rightarrow \|f - f_s\|_\infty < \epsilon$ (uniform continuity) and Λ with $\int_{\mathbb{R}\setminus[-\delta,\delta]} h_\lambda < \epsilon/(2\|f\|_\infty)$ for $\lambda \geq \Lambda$. Then for every t,

$$|f(t) - (h_\lambda * f)(t)| \leq \int_{\mathbb{R}} h_\lambda(s)|f(t) - f_s(t)|\,ds \leq \epsilon + 2\|f\|_\infty \cdot \frac{\epsilon}{2\|f\|_\infty} = 2\epsilon.$$

□

If $h : \mathbb{R} \to \mathbb{C}$ satisfies

$$h \geq 0, \qquad \int_{\mathbb{R}} h = 1, \tag{11}$$

put

$$h_\lambda(x) = \lambda h(\lambda x), \qquad x \in \mathbb{R},\ \lambda > 0. \tag{12}$$

Then $\{h_\lambda : \lambda > 0\}$ satisfies (10): $h_\lambda \geq 0$ and $\int h_\lambda = 1$ are clear, and for fixed $\delta > 0$, as $\lambda \to \infty$,

$$\int_{\mathbb{R}\setminus[-\delta,\delta]} h_\lambda(x)\,dx = \int_{\mathbb{R}\setminus[-\lambda\delta,\lambda\delta]} h(x)\,dx \to 0.$$

Exercise 5.2.4

Show that $\widehat{h_\lambda}(\alpha) = \widehat{h}(\frac{\alpha}{\lambda})$ for every $\alpha \in \mathbb{R}$.

Solution [Supplement]. Substituting $y = \lambda x$: $\widehat{h_\lambda}(\alpha) = \int_{\mathbb{R}} \lambda h(\lambda x)e^{-i\alpha x}\,dx = \int_{\mathbb{R}} h(y)e^{-i(\alpha/\lambda)y}\,dy = \widehat{h}(\alpha/\lambda)$. □

The case where h satisfying (11) has the form

$$h(x) = \frac{1}{2\pi} \int_{\mathbb{R}} p(\alpha)u_\alpha(x)\,d\alpha \tag{13}$$

is very useful. Here $p : \mathbb{R} \to \mathbb{C}$ is assumed to satisfy

$$0 \leq p \leq 1, \qquad p(0) = \lim_{\alpha\to 0} p(\alpha) = 1, \qquad p \in L^1(\mathbb{R}). \tag{14}$$

A simple example satisfying all requirements:

$$H(x) = \frac{1}{2\pi}\int_{\mathbb{R}} e^{-|\alpha|}e^{i\alpha x}\,d\alpha = \frac{1}{\pi}\,\frac{1}{1+x^2}, \tag{15}$$

and then

$$H_\lambda(x) = \frac{1}{2\pi}\int_{\mathbb{R}} e^{-|\alpha/\lambda|}e^{i\alpha x}\,d\alpha = \frac{1}{2\pi}\,\frac{2\lambda}{1+\lambda^2x^2}.$$

$\{H_\lambda : \lambda > 0\}$ is the **Poisson kernel**.

5.2.5 Smooth functions are dense

Let $C_c^n(\mathbb{R})$ be the $f \in C_c(\mathbb{R})$ with continuous $f^{(n)}$, and $C_c^\infty(\mathbb{R}) = \bigcap_n C_c^n(\mathbb{R})$. If $h \in C_c^\infty(\mathbb{R})$ satisfies (11), then $h_\lambda \in C_c^\infty(\mathbb{R})$ and, by Proposition 4.3.2 repeatedly, $h_\lambda * f \in C_c^\infty(\mathbb{R})$ for $f \in C_c(\mathbb{R})$. By Theorem 5.2.4

$$\lim_{\lambda\to\infty}\|f - h_\lambda * f\|_1 = \lim_{\lambda\to\infty}\|f - h_\lambda * f\|_\infty = 0.$$

Since $C_c(\mathbb{R})$ is dense in $L^1(\mathbb{R})$ and in $C_0(\mathbb{R})$:

Proposition 5.2.5

$C_c^\infty(\mathbb{R})$ is dense in the normed spaces $L^1(\mathbb{R})$ and $C_0(\mathbb{R})$.

Exercise 5.2.5

Give an example of $h \in C_c^\infty(\mathbb{R})$ satisfying (11).

Solution [Supplement]. Let $\varphi(x) = \exp(-\frac{1}{1-x^2})$ for $|x| < 1$, 0 otherwise. $\psi(t) = e^{-1/t}$ $(t > 0)$, 0 $(t \le 0)$ is C^∞ (derivatives $P(1/t)e^{-1/t} \to 0$ as $t \downarrow 0$), so $\varphi = \psi(1-x^2) \in C_c^\infty(\mathbb{R})$, $\varphi \ge 0$, $\int\varphi > 0$. Take $h = \varphi/\int_{\mathbb{R}}\varphi$. □

5.2.6 The inversion formula

For h given by (13) and $\{h_\lambda\}$ by (12), compute $h_\lambda * f$ for $f \in L^1(\mathbb{R})$ using Fubini:

$$\begin{aligned}(h_\lambda * f)(x) &= \int_{\mathbb{R}} f(x-y)\frac{\lambda}{2\pi}\int_{\mathbb{R}} p(\alpha)e^{i\alpha\lambda y}\,d\alpha\,dy = \frac{1}{2\pi}\int_{\mathbb{R}} f(x-y)\int_{\mathbb{R}} p(\tfrac{\alpha}{\lambda})e^{i\alpha y}\,d\alpha\,dy\\ &= \frac{1}{2\pi}\int_{\mathbb{R}} p(\tfrac{\alpha}{\lambda})\int_{\mathbb{R}} f(x-y)e^{i\alpha y}\,dy\,d\alpha = \frac{1}{2\pi}\int_{\mathbb{R}} p(\tfrac{\alpha}{\lambda})\int_{\mathbb{R}} f(y)e^{i\alpha(x-y)}\,dy\,d\alpha\\ &= \frac{1}{2\pi}\int_{\mathbb{R}} p(\tfrac{\alpha}{\lambda})\widehat{f}(\alpha)e^{i\alpha x}\,d\alpha.\end{aligned}$$

(Fubini applies since $\iint|f(x-y)||p(\alpha/\lambda)|\,d\alpha\,dy = \lambda\|f\|_1\|p\|_1 < \infty$.)

From now on assume p, h satisfy (14), (11) and (13). Then

$$(h_\lambda * f)(x) = \frac{1}{2\pi}\int_{\mathbb{R}} p\Big(\frac{\alpha}{\lambda}\Big)\widehat{f}(\alpha)e^{i\alpha x}\,d\alpha.$$

Since

$$\Big|p\Big(\frac{\alpha}{\lambda}\Big)\widehat{f}(\alpha)e^{i\alpha x}\Big| \le |\widehat{f}(\alpha)|, \qquad \lim_{\lambda\to\infty} p\Big(\frac{\alpha}{\lambda}\Big) = 1,$$

if $\widehat{f} \in L^1(\mathbb{R})$ the Lebesgue convergence theorem gives

$$\lim_{\lambda\to\infty}(h_\lambda * f)(x) = \frac{1}{2\pi}\int_{\mathbb{R}}\widehat{f}(\alpha)e^{i\alpha x}\,d\alpha, \qquad x \in \mathbb{R}.$$

On the other hand $\lim_{\lambda\to\infty}\|h_\lambda * f - f\|_1 = 0$ by Theorem 5.2.4, so by Fact B.18 there is a sequence $\lambda_n \to \infty$ with

$$\lim_{n\to\infty}(h_{\lambda_n} * f)(x) = f(x)$$

for almost every x. This proves the analogue of Corollary 5.1.9 (with, e.g., $p(\alpha) = e^{-|\alpha|}$ and $h = H$ from (15)):

Theorem 5.2.6 (inversion formula)

If $f \in L^1(\mathbb{R})$ and $\widehat{f} \in L^1(\mathbb{R})$, then

$$f(x) = \frac{1}{2\pi}\int_{\mathbb{R}} \widehat{f}(\alpha)e^{i\alpha x}\,d\alpha \tag{16}$$

for almost every x.

The right side of (16) is continuous in x, so for continuous f, (16) holds everywhere.

Example 2

Applying the inversion formula to Example 1 (with $a = 1$),

$$e^{-|x|} = \int_{-\infty}^{\infty}\frac{1}{\pi(1+\alpha^2)}e^{i\alpha x}\,d\alpha = \int_{-\infty}^{\infty}\frac{\cos\alpha x}{\pi(1+\alpha^2)}\,d\alpha + i\int_{-\infty}^{\infty}\frac{\sin\alpha x}{\pi(1+\alpha^2)}\,d\alpha.$$

Since the left side is real (and the cosine integrand is even),

$$\int_0^{\infty}\frac{\cos\alpha x}{1+\alpha^2}\,d\alpha = \frac{\pi}{2}e^{-|x|}.$$

If $\widehat{f} = 0$, then $\widehat{f} \in L^1$ and (16) gives $f = 0$ almost everywhere, the analogue of Theorem 5.1.8:

Theorem 5.2.7 (uniqueness)

If $f \in L^1(\mathbb{R})$ and $\widehat{f} = 0$, then $f = 0$ almost everywhere.

5.2.7 Fejér and Gauss kernels

Two more p satisfying (14), (11), (13). With the Fejér kernel (8), for $\lambda > 0$ put

$$K_\lambda(x) = \frac{1}{2\pi}\int_{\mathbb{R}}\max\left\{1-\frac{|\alpha|}{\lambda},0\right\}u_\alpha(x)\,d\alpha = \frac{1}{2\pi}\int_{-\lambda}^{\lambda}\left(1-\frac{|\alpha|}{\lambda}\right)e^{i\alpha x}\,d\alpha.$$

Then

$$\lambda K_1(\lambda x) = \frac{1}{2\pi}\int_{-1}^{1}(1-|\alpha|)e^{i\alpha\lambda x}\lambda\,d\alpha = K_\lambda(x),$$

so with $\lambda = 1$ in mind define

$$K(x) = \frac{1}{2\pi}\int_{-1}^{1}(1-|\alpha|)e^{i\alpha x}\,d\alpha.$$

A direct computation (integration by parts) gives

$$K_\lambda(x) = \frac{1}{\pi}\int_0^{\lambda}\left(1-\frac{\alpha}{\lambda}\right)\cos\alpha x\,d\alpha = \frac{1}{\pi\lambda x^2}(1-\cos\lambda x) = \frac{1}{2\pi\lambda}\frac{\sin^2(\frac{\lambda x}{2})}{(\frac{x}{2})^2},$$

so $K \geq 0$. As $\alpha \mapsto -\alpha$ leaves K_λ unchanged, $K_\lambda = \frac{1}{2\pi}\widehat{f_\lambda}$ with $f_\lambda(\alpha) = \max\{1 - \frac{|\alpha|}{\lambda}, 0\}$; both are integrable ($\widehat{f_\lambda} = O(x^{-2})$), so inversion gives

$$\int_{-\infty}^{\infty} K_\lambda(x) e^{i\alpha x}\, dx = \frac{1}{2\pi}\int_{-\infty}^{\infty} \widehat{f_\lambda}(x) e^{i\alpha x}\, dx = f_\lambda(\alpha) = \max\Big\{1 - \frac{|\alpha|}{\lambda}, 0\Big\}.$$

Since f_λ is continuous, this holds for every $\alpha \in \mathbb{R}$; in particular for $\lambda = 1$, $\alpha = 0$,

$$\int_{-\infty}^{\infty} K(x)\, dx = 1.$$

The family $\{K_\lambda : \lambda > 0\}$ obtained this way is also called the **Fejér kernel**.

Finally let

$$G(x) = \frac{1}{2\pi}\int_{\mathbb{R}} e^{-\alpha^2/2} u_\alpha(x)\, d\alpha.$$

By Fact B.11 (differentiation under the integral) and integration by parts,

$$G'(x) = \frac{1}{2\pi}\int_{\mathbb{R}} i\alpha e^{-\alpha^2/2} e^{i\alpha x}\, d\alpha = -xG(x)$$

(indeed $\int i\alpha e^{-\alpha^2/2} e^{i\alpha x}\, d\alpha = -i\int (e^{-\alpha^2/2})' e^{i\alpha x}\, d\alpha = i \cdot ix \int e^{-\alpha^2/2} e^{i\alpha x}\, d\alpha$). Since $G(0) = \frac{1}{2\pi}\sqrt{2\pi} = \frac{1}{\sqrt{2\pi}}$, solving the ODE gives

$$G(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2},$$

so $G \geq 0$ and $\int_{\mathbb{R}} G = 1$. The family $\{G_\lambda : \lambda > 0\}$ is the **Gauss kernel**.

5.2.8 Transforms of the kernels

The inversion formula says that if g and $\widehat{g}$ are continuous L^1-functions, then

$$g(x) = \frac{1}{2\pi}\widehat{\widehat{g}}(-x), \qquad x \in \mathbb{R}. \tag{17}$$

Suppose $f, g \in L^1(\mathbb{R})$, g continuous, satisfy

$$f(x) = \frac{1}{2\pi}\int_{\mathbb{R}} g(\alpha) e^{i\alpha x}\, d\alpha, \qquad x \in \mathbb{R}.$$

Then $f(x) = \frac{1}{2\pi}\widehat{g}(-x)$, so $\widehat{g} \in L^1(\mathbb{R})$. By the inversion formula for g (valid everywhere, g being continuous),

$$g(y) = \frac{1}{2\pi}\int_{\mathbb{R}} \widehat{g}(\alpha) e^{i\alpha y}\, d\alpha = \int_{\mathbb{R}} f(-\alpha) e^{i\alpha y}\, d\alpha = \int_{\mathbb{R}} f(\beta) e^{-i\beta y}\, d\beta = \widehat{f}(y).$$

So $\widehat{f} = g$. Applying this to the three kernels:

$$\widehat{K_\lambda}(\alpha) = \max\Big\{1 - \frac{|\alpha|}{\lambda}, 0\Big\}, \qquad \widehat{P_\lambda}(\alpha) = e^{-|\alpha/\lambda|}, \qquad \widehat{G_\lambda}(\alpha) = e^{-\alpha^2/2\lambda^2},$$

where $P_\lambda = H_\lambda$ is the Poisson kernel.

Exercise 5.2.6

For $f \in L^1(\mathbb{R})$ express $\widehat{K_\lambda * f}$ in terms of $\widehat{f}$. Use this to show that $\{f \in L^1(\mathbb{R}) : \widehat{f} \in C_c(\mathbb{R})\}$ is dense in $L^1(\mathbb{R})$, and compare with Theorem 5.1.7 for Fourier series.

Solution [Supplement]**.** By Proposition 5.2.1,

$$\widehat{K_\lambda * f}(\alpha) = \widehat{K_\lambda}(\alpha)\widehat{f}(\alpha) = \max\left\{1 - \frac{|\alpha|}{\lambda}, 0\right\}\widehat{f}(\alpha),$$

continuous and zero outside $[-\lambda, \lambda]$, so $K_\lambda * f$ lies in the set. As $K_\lambda(x) = \lambda K(\lambda x)$, $K \geq 0$, $\int K = 1$, (10) holds and Theorem 5.2.4 gives $\|K_\lambda * f - f\|_1 \to 0$. On $\mathbb{T}$, $\widehat{K_n * f}(k) = (1 - \frac{|k|}{n+1})\widehat{f}(k)$ vanishes for $|k| > n$: compactly supported transforms on $\mathbb{R}$ correspond to trigonometric polynomials on $\mathbb{T}$, and both density results use the Fejér kernel. □

5.2.9 The Fourier transform as a homomorphism

$f \mapsto \widehat{f}$ is an injective linear map $L^1(\mathbb{R}) \to C_0(\mathbb{R})$. For $f \in C_c^2(\mathbb{R})$, Proposition 5.2.2(b) gives $\widehat{f''} = -\alpha^2\widehat{f}$, so $|\widehat{f}(\alpha)| \leq \min\{\|f\|_1, \|f''\|_1/\alpha^2\}$ and $\widehat{f} \in L^1(\mathbb{R})$; by (17), $f = \widehat{k}$ with $k(\alpha) = \frac{1}{2\pi}\widehat{f}(-\alpha)$. So $C_c^2(\mathbb{R}) \supset C_c^\infty(\mathbb{R})$ lies in the range and is dense in $C_0(\mathbb{R})$ (Proposition 5.2.5):

Theorem 5.2.8

The Fourier transform $f \mapsto \widehat{f}$ is an injective, norm-decreasing homomorphism from the Banach algebra $L^1(\mathbb{R})$ into $C_0(\mathbb{R})$, and its range is dense.

Here $C_0(\mathbb{R})$ has the pointwise product. $f \mapsto \widehat{f} : L^1(\mathbb{T}) \to c_0(\mathbb{Z})$ (Theorem 5.1.11) is also a Fourier transform in this sense.

Exercise 5.2.7

Show that the range of $f \mapsto \widehat{f} : L^1(\mathbb{T}) \to c_0(\mathbb{Z})$ is dense.

Solution [Supplement]**.** Each finitely supported a is the transform of the trigonometric polynomial $\sum_n a(n)u_n$, and for $a \in c_0(\mathbb{Z})$, $a_N = a\chi_{[-N,N]}$ has $\|a - a_N\|_\infty = \sup_{|n|>N}|a(n)| \to 0$: the range is dense. □

5.2.10 Poisson summation

How are Fourier series and Fourier integrals related? Given $f \in L^1(\mathbb{R})$, as in Fact B.12 the series

$$F(t) = \sum_{n=-\infty}^{\infty} f(t - 2n\pi) \tag{18}$$

converges absolutely a.e., and $F \in L^1(\mathbb{T})$ ($\int_{\mathbb{T}} \sum_n |f(t - 2n\pi)|\, dt = \frac{1}{2\pi}\|f\|_1$). By Levi's theorem,

$$\widehat{F}(n) = \int_{\mathbb{T}} F(t)e^{-int}\, dt = \sum_{k=-\infty}^{\infty} \frac{1}{2\pi}\int_{-\pi}^{\pi} f(t + 2k\pi)e^{-int}\, dt = \frac{1}{2\pi}\int_{\mathbb{R}} f(t)e^{-int}\, dt = \frac{1}{2\pi}\widehat{f}(n). \tag{19}$$

Hence if the Fourier series of F converges at $t \in \mathbb{T}$ (to $F(t)$), we get the **Poisson summation formula**

$$\sum_{n=-\infty}^{\infty} f(t - 2n\pi) = \frac{1}{2\pi}\sum_{n=-\infty}^{\infty} \widehat{f}(n)e^{int}.$$

Exercise 5.2.8

Use $f(x) = e^{-|x|}$ to prove

$$\sum_{n=-\infty}^{\infty} \frac{1}{n^2+1} = \pi \frac{e^{2\pi}+1}{e^{2\pi}-1}.$$

Solution [Supplement]. $\widehat{f}(\alpha) = \frac{2}{1+\alpha^2}$ (Example 1), so $\widehat{F}(n) = \frac{1}{\pi}\frac{1}{1+n^2} \in \ell^1(\mathbb{Z})$. $F(t) = \sum_n e^{-|t-2n\pi|}$ converges uniformly on $[-\pi,\pi]$ (terms $\le e^{-(2|n|-1)\pi}$), so F is continuous, and by Corollary 5.1.9 and Exercise 5.1.1 $F(t) = \sum_n \widehat{F}(n)e^{int}$ for every t. At $t = 0$:

$$\sum_n e^{-2\pi|n|} = \frac{1}{\pi}\sum_n \frac{1}{1+n^2}.$$

The left side is $1 + \frac{2e^{-2\pi}}{1-e^{-2\pi}} = \frac{1+e^{-2\pi}}{1-e^{-2\pi}} = \frac{e^{2\pi}+1}{e^{2\pi}-1}$, which gives the formula. □

Exercise 5.2.9

For $f \in L^1(\mathbb{R})$ and $\lambda > 0$ define, as in (18),

$$F_\lambda(t) = \sum_{n=-\infty}^{\infty} f_\lambda(t - 2n\pi), \qquad f_\lambda(x) = \lambda f(\lambda x).$$

Show that

$$\widehat{F_\lambda}(n) = \frac{1}{2\pi}\widehat{f}\Big(\frac{n}{\lambda}\Big), \qquad \lim_{\lambda\to\infty} \|F_\lambda\|_{L^1(\mathbb{T})} = \frac{1}{2\pi}\|f\|_{L^1(\mathbb{R})}.$$

Use this to prove Theorem 5.2.7.

Solution [Supplement]. By (19) and Exercise 5.2.4, $\widehat{F_\lambda}(n) = \frac{1}{2\pi}\widehat{f_\lambda}(n) = \frac{1}{2\pi}\widehat{f}(n/\lambda)$. Upper bound: $\|F_\lambda\|_{L^1(\mathbb{T})} \le \frac{1}{2\pi}\sum_n \int_{-\pi}^{\pi} |f_\lambda(t-2n\pi)|\,dt = \frac{1}{2\pi}\|f_\lambda\|_1 = \frac{1}{2\pi}\|f\|_1$. Lower bound: separating the $n = 0$ term,

$$\|F_\lambda\|_{L^1(\mathbb{T})} \ge \frac{1}{2\pi}\Big[\int_{-\pi}^{\pi} |f_\lambda(t)|\,dt - \sum_{n\neq 0}\int_{-\pi}^{\pi} |f_\lambda(t-2n\pi)|\,dt\Big] = \frac{1}{2\pi}\Big[\int_{|x|\le\pi} |f_\lambda| - \int_{|x|>\pi} |f_\lambda|\Big]$$
$$= \frac{1}{2\pi}\Big[\int_{|y|\le\lambda\pi} |f| - \int_{|y|>\lambda\pi} |f|\Big] \longrightarrow \frac{1}{2\pi}\|f\|_1 \quad (\lambda\to\infty).$$

Proof of Theorem 5.2.7: if $\widehat{f} = 0$, then $\widehat{F_\lambda}(n) = 0$ for all n and λ, so $F_\lambda = 0$ a.e. by Theorem 5.1.8, i.e. $\|F_\lambda\|_{L^1(\mathbb{T})} = 0$. Letting $\lambda\to\infty$ gives $\|f\|_1 = 0$. □

Chapter 6. Normed Spaces

Overview

Results from Chapters 1–3 are cited as *Fact B.k* (see the *Background* notes). This chapter studies general normed spaces: dual spaces (of ℓ^p and of Hilbert spaces), and bounded linear maps between Banach spaces (Baire category, open mapping, closed graph, uniform boundedness), with applications to Fourier transforms.

6.1 Dual Spaces

Motivation. A normed space is understood through its continuous linear functionals; identifying duals, and having enough functionals to separate points (Hahn–Banach), is the basic tool.

6.1.1 Bounded linear maps

Definition (operator norm)

For a linear map $T : X \to Y$ between normed spaces put

$$\|T\| = \sup\{\|T(x)\| : x \in X,\ \|x\| \le 1\}.$$

If $\|T\| < \infty$, T is a **bounded linear map** and $\|T\|$ is its **norm**.

The norm of T measures how much T stretches the unit ball $\{x \in X : \|x\| \le 1\}$ of X inside Y. For example, $z \mapsto \alpha z : \mathbb{C} \to \mathbb{C}$ has norm $|\alpha|$.

Exercise 6.1.1

Show that a linear map $T : X \to Y$ is bounded if and only if there is $M > 0$ with

$$x \in X \implies \|T(x)\| \le M\|x\|,$$

and that in this case $\|T\| \le M$.

Solution [Supplement]. If T is bounded, take $M = \|T\|$ (or $\|T\|+1$): $\|T(x)\| = \|x\|\,\|T(x/\|x\|)\| \le \|T\|\|x\|$ for $x \neq 0$. Conversely $\|T(x)\| \le M\|x\|$ gives $\|T\| \le M$. □

Exercise 6.1.2

Prove that $\|T\| = \sup\left\{\dfrac{\|T(x)\|}{\|x\|} : x \in X,\ x \neq 0\right\}$.

Solution [Supplement]. Let S be the right side. For $x \neq 0$, $\|T(x)\|/\|x\| = \|T(x/\|x\|)\| \le \|T\|$, so $S \le \|T\|$. Conversely, if $0 < \|x\| \le 1$ then $\|T(x)\| \le \|T(x)\|/\|x\| \le S$, and $T(0) = 0$; so $\|T\| \le S$. □

Exercise 6.1.3

Show that the set of all bounded linear maps between two normed spaces is a vector space, and that $T \mapsto \|T\|$ is a norm on it.

Solution [Supplement]. If S, T are bounded and $a \in \mathbb{C}$, then for $\|x\| \le 1$

$$\|(S+T)(x)\| \le \|S(x)\| + \|T(x)\| \le \|S\| + \|T\|, \qquad \|(aT)(x)\| = |a|\|T(x)\| \le |a|\|T\|,$$

so $S+T$, aT are bounded with $\|S+T\| \le \|S\| + \|T\|$, $\|aT\| \le |a|\|T\|$, and equality in the latter via a^{-1}. $\|T\| = 0$ forces $T = 0$ on the unit ball, hence everywhere. □

Proposition 6.1.1

For a linear map $T : X \to Y$ between normed spaces, the following are equivalent:

(a) T is bounded;

(b) T is continuous;

(c) T is continuous at some point $x_0 \in X$.

Proof. $\|T(x) - T(y)\| \le \|T\|\|x - y\|$ gives (a)⇒(b); (b)⇒(c) is trivial. If T is continuous at x_0, $\|x - x_0\| < \delta \Rightarrow \|T(x) - T(x_0)\| < 1$, i.e. $\|y\| < \delta \Rightarrow \|T(y)\| < 1$, so $\|T\| \le 1/\delta$. ■

Proposition 6.1.2

For normed spaces X, Y, the set $B(X, Y)$ of bounded linear maps from X to Y is a normed space. If Y is a Banach space, so is $B(X, Y)$.

Proof. $B(X, Y)$ is normed (Exercise 6.1.3). For Y Banach and $\langle T_n \rangle$ Cauchy, $\|T_n(x) - T_m(x)\| \le \|T_n - T_m\|\|x\|$, so $T(x) = \lim T_n(x)$ exists, and $T \in B(X, Y)$ with $\|T_n - T\| \to 0$ (Exercise 6.1.4). ■

Exercise 6.1.4

Complete the proof of Proposition 6.1.2.

Solution [Supplement]. T is linear. If $\|T_n - T_m\| < \epsilon$ for $m, n \ge N$, then $\|T_n(x) - T_m(x)\| < \epsilon$ for $\|x\| \le 1$, and $m \to \infty$ gives $\|T_n(x) - T(x)\| \le \epsilon$. So $T = T_N - (T_N - T) \in B(X, Y)$ and $\|T_n - T\| \le \epsilon$ for $n \ge N$. □

Definition (dual space)

A scalar-valued linear map is a **linear functional**; $X^* = B(X, \mathbb{C})$, the **dual space**, is a Banach space by Proposition 6.1.2.

Example 1

$f \mapsto \int_0^1 f$ is a bounded linear functional on $(C[0,1], \|\cdot\|_{\sup})$. However

$$f \mapsto \int_{\mathbb{R}} f(x)\,dx \tag{1}$$

is not even defined on $C_0(\mathbb{R})$ (integrals may diverge), and on $(C_c(\mathbb{R}), \|\cdot\|_{\sup})$ it is unbounded: $f_n \in C_c(\mathbb{R})$, $0 \le f_n \le 1$, $f_n = 1$ on $[-n, n]$ give $\|f_n\|_{\sup} = 1$, $\int f_n \to \infty$. On $(L^1(\mathbb{R}), \|\cdot\|_1)$ it is bounded by Fact B.9.

Example 2

Let (X, μ) be a measure space, p, q conjugate exponents, and $g \in L^q(\mu)$ $(1 \le q \le \infty)$. Consider

$$\varphi_g : f \mapsto \int_X fg\, d\mu, \qquad f \in L^p(\mu).$$

By Hölder, $|\varphi_g(f)| \le \|g\|_q \|f\|_p$, so φ_g is a bounded linear functional on $L^p(\mu)$ with

$$\|\varphi_g\| \le \|g\|_q, \qquad g \in L^q(\mu).$$

In fact equality holds.
Case $p = \infty$, $q = 1$. For fixed $g \in L^1(\mu)$ we want $f \in L^\infty(\mu)$ with $\|f\|_\infty \|g\|_1 = |\int_X gf|$. Take

$$f(x) = \begin{cases} \dfrac{|g(x)|}{g(x)}, & g(x) \neq 0, \\ 0, & g(x) = 0; \end{cases}$$

this function is written $\operatorname{sgn} g$. Then $gf = |g|$, so $\int_X gf = \|g\|_1 = \|f\|_\infty \|g\|_1$ (if $g \neq 0$).
Case $p = 1$, $q = \infty$. For $g \in L^\infty(\mu)$, $\|g\|_\infty \neq 0$, and small $\epsilon > 0$, $\{|g| > \|g\|_\infty - \epsilon\}$ has positive measure; take E in it with $0 < \mu(E) < \infty$ (σ-finiteness) and $f = \chi_E \cdot \operatorname{sgn} g \in L^1(\mu)$. Then

$$\int_X gf = \int_E g \cdot \operatorname{sgn} g = \int_E |g| \ge \mu(E)(\|g\|_\infty - \epsilon) = \|f\|_1(\|g\|_\infty - \epsilon),$$

so $\|\varphi_g\| \ge \|g\|_\infty - \epsilon$, and hence $\|\varphi_g\| = \|g\|_\infty$.
Case $1 < q < \infty$. For fixed $g \in L^q(\mu)$ put $f = |g|^{q/p} \operatorname{sgn} g$. Then $|f|^p = |g|^q$, so $\|g\|_q^{q-1} = \|f\|_p$ (as $q/p = q - 1$), and $fg = |g|^q$. Hence

$$\int_X fg = \int |g|^q = \|g\|_q^q = \|g\|_q \|f\|_p,$$

and $\|\varphi_g\| = \|g\|_q$.

So for any measure space (X, μ) and $q \in [1, \infty]$,

$$g \mapsto \varphi_g : L^q(\mu) \to L^p(\mu)^* \tag{2}$$

is a norm-preserving linear map, an **isometry** (hence injective) (for $p = 1$, $q = \infty$ assume σ-finiteness).

Exercise 6.1.5

For the bounded function $g(x) = x$ on $[0, 1]$, show that there is no $f \in L^1[0, 1]$, $f \neq 0$, with $\|f\|_1 \|g\|_\infty = \int_0^1 fg$.

Solution [Supplement]**.** Here $\|g\|_\infty = 1$. For any $f \in L^1[0, 1]$,

$$\Big| \int_0^1 f(x) x\, dx \Big| \le \int_0^1 |f(x)| x\, dx \le \int_0^1 |f(x)|\, dx = \|f\|_1 \|g\|_\infty.$$

Equality would force $\int_0^1 |f(x)|(1 - x)\, dx = 0$, so $|f(x)|(1 - x) = 0$ a.e.; since $1 - x > 0$ on $[0, 1)$, $f = 0$ a.e. So the supremum $\|\varphi_g\| = 1$ is not attained. □

6.1.2 The duals of ℓ^p and c_0

In many cases (2) is surjective; we examine ℓ^p. For $n = 1, 2, \dots$ define the sequence

$$e_n(m) = \begin{cases} 1, & m = n, \\ 0, & m \neq n. \end{cases} \tag{3}$$

If $\varphi \in (\ell^p)^*$ and $\varphi = \varphi_g$, then

$$\varphi(e_n) = \varphi_g(e_n) = \sum_{i=1}^{\infty} g(i)e_n(i) = g(n), \qquad n \in \mathbb{N}.$$

So for arbitrary $\varphi \in (\ell^p)^*$ define $g(n) = \varphi(e_n)$. Assume $1 \leq p < \infty$; it suffices to show $g \in \ell^q$ (and $\varphi = \varphi_g$) to prove (2) is surjective.

$p = 1$, $q = \infty$. If $\varphi \in (\ell^1)^*$, then since $e_n \in \ell^1$, $|\varphi(e_n)| \leq \|\varphi\|\|e_n\|_1 = \|\varphi\|$ for each n, so $g \in \ell^\infty$ and $\|g\|_\infty \leq \|\varphi\|$.

$1 < p < \infty$. Fix $N \in \mathbb{N}$ and, as in Example 2, put

$$f(n) = \begin{cases} |g(n)|^{q-1} \cdot \operatorname{sgn} g(n), & n \leq N, \\ 0, & n > N. \end{cases}$$

Then $\|f\|_p^p = \sum_{n=1}^N |f(n)|^p = \sum_{n=1}^N |g(n)|^q$, so

$$\sum_{n=1}^{N} |g(n)|^q = \sum_{n=1}^{\infty} f(n)g(n) = \varphi(f) \leq \|\varphi\|\|f\|_p = \|\varphi\|\Big(\sum_{n=1}^{N} |g(n)|^q\Big)^{1/p}.$$

Thus $\big(\sum_{n=1}^N |g(n)|^q\big)^{1/q} \leq \|\varphi\|$ for every N, so $\|g\|_q \leq \|\varphi\|$ and $g \in \ell^q$.

Finally $\varphi(e_n) = g(n) = \varphi_g(e_n)$, so $\varphi = \varphi_g$ on the finite combinations of the e_n, dense in ℓ^p ($p < \infty$); both are continuous, so $\varphi_g = \varphi$.

Theorem 6.1.3

If $1 \leq p < \infty$ and p, q are conjugate exponents, then $g \mapsto \varphi_g : \ell^q \to (\ell^p)^*$ is a surjective isometry.

By Hahn–Banach (below), $g \mapsto \varphi_g$ is not surjective for $p = \infty$: $(\ell^\infty)^* \supsetneq \ell^1$. Restricting functionals of ℓ^∞ to $c_0 = c_0(\mathbb{N})$, consider

$$g \mapsto \varphi_g : \ell^1 \to (c_0)^*. \tag{4}$$

Exercise 6.1.6

Show that (4) is a surjective isometry.

Solution [Supplement]. *Isometry.* $|\varphi_g(x)| = |\sum g(n)x(n)| \leq \|g\|_1\|x\|_\infty$, so $\|\varphi_g\| \leq \|g\|_1$. For $N \in \mathbb{N}$ let $x_N(n) = \operatorname{sgn} g(n)$ for $n \leq N$ and 0 for $n > N$. Then $x_N \in c_0$, $\|x_N\|_\infty \leq 1$, and $\varphi_g(x_N) = \sum_{n \leq N} |g(n)|$. Letting $N \to \infty$ gives $\|\varphi_g\| \geq \|g\|_1$. *Surjectivity.* For $\varphi \in (c_0)^*$ put $g(n) = \varphi(e_n)$; with x_N as above, $\sum_{n \leq N} |g(n)| = \varphi(x_N) \leq \|\varphi\|$, so $\|g\|_1 \leq \|\varphi\|$. φ and φ_g agree on finite combinations of the e_n, dense in c_0 ($\|x - x\chi_{[1,N]}\|_\infty = \sup_{n>N} |x(n)| \to 0$), so $\varphi = \varphi_g$. □

Summarizing:

$$(c_0)^* = \ell^1, \qquad (\ell^p)^* = \ell^q \quad (1 \leq p < \infty),$$

and therefore in many cases $X = X^{**}$.

6.1.3 The canonical embedding $X \to X^{**}$

For each $x \in X$ define

$$T_x(\varphi) = \varphi(x), \qquad \varphi \in X^*. \tag{5}$$

Clearly T_x is a linear functional on X^* and $\|T_x\| \le \|x\|$, so $T_x \in X^{**}$. To show that $x \mapsto T_x$ is injective we must show

$$x \neq 0 \implies T_x \neq 0,$$

i.e. for $x \neq 0$ some $\varphi \in X^*$ has $\varphi(x) \neq 0$: does X have enough functionals to separate points? The next theorem says yes.

Theorem 6.1.4 (Hahn–Banach)

Let X be a normed space and Y a subspace. For every $\psi \in Y^*$ there is $\varphi \in X^*$ with

$$\varphi|_Y = \psi, \qquad \|\varphi\| = \|\psi\|.$$

Proof. Prove it first for real spaces: X, Y real, ψ real. If $\psi = 0$ take $\varphi = 0$; else normalize $\|\psi\| = 1$. Step 1 (the heart) adds one dimension; Step 2 iterates it via the axiom of choice; Step 3 passes to complex spaces.

Step 1 (one more dimension). For $x_0 \in X \setminus Y$ consider

$$Y_1 = \{ax_0 + y : a \in \mathbb{R},\ y \in Y\},$$

and extend ψ to $\psi_1 : Y_1 \to \mathbb{R}$. Only $\psi_1(x_0)$ is to be chosen, so that $\|\psi_1\| = 1$, i.e.

$$|a\psi_1(x_0) + \psi(y)| \le \|ax_0 + y\|, \qquad a \in \mathbb{R},\ y \in Y. \tag{6}$$

For $y, y' \in Y$,

$$\psi(y) - \psi(y') = \psi(y - y') \le \|y - y'\| \le \|y - x_0\| + \|y' - x_0\|,$$

so

$$\psi(y) - \|y - x_0\| \le \psi(y') + \|y' - x_0\|, \qquad y, y' \in Y.$$

Hence with

$$A = \sup\{\psi(y) - \|y - x_0\| : y \in Y\}, \qquad B = \inf\{\psi(y') + \|y' - x_0\| : y' \in Y\},$$

we have $A \le B$. Choose any number in $[A, B]$ as $\psi_1(x_0)$. Then for all $y, y' \in Y$

$$\psi(y) - \|y - x_0\| \le \psi_1(x_0) \le \psi(y') + \|y' - x_0\|,$$

equivalent to $|\psi_1(x_0) - \psi(y)| \le \|x_0 - y\|$ for $y \in Y$; applied to $-\frac{y}{a}$ ($a \neq 0$) and multiplied by $|a|$, it gives (6) ($a = 0$ is trivial).

Step 2 (Zorn). Define

$$\mathcal{P} = \{(Y_\iota, \psi_\iota) :\ Y_\iota \text{ a subspace of } X \text{ containing } Y,\ \psi_\iota \in Y_\iota^*,\ \psi_\iota|_Y = \psi,\ \|\psi_\iota\| = 1\}.$$

Order $\mathcal{P}$ by $(Y_1, \psi_1) \le (Y_2, \psi_2)$ if $Y_1 \subset Y_2$ and $\psi_1 = \psi_2|_{Y_1}$. By the axiom of choice (Hausdorff maximality) there is a maximal totally ordered $\Omega \subset \mathcal{P}$ (no totally ordered $\mathcal{N} \supsetneq \Omega$). Then

$$Z = \bigcup\{Y_\iota : (Y_\iota, \psi_\iota) \in \Omega\}$$

is a subspace, and $\rho(x) = \psi_\iota(x)$ for $x \in Y_\iota$ defines a linear functional on Z with $\|\rho\| = 1$ (Exercise 6.1.7). If $Z \subsetneq X$, Step 1 gives $(Z_1, \rho_1) \in \mathcal{P}$, $Z_1 \supsetneq Z$, and $\Omega \cup \{(Z_1, \rho_1)\}$ is totally ordered, contradicting maximality. So $Z = X$.

Step 3 (complex case). For complex X, Y, ψ, regard them as real spaces and write $\psi = u + iv$; u is real linear on Y with $\|u\| \le \|\psi\|$, and the real case gives U on X with $U|_Y = u$, $\|U\| = \|u\|$. For $x \in X$ define

$$\varphi(x) = U(x) - iU(ix).$$

Clearly $\varphi(x+y) = \varphi(x) + \varphi(y)$, and

$$\varphi(ix) = U(ix) - iU(-x) = i\big[U(x) - iU(ix)\big] = i\varphi(x),$$

so φ is complex linear. For $y \in Y$,

$$\varphi(y) = U(y) - iU(iy) = u(y) - iu(iy),$$

so to show $\varphi|_Y = \psi$ it suffices that $\psi(y) = u(y) - iu(iy)$. For any $\alpha \in \mathbb{C}$ one checks immediately $\alpha = \operatorname{Re}\alpha - i\operatorname{Re}(i\alpha)$; since ψ is complex linear,

$$\psi(y) = \operatorname{Re}\psi(y) - i\operatorname{Re}(i\psi(y)) = u(y) - i\operatorname{Re}(\psi(iy)) = u(y) - iu(iy),$$

so $\varphi|_Y = \psi$.

Finally $\|\varphi\| = \|\psi\|$: $\|\varphi\| \ge \|\psi\|$ as φ extends ψ, and it suffices that $\|\varphi\| \le \|U\| \le \|\psi\|$. For $x \in X$ take $|\alpha| = 1$ with $\alpha\varphi(x) = |\varphi(x)|$; then $\varphi(\alpha x)$ is real, so

$$|\varphi(x)| = \varphi(\alpha x) = U(\alpha x) \le \|U\|\|\alpha x\| = \|U\|\|x\|.$$

Thus $\|\varphi\| \le \|U\|$, completing the proof. ∎

Exercise 6.1.7

In the proof of the Hahn–Banach theorem, show that Z is a subspace and that ρ is a linear functional on Z with $\|\rho\| = 1$.

Solution [Supplement]. *Z is a subspace.* Let $x, x' \in Z$, say $x \in Y_\iota$, $x' \in Y_\kappa$ with $(Y_\iota, \psi_\iota), (Y_\kappa, \psi_\kappa) \in \Omega$. Since Ω is totally ordered, one contains the other, say $Y_\iota \subset Y_\kappa$; then $ax + bx' \in Y_\kappa \subset Z$. *$\rho$ is well defined.* If $x \in Y_\iota \cap Y_\kappa$ and $Y_\iota \subset Y_\kappa$, then $\psi_\iota = \psi_\kappa|_{Y_\iota}$, so $\psi_\iota(x) = \psi_\kappa(x)$. *Linearity.* With $x, x' \in Y_\kappa$ as above, $\rho(ax + bx') = \psi_\kappa(ax + bx') = a\psi_\kappa(x) + b\psi_\kappa(x') = a\rho(x) + b\rho(x')$. *Norm.* For $x \in Y_\iota$, $|\rho(x)| = |\psi_\iota(x)| \le \|x\|$, so $\|\rho\| \le 1$; and $\rho|_Y = \psi$ with $\|\psi\| = 1$ gives $\|\rho\| \ge 1$. □

For $x \in X$, $x \neq 0$, apply Hahn–Banach to the one-dimensional subspace $Y = \{\alpha x : \alpha \in \mathbb{C}\}$ and the functional $\psi(\alpha x) = \alpha\|x\|$ (which has $\|\psi\| = 1$):

Corollary 6.1.5

If $x \neq 0$ is a point of a normed space X, there is $\varphi \in X^*$ with $\varphi(x) = \|x\|$ and $\|\varphi\| = 1$.

By Corollary 6.1.5, $\|T_x\| = \|x\|$ in (5): $x \mapsto T_x$ embeds X isometrically in X^{**}. If it is surjective, X is **reflexive**; e.g. ℓ^p $(1 < p < \infty)$, but not c_0, as $(c_0)^{**} = \ell^\infty$.

The next corollary is useful for deciding whether a given subspace is closed. The closure of a subset Y is written $\overline{Y}$.

Corollary 6.1.6

For a subspace Y of a normed space X and a point $x \in X$, the following are equivalent:

(a) $x \in \overline{Y}$;

(b) if $\varphi \in X^*$ and $\varphi|_Y = 0$, then $\varphi(x) = 0$.

Proof. Every $\varphi \in X^*$ is continuous, so (a)⇒(b) is clear. If $x \notin \overline{Y}$, there is $\delta > 0$ with

$$y \in Y \implies \|x - y\| > \delta.$$

Consider the subspace $Z = \{\alpha x + y : \alpha \in \mathbb{C},\ y \in Y\}$ spanned by x and Y, and define $\psi(\alpha x + y) = \alpha$ (well defined since $x \notin Y$); ψ is a linear functional on Z. For $\alpha x + y \in Z$ with $\alpha \neq 0$,

$$|\psi(\alpha x + y)|\delta = |\alpha|\delta \leq |\alpha|\left\|x + \frac{1}{\alpha}y\right\| = \|\alpha x + y\|,$$

so $\psi \in Z^*$ with $\|\psi\| \leq \frac{1}{\delta}$. Also $\psi(x) = 1 \neq 0$. Extending ψ to X by Hahn–Banach gives $\varphi \in X^*$ with $\varphi|_Y = 0$ but $\varphi(x) \neq 0$. ■

Exercise 6.1.8

Show that there is $\varphi \in (\ell^\infty)^*$ with $\varphi|_{c_0} = 0$ but $\varphi \neq 0$. Show that $(\ell^\infty)^* \supsetneq \ell^1$, and that the Banach space ℓ^1 is not reflexive.

Solution [Supplement]**.** c_0 is a closed subspace of ℓ^∞ (Exercise 5.1.9) and the constant sequence $\mathbf{1} \notin c_0$, so $\mathbf{1} \notin \overline{c_0}$. By Corollary 6.1.6 there is $\varphi \in (\ell^\infty)^*$ with $\varphi|_{c_0} = 0$ and $\varphi(\mathbf{1}) \neq 0$. Such φ is not of the form φ_g, $g \in \ell^1$: otherwise $g(n) = \varphi_g(e_n) = \varphi(e_n) = 0$ for all n (as $e_n \in c_0$), so $g = 0$ and $\varphi = 0$. Hence the isometry $g \mapsto \varphi_g : \ell^1 \to (\ell^\infty)^*$ is not surjective: $(\ell^\infty)^* \supsetneq \ell^1$. ℓ^1 not reflexive: $(\ell^1)^* = \ell^\infty$ (Theorem 6.1.3), so $(\ell^1)^{**} = (\ell^\infty)^*$. If $x \mapsto T_x$ were onto, then $\varphi = T_g$ for some $g \in \ell^1$, i.e. $\varphi(h) = T_g(\varphi_h) = \sum_n g(n)h(n)$ for $h \in \ell^\infty$ (identifying h with $\varphi_h \in (\ell^1)^*$); so $\varphi = \varphi_g$, which we just ruled out. □

Exercise 6.1.9

Show that a Banach space X is reflexive if and only if X^* is reflexive. Show that the Banach space ℓ^∞ is not reflexive.

Solution [Supplement]**.** Write $J_X : X \to X^{**}$ and $J_{X^*} : X^* \to X^{***}$ for the canonical isometries.

(⇒) Let X be reflexive and $\Phi \in X^{***}$. Put $\varphi = \Phi \circ J_X \in X^*$. For $\xi \in X^{**}$ write $\xi = J_X(x)$; then $\Phi(\xi) = \Phi(J_X x) = \varphi(x) = J_X(x)(\varphi) = \xi(\varphi) = J_{X^*}(\varphi)(\xi)$. So $\Phi = J_{X^*}(\varphi)$ and X^* is reflexive.

(⇐) If X^* is reflexive and $J_X(X) \subsetneq X^{**}$, then $J_X(X)$ is closed (X complete), so Corollary 6.1.6 gives $0 \neq \Phi \in X^{***}$ vanishing on it, $\Phi = J_{X^*}(\varphi)$. Then $\varphi(x) = \Phi(J_X x) = 0$ for all x, so $\varphi = 0$ and $\Phi = 0$: a contradiction. Finally $\ell^\infty = (\ell^1)^*$ and ℓ^1 is not reflexive (Exercise 6.1.8), so ℓ^∞ is not reflexive. □

6.2 Linear Maps between Banach Spaces

Motivation. Via Baire's theorem, completeness makes bijective bounded maps invertible, closed graphs bounded, and pointwise bounded families uniformly bounded; one application is a continuous function with divergent Fourier series.

Completeness now matters: the domain or range is a Banach space. The key property of complete metric spaces is Baire's theorem. For x in a metric space (X, d) and $r > 0$ write

$$N_X(x, r) = \{y \in X : d(x, y) < r\}$$

for the neighborhood of x, or just $N(x, r)$. When X is a normed space, $N_X(0, r)$ is written X_r.

Theorem 6.2.1 (Baire)

If a sequence $\langle C_n\rangle$ of closed subsets of a complete metric space X has union X, then some C_n contains a nonempty open set.

Proof. Suppose not. As $C_1 \neq X$ is closed, some $N(x_1, r_1)$ has $C_1 \cap \overline{N(x_1, r_1)} = \emptyset$ ($\overline{A}$ the closure). As $N(x_1, r_1) \not\subset C_2$, there is $N(x_2, r_2)$ with

$$\overline{N(x_2, r_2)} \subset N(x_1, r_1) \setminus C_2$$

($N(x_1, r_1) \setminus C_2$ is a nonempty open set). Repeating, we get neighborhoods $\langle N(x_n, r_n)\rangle$ with

$$N(x_{n+1}, r_{n+1}) \subset N(x_n, r_n), \qquad \overline{N(x_n, r_n)} \cap \Big[\bigcup_{k=1}^{n} C_k\Big] = \emptyset$$

for each n. We may also arrange $r_n \to 0$; then $\langle x_n\rangle$ is a Cauchy sequence (for $m > n$, $x_m \in N(x_n, r_n)$). Its limit x lies in every $\overline{N(x_n, r_n)}$, so $x \notin \bigcup_{n=1}^{\infty} C_n = X$, a contradiction. ■

Corollary 6.2.2

In a complete metric space, the intersection of a sequence of dense open sets is nonempty.

Exercise 6.2.1

Prove Corollary 6.2.2.

Solution [Supplement]. $C_n = U_n^c$ is closed with no interior. If $\bigcap_n U_n = \emptyset$, then $\bigcup_n C_n = X$ and Baire gives some C_n with interior: a contradiction. (Applied to $\overline{N(x, r)}$, this shows $\bigcap U_n$ is dense.) □

The open mapping theorem and uniform boundedness principle, with Hahn–Banach, are the three basic tools for Banach spaces.

6.2.1 The open mapping theorem

Theorem 6.2.3 (open mapping theorem)

Let $T : X \to Y$ be a surjective bounded linear map between Banach spaces. Then there is $\delta > 0$ with $T(X_1) \supset Y_\delta$.

Proof. First, $X = \bigcup_n X_n$ and T is surjective, so $Y = \bigcup_n \overline{T(X_n)}$. By Baire's theorem there are $n \in \mathbb{N}$ and a neighborhood $N_Y(y_0, r)$ with $N_Y(y_0, r) \subset \overline{T(X_n)}$. If $\|y\| < r$, then $y + y_0 \in N_Y(y_0, r)$, so $y + y_0 \in \overline{T(X_n)}$. Choosing x_0 with $T(x_0) = y_0$,

$$y \in \overline{T(X_n)} - T(x_0) \subset \overline{T(X_{n+\|x_0\|})}.$$

Putting $\delta = \dfrac{r}{n + \|x_0\|}$ we get $Y_\delta \subset \overline{T(X_1)}$, and by homogeneity

$$Y_s \subset \overline{T(X_{s/\delta})}, \qquad s \in (0, \infty). \tag{7}$$

By (7) we show $Y_\delta \subset T(X_1)$ up to a factor. Fix $\epsilon \in (0, 1)$. For $y \in Y_1$, (7) gives $x_1 \in X_{1/\delta}$ with $\|y - T(x_1)\| < \epsilon$; as $y - T(x_1) \in Y_\epsilon$, (7) with $s = \epsilon$ gives $x_2 \in X_{\epsilon/\delta}$ with $\|y - T(x_1) - T(x_2)\| < \epsilon^2$. Continuing,

$$\|x_n\| < \frac{\epsilon^{n-1}}{\delta}, \qquad \|y - T(x_1) - T(x_2) - \cdots - T(x_n)\| < \epsilon^n.$$

Hence $\sum_{n=1}^{\infty} x_n$ converges (absolutely, in the Banach space X) to some $x \in X$ with $\|x\| \le \frac{1}{\delta(1-\epsilon)}$, and

$$y = \lim_{n\to\infty} T(x_1 + x_2 + \cdots + x_n) = T(x).$$

So $y \in T(X_{1/\delta(1-\epsilon)})$; thus $Y_1 \subset T(X_{1/\delta(1-\epsilon)})$, i.e. $Y_{\delta(1-\epsilon)} \subset T(X_1)$. ■

$f : X \to Y$ is an **open map** if $f(U)$ is open for every open $U \subset X$; hence the name:

Corollary 6.2.4

A surjective bounded linear map $T : X \to Y$ between Banach spaces is always an open map.

Exercise 6.2.2

Prove Corollary 6.2.4.

Solution [Supplement]. Let $U \subset X$ be open and $y = T(x) \in T(U)$ with $x \in U$. Choose $r > 0$ with $N_X(x, r) = x + X_r \subset U$. By Theorem 6.2.3, $T(X_r) = rT(X_1) \supset rY_\delta = Y_{r\delta}$. Hence

$$N_Y(y, r\delta) = y + Y_{r\delta} \subset T(x) + T(X_r) = T(x + X_r) \subset T(U),$$

so $T(U)$ is open. □

Corollary 6.2.5

If a bounded linear map $T : X \to Y$ between Banach spaces is bijective, then its inverse $T^{-1} : Y \to X$ is also bounded.

Proof. Choose $\delta > 0$ with $Y_\delta \subset T(X_1)$. Then $T^{-1}(Y_\delta) \subset X_1$, so $\|T^{-1}\| \le \frac{1}{\delta}$. ■

6.2.2 The Fourier transform $L^1(\mathbb{T}) \to c_0(\mathbb{Z})$ is not onto

For $n = 1, 2, \ldots$ let

$$D_n(t) = \sum_{k=-n}^{n} u_k(t) = \frac{\sin(n + \frac{1}{2})t}{\sin \frac{1}{2}t}, \qquad t \in \mathbb{T}. \tag{8}$$

$\langle D_n \rangle$ is the **Dirichlet kernel**.

Exercise 6.2.3

Prove the second equality in (8).

Solution [Supplement]. For $t \neq 0 \pmod{2\pi}$, the geometric sum gives

$$\sum_{k=-n}^{n} e^{ikt} = e^{-int}\frac{e^{i(2n+1)t} - 1}{e^{it} - 1} = \frac{e^{i(n+1)t} - e^{-int}}{e^{it} - 1} = \frac{e^{i(n+\frac{1}{2})t} - e^{-i(n+\frac{1}{2})t}}{e^{it/2} - e^{-it/2}} = \frac{\sin(n + \frac{1}{2})t}{\sin \frac{1}{2}t},$$

after multiplying numerator and denominator by $e^{-it/2}$. At $t = 0$ both sides equal $2n + 1$ (as a limit). □

Then $\widehat{D_n} = \sum_{k=-n}^{n} \chi_{\{k\}}$, so $\|\widehat{D_n}\|_\infty = 1$. On the other hand, since $|\sin t| \le |t|$,

$$\|D_n\|_1 \ge \frac{1}{2\pi}\int_{-\pi}^{\pi} \left|\frac{\sin(n + \frac{1}{2})t}{\frac{t}{2}}\right| dt = \frac{2}{\pi}\int_0^{\pi} \frac{1}{t}\left|\sin\left(n + \frac{1}{2}\right)t\right| dt = \frac{2}{\pi}\int_0^{(n+\frac{1}{2})\pi} \frac{1}{t}|\sin t|\, dt$$

$$\geq \frac{2}{\pi}\sum_{k=1}^{n}\frac{1}{k\pi}\int_{(k-1)\pi}^{k\pi}|\sin t|\,dt = \frac{4}{\pi^2}\sum_{k=1}^{n}\frac{1}{k},$$

so $\|D_n\|_1 \to \infty$, and $E_n = D_n/\|D_n\|_1$ has $\|E_n\|_1 = 1$, $\|\widehat{E_n}\|_\infty \to 0$. So no $\delta > 0$ has $\|\widehat{f}\|_\infty \geq \delta\|f\|_1$, as Corollary 6.2.5 would give for the injective $f \mapsto \widehat{f}$: *the Fourier transform $L^1(\mathbb{T}) \to c_0(\mathbb{Z})$ is not surjective.*

6.2.3 The closed graph theorem

The closed graph theorem, from the open mapping theorem, proves boundedness; the hypothesis says the graph of T is closed.

Theorem 6.2.6 (closed graph theorem)

A linear map $T : X \to Y$ between Banach spaces satisfying

$$x_n \in X,\ \lim_{n\to\infty} x_n = x,\ \lim_{n\to\infty} T(x_n) = y \implies T(x) = y$$

is bounded.

Proof. $\|x\|' = \|x\| + \|T(x)\|$ is a norm on X. If $\langle x_n\rangle$ is $\|\cdot\|'$-Cauchy, then $\langle x_n\rangle$ and $\langle T(x_n)\rangle$ are Cauchy in the Banach spaces X, Y, so there are x, y with

$$\lim_{n\to\infty}\|x_n - x\| = 0, \qquad \lim_{n\to\infty}\|T(x_n) - y\| = 0.$$

By hypothesis $T(x) = y$, so

$$\lim_{n\to\infty}\|x_n - x\|' = \lim_{n\to\infty}\|x_n - x\| + \lim_{n\to\infty}\|T(x_n) - T(x)\| = 0,$$

so $(X, \|\cdot\|')$ is Banach. As $\|x\| \leq \|x\|'$, the identity $(X, \|\cdot\|') \to (X, \|\cdot\|)$ is a bounded bijection, with bounded inverse by Corollary 6.2.5: $\|x\|' \leq M\|x\|$, and

$$\|Tx\| + \|x\| = \|x\|' \leq M\|x\|, \qquad x \in X,$$

so $\|Tx\| \leq (M-1)\|x\|$. ■

Exercise 6.2.4

In the proof of the closed graph theorem, show that $x \mapsto \|x\|'$ is a norm.

Solution [Supplement]. $\|x\|' \geq 0$, and $\|x\|' = 0$ forces $\|x\| = 0$, i.e. $x = 0$. By linearity of T, $\|ax\|' = |a|\|x\| + |a|\|Tx\| = |a|\|x\|'$ and

$$\|x + z\|' = \|x + z\| + \|Tx + Tz\| \leq \|x\| + \|z\| + \|Tx\| + \|Tz\| = \|x\|' + \|z\|'.$$

□

6.2.4 The uniform boundedness principle

Another application of Baire's theorem is the uniform boundedness principle, also called the Banach–Steinhaus theorem.

Theorem 6.2.7 (uniform boundedness principle)

Let $\{T_\alpha : \alpha \in A\}$ be a family of bounded linear maps from a Banach space X to a normed space Y. If for every $x \in X$

$$M(x) = \sup\{\|T_\alpha(x)\| : \alpha \in A\} < \infty,$$

then $\sup\{\|T_\alpha\| : \alpha \in A\} < \infty$.

Proof. For $n = 1, 2, \ldots$ put

$$C_n = \bigcap_{\alpha \in A} \{x \in X : \|T_\alpha(x)\| \le n\}.$$

Each $x \mapsto \|T_\alpha(x)\|$ is continuous, so C_n is closed, and by hypothesis $X = \bigcup_n C_n$. By Baire's theorem some C_n contains a neighborhood $N(x_0, r)$, i.e.

$$\alpha \in A,\ \|x - x_0\| < r \implies \|T_\alpha(x)\| \le n.$$

If $\|x\| < r$, then $\|(x + x_0) - x_0\| < r$, so for every $\alpha \in A$

$$\|T_\alpha(x)\| - \|T_\alpha(x_0)\| \le \|T_\alpha(x + x_0)\| \le n,$$

hence $\|T_\alpha(x)\| \le n + \|T_\alpha(x_0)\| \le n + M(x_0)$. Therefore

$$\sup_{\alpha \in A} \|T_\alpha\| \le \frac{1}{r}(n + M(x_0)).$$

■

The next corollary is very useful for defining new bounded linear maps from a given sequence.

Corollary 6.2.8

Let $T_n : X \to Y$ be bounded linear maps from a Banach space to a normed space with $\langle T_n(x) \rangle$ convergent for every x. Then $T(x) = \lim_n T_n(x)$ is a bounded linear map.

Proof. T is linear, and convergent sequences are bounded, so uniform boundedness gives $M = \sup_n \|T_n\| < \infty$ and

$$\|T(x)\| = \lim_{n \to \infty} \|T_n(x)\| \le \sup_n \|T_n\| \|x\| = M\|x\|,$$

so $\|T\| \le M$. ■

6.2.5 A continuous function whose Fourier series diverges

By uniform boundedness, Fourier series of continuous functions need not converge pointwise. By (2) of §5.1 the partial sums are

$$\sum_{k=-n}^{n} \widehat{f}(k) u_k = \sum_{k=-n}^{n} u_k * f = D_n * f.$$

Fix $t_0 \in \mathbb{T}$ and n, and consider the linear functional on the Banach space $C(\mathbb{T})$

$$T_n : f \mapsto (D_n * f)(t_0), \qquad f \in C(\mathbb{T}).$$

Then

$$|T_n(f)| \le \int_{\mathbb{T}} |f(t) D_n(t_0 - t)|\, dt \le \|f\|_\infty \|D_n\|_1, \qquad f \in C(\mathbb{T}),$$

so $\|T_n\| \le \|D_n\|_1$. $\operatorname{sgn} D_n(t_0 - t)$ has finitely many discontinuities, so continuous h_i with $\|h_i\|_\infty \le 1$ converge to it except at finitely many points, and by dominated convergence

$$T_n(h_i) = \int_{\mathbb{T}} h_i(t) D_n(t_0 - t)\, dt \to \int_{\mathbb{T}} |D_n(t_0 - t)|\, dt = \|D_n\|_1,$$

so $\|T_n\| = \|D_n\|_1 \to \infty$. If every $f \in C(\mathbb{T})$ had Fourier series converging at t_0, uniform boundedness would bound $\|T_n\|$. So *some continuous function has a Fourier series diverging at* t_0.

6.3 Hilbert Spaces

Motivation. Inner products bring Euclidean geometry to infinite dimensions: functionals are inner products, vectors expand in orthonormal bases, and $L^2(\mathbb{T})$ gives Fourier series.

6.3.1 Inner products

Definition (inner product)

For a complex vector space X, a function $(x, y) \mapsto \langle x, y\rangle : X \times X \to \mathbb{C}$ satisfying

(I1) $x \mapsto \langle x, y\rangle : X \to \mathbb{C}$ is a linear functional for each $y \in X$;

(I2) $\langle x, y\rangle = \overline{\langle y, x\rangle}$ for all $x, y \in X$;

(I3) $\langle x, x\rangle \ge 0$ for all $x \in X$;

(I4) $x = 0 \iff \langle x, x\rangle = 0$

is an **inner product**, and a vector space with an inner product is an **inner product space**.

Example 1

On $L^2(X, \mu)$,

$$\langle f, g\rangle = \int_X f\overline{g}\, d\mu, \qquad f, g \in L^2(X, \mu),$$

is an inner product.

For x, y in an inner product space X, choose $\alpha \in \mathbb{C}$ with $|\alpha| = 1$ and $\alpha\langle y, x\rangle = |\langle x, y\rangle|$. Then for every real t,

$$0 \le \langle x - t\alpha y, x - t\alpha y\rangle = \langle x, x\rangle - t\alpha\langle y, x\rangle - t\overline{\alpha}\langle x, y\rangle + t^2\langle y, y\rangle = \langle x, x\rangle - 2t|\langle x, y\rangle| + t^2\langle y, y\rangle.$$

A nonnegative real quadratic in t has nonpositive discriminant, so

$$|\langle x, y\rangle|^2 \le \langle x, x\rangle\langle y, y\rangle, \qquad x, y \in X.$$

This is the **Schwarz inequality**; for $L^2(X, \mu)$ it is a special case of Hölder's inequality. For $x \in X$ put

$$\|x\| = \langle x, x\rangle^{1/2}; \tag{9}$$

$x \mapsto \|x\|$ is a norm, the triangle inequality following from Schwarz: $\|x + y\|^2 = \|x\|^2 + 2\operatorname{Re}\langle x, y\rangle + \|y\|^2 \le (\|x\| + \|y\|)^2$. A complete inner product space is a **Hilbert space**. The norm determines the inner product by the **polarization identity**

$$\langle x, y\rangle = \frac{1}{4}\sum_{k=0}^{3} i^k \|x + i^k y\|^2, \tag{10}$$

and the norm of an inner product space satisfies the **parallelogram law**

$$\|x+y\|^2+\|x-y\|^2=2\|x\|^2+2\|y\|^2. \tag{11}$$

Exercise 6.3.1

Prove (10) and (11) for vectors x, y of an inner product space X.

Solution [Supplement]. For $c\in\mathbb{C}$ with $|c|=1$,

$$\|x+cy\|^2=\|x\|^2+\overline{c}\langle x,y\rangle+c\langle y,x\rangle+\|y\|^2.$$

(11): add the cases $c=1$ and $c=-1$; the cross terms cancel. (10): multiply the case $c=i^k$ by i^k and sum over $k=0,1,2,3$:

$$\sum_k i^k\|x+i^k y\|^2=\Big(\sum_k i^k\Big)(\|x\|^2+\|y\|^2)+\Big(\sum_k i^k\overline{i^k}\Big)\langle x,y\rangle+\Big(\sum_k i^{2k}\Big)\langle y,x\rangle=0+4\langle x,y\rangle+0,$$

since $\sum_{k=0}^{3} i^k=0$, $i^k\overline{i^k}=1$, and $\sum_k(-1)^k=0$. □

6.3.2 The dual of a Hilbert space

We now determine the dual of a Hilbert space H. By the Schwarz inequality, for each $y\in H$

$$\phi_y: x\mapsto\langle x,y\rangle,\qquad x\in H,$$

is a bounded linear functional with $\|\phi_y\|=\|y\|$ (take $x=y$ for equality). Hence

$$y\mapsto\phi_y: H\to H^* \tag{12}$$

is an isometric injection, not linear but **conjugate linear**: $\phi_{\alpha y}=\overline{\alpha}\phi_y$.

To show (12) is onto, for $\phi\in H^*$ we need y with $\phi(x)=\langle x,y\rangle$, i.e. a vector perpendicular to $\ker\phi$; the next theorem supplies one.

Theorem 6.3.1 (closest point)

Let C be a closed convex subset of a Hilbert space H and $x_0\in H\setminus C$. Put

$$d=\inf\{\|x_0-y\|: y\in C\}.$$

Then there is a unique $y_0\in C$ with $\|x_0-y_0\|=d$.

Proof. We may assume $x_0=0$ (translate C by $-x_0$). Replacing x, y by $\frac{1}{2}x, \frac{1}{2}y$ in the parallelogram law gives

$$\frac{1}{4}\|x-y\|^2=\frac{1}{2}\|x\|^2+\frac{1}{2}\|y\|^2-\Big\|\frac{x+y}{2}\Big\|^2.$$

If $x,y\in C$ then $\frac{x+y}{2}\in C$ by convexity, so $\|\frac{x+y}{2}\|\geq d$ and

$$\|x-y\|^2\leq 2\|x\|^2+2\|y\|^2-4d^2,\qquad x,y\in C. \tag{13}$$

Uniqueness. If $x,y\in C$ and $\|x\|=\|y\|=d$, (13) gives $x=y$. *Existence.* Take a sequence $\langle y_n\rangle$ in C with $\|y_n\|\to d$. By (13),

$$\|y_n-y_m\|^2\leq 2\|y_n\|^2+2\|y_m\|^2-4d^2\to 0,$$

so $\langle y_n\rangle$ is Cauchy. As H is complete and C closed, $y_0=\lim_n y_n\in C$, and $\|y_0\|=\lim_n\|y_n\|=d$. ■

Now let E be a closed subspace of H and $x_0 \in H \setminus E$. By Theorem 6.3.1 there is $y_0 \in E$ with

$$y \in E \implies \|x_0 - y_0\| \le \|x_0 - y\|.$$

Put $z_0 = x_0 - y_0$. For every $y \in E$ and $\alpha \in \mathbb{C}$,

$$\langle z_0, z_0\rangle = \|x_0 - y_0\|^2 \le \|x_0 - y_0 - \alpha y\|^2 = \langle z_0 - \alpha y, z_0 - \alpha y\rangle,$$

so

$$0 \le -\alpha\langle y, z_0\rangle - \overline{\alpha}\langle z_0, y\rangle + |\alpha|^2\langle y, y\rangle.$$

Substituting $\alpha = t\langle z_0, y\rangle$ $(t \in \mathbb{R})$,

$$0 \le -2t|\langle y, z_0\rangle|^2 + t^2|\langle y, z_0\rangle|^2\langle y, y\rangle$$

for every $t \in \mathbb{R}$; small $t > 0$ forces $\langle y, z_0\rangle = 0$ for $y \in E$. x, y are **orthogonal**, $x \perp y$, if $\langle x, y\rangle = 0$; $E^\perp$ is the set of vectors orthogonal to E. As $z_0 \neq 0$ $(x_0 \notin E)$ and $z_0 \in E^\perp$:

Corollary 6.3.2

If E is a closed proper subspace of a Hilbert space H, then $E^\perp \supsetneq \{0\}$.

Return to $\phi \in H^*$. If $\phi \neq 0$, apply Corollary 6.3.2 to the closed proper subspace $\ker\phi$ to get $z \in (\ker\phi)^\perp$ with $\|z\| = 1$. Since $\phi(x)z - \phi(z)x \in \ker\phi$, for every $x \in H$

$$0 = \langle \phi(x)z - \phi(z)x, z\rangle = \phi(x) - \phi(z)\langle x, z\rangle = \phi(x) - \langle x, \overline{\phi(z)}z\rangle.$$

So with $y = \overline{\phi(z)}z$,

$$\phi(x) = \langle x, y\rangle, \qquad x \in H.$$

This proves:

Theorem 6.3.3 (Riesz representation)

For a Hilbert space H, the map $y \mapsto \phi_y$ defined by (12) is a surjective, isometric, conjugate linear map $H \to H^*$.

Exercise 6.3.2

Show that every Hilbert space is reflexive.

Solution [Supplement]. By Theorem 6.3.3, H^* is a Hilbert space with $\langle \phi_y, \phi_z\rangle_{H^*} = \langle z, y\rangle$ (inducing the dual norm, as $y \mapsto \phi_y$ is a conjugate-linear isometric bijection). For $\Lambda \in H^{**}$, Theorem 6.3.3 on H^* gives ϕ_z with $\Lambda(\phi) = \langle \phi, \phi_z\rangle_{H^*}$. Then for every y,

$$\Lambda(\phi_y) = \langle \phi_y, \phi_z\rangle_{H^*} = \langle z, y\rangle = \phi_y(z) = T_z(\phi_y).$$

Since every element of H^* is some ϕ_y, $\Lambda = T_z$. So $x \mapsto T_x$ is onto. □

Exercise 6.3.3

Show that $E^\perp$ is a closed subspace for any subset E of a Hilbert space. Show that $\overline{E} = E^{\perp\perp}$ for any subspace E.

Solution [Supplement]. $E^\perp = \bigcap_{y\in E} \ker\phi_y$ is an intersection of kernels of bounded linear functionals, hence a closed subspace. $E \subset E^{\perp\perp}$, closed, so $\overline{E} \subset E^{\perp\perp}$. Conversely, for $x \in E^{\perp\perp} \setminus \overline{E}$ the closest-point argument gives $y_0 \in \overline{E}$ with $z_0 = x - y_0 \neq 0$, $z_0 \in E^\perp$; also $z_0 \in E^{\perp\perp}$ (as $x, y_0 \in E^{\perp\perp}$), so $z_0 \perp z_0$, $z_0 = 0$: a contradiction. So $E^{\perp\perp} = \overline{E}$. □

6.3.3 Orthonormal sets

A subset $\{u_\alpha : \alpha \in A\}$ of an inner product space H satisfying

$$\langle u_\alpha, u_\beta \rangle = \begin{cases} 1, & \alpha = \beta, \\ 0, & \alpha \neq \beta \end{cases}$$

is an **orthonormal set**; the subspace it spans (finite linear combinations) is written P_A.

Proposition 6.3.4

Let $\{u_\alpha : \alpha \in F\}$ be a finite subset of an orthonormal set $\{u_\alpha : \alpha \in A\}$ in an inner product space H. For fixed $x \in H$,

$$\Big\| x - \sum_{\alpha \in F} \langle x, u_\alpha \rangle u_\alpha \Big\| \leq \|x - y\|, \qquad y \in P_F, \tag{14}$$

with equality only for $y = \sum_{\alpha \in F} \langle x, u_\alpha \rangle u_\alpha$. Moreover

$$\sum_{\alpha \in F} |\langle x, u_\alpha \rangle|^2 \leq \|x\|^2, \qquad x \in H.$$

Proof. Write $z = \sum_{\alpha \in F} \langle x, u_\alpha \rangle u_\alpha$. For each $\alpha \in F$, $\langle x - z, u_\alpha \rangle = \langle x, u_\alpha \rangle - \langle x, u_\alpha \rangle = 0$. If $y \in P_F$ then $y - z \in P_F$, so $\langle x - z, y - z \rangle = 0$, and

$$\|x - y\|^2 = \|x - z\|^2 + \|z - y\|^2, \qquad y \in P_F,$$

which proves the first part (with equality iff $y = z$). Also $\|z\|^2 = \sum_{\alpha \in F} |\langle x, u_\alpha \rangle|^2$, so putting $y = 0$ gives $\|x\|^2 = \|x - z\|^2 + \|z\|^2 \geq \|z\|^2$, the second inequality. ∎

Inequality (14) says that the coefficients c_α making $\sum_\alpha c_\alpha u_\alpha$ closest to x are $c_\alpha = \langle x, u_\alpha \rangle$. Since the second inequality holds for every finite $F \subset A$, Fact B.28 gives:

Corollary 6.3.5 (Bessel's inequality)

For any orthonormal set $\{u_\alpha : \alpha \in A\}$ in an inner product space H,

$$\sum_{\alpha \in A} |\langle x, u_\alpha \rangle|^2 \leq \|x\|^2, \qquad x \in H.$$

When is Bessel's inequality an equality? In finite dimensions, for a basis; here is the general analogue.

Theorem 6.3.6

For an orthonormal set $\mathcal{U} = \{u_\alpha : \alpha \in A\}$ in a Hilbert space H the following are equivalent:

(a) for every $x \in H$,
$$\sum_{\alpha \in A} |\langle x, u_\alpha \rangle|^2 = \|x\|^2; \tag{15}$$

(b) for all $x, y \in H$,
$$\sum_{\alpha \in A} \langle x, u_\alpha \rangle \overline{\langle y, u_\alpha \rangle} = \langle x, y \rangle; \tag{16}$$

(c) if $\mathcal{W} \supset \mathcal{U}$ is an orthonormal set, then $\mathcal{W} = \mathcal{U}$ (maximality);

(d) P_A is dense in H.

Proof. For each $x \in H$ define $\widehat{x} : A \to \mathbb{C}$ by

$$\widehat{x}(\alpha) = \langle x, u_\alpha \rangle, \qquad \alpha \in A.$$

By Bessel's inequality $\widehat{x} \in \ell^2(A)$, and we get a bounded linear map

$$\Phi : x \mapsto \widehat{x} : H \to \ell^2(A). \tag{17}$$

(a) says Φ preserves norms; such a map preserves inner products by the polarization identity (10). The left side of (16) is exactly $\langle \widehat{x}, \widehat{y} \rangle$, so (a)⇒(b).

(b)⇒(c): if $\mathcal{W} \supsetneq \mathcal{U}$ is orthonormal, pick $u \in \mathcal{W} \setminus \mathcal{U}$; then (16) with $x = y = u$ reads $0 = \sum_\alpha |\langle u, u_\alpha \rangle|^2 = \|u\|^2 = 1$, a contradiction.

(c)⇒(d): if P_A is not dense in H, Corollary 6.3.2 gives a unit vector $u \in (P_A)^\perp = (\overline{P_A})^\perp$, and $\mathcal{U} \cup \{u\}$ is a larger orthonormal set, so (c) fails.

(d)⇒(a): (15) holds on the dense subspace P_A of finite combinations, so Φ is an isometry there. For $x \in H$ take $x_n \in P_A$, $x_n \to x$; as Φ is bounded,

$$\sum_{\alpha \in A} |\langle x, u_\alpha \rangle|^2 = \|\widehat{x}\|_2^2 = \lim_{n\to\infty} \|\widehat{x_n}\|_2^2 = \lim_{n\to\infty} \|x_n\|^2 = \|x\|^2.$$

■

(16) is **Parseval's identity**, and an orthonormal set as in Theorem 6.3.6 is an **orthonormal basis**; then Φ in (17) preserves inner products, a **Hilbert space isomorphism** if onto. It is onto: for $y \in \ell^2(A)$ take finitely supported $y_n \to y$, $y_n = \Phi(x_n)$, $x_n \in P_A$; $\|x_n - x_m\| = \|y_n - y_m\|_2$, so $x_n \to x$ and $\Phi(x) = \lim_n y_n = y$. Hence:

Theorem 6.3.7

If $\{u_\alpha : \alpha \in A\}$ is an orthonormal basis of a Hilbert space H, then $x \mapsto \widehat{x}$ is a bijective Hilbert space isomorphism $H \to \ell^2(A)$.

So a Hilbert space is determined by the cardinality of an orthonormal basis; which basis is convenient depends on the situation.

6.3.4 Fourier series in $L^2(\mathbb{T})$

$\{u_n : n \in \mathbb{Z}\}$ (§5.1 (1)) is orthonormal in $L^2(\mathbb{T})$ with $\widehat{f}(n) = \langle f, u_n \rangle$, and Corollary 5.1.13 is Theorem 6.3.6(d): an orthonormal basis of $L^2(\mathbb{T})$. Explicitly:

$$\sum_{n=-\infty}^{\infty} \widehat{f}(n)\overline{\widehat{g}(n)} = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(t)\overline{g(t)}\, dt, \qquad f, g \in L^2(\mathbb{T}),$$

also called **Parseval's identity** (the series converges absolutely for $f, g \in L^2(\mathbb{T})$, so the limit is unambiguous). Moreover

$$\lim_{n\to\infty} \Big\| f - \sum_{k=-n}^{n} \widehat{f}(k) u_k \Big\|_2 = 0, \qquad f \in L^2(\mathbb{T}). \tag{18}$$

(The partial sums converge in $\|\cdot\|_2$, so by Fact B.18 a subsequence converges a.e. In fact the Fourier series of an L^2 function converges a.e. (Carleson), while some L^1 function has an everywhere divergent Fourier series (Kolmogorov).)

Completeness played no big role in Theorem 6.3.6, but Theorem 6.3.7 for Fourier series gives, for every $a \in \ell^2(\mathbb{Z})$, an L^2 function with coefficients a; here completeness of $L^2(\mathbb{T})$ is decisive. With Fact B.17 this is the **Riesz–Fischer theorem**, unavailable for improper Riemann integrals: a main reason the Lebesgue integral is more useful.

6.3.5 The Plancherel transform

As $L^2(\mathbb{T}) \subset L^1(\mathbb{T})$ (Fact B.29), restricting gives the isomorphism $f \mapsto \widehat{f}$ and Parseval (18). As $L^2(\mathbb{R}) \not\subset L^1(\mathbb{R})$, on $\mathbb{R}$ we start from $C_c(\mathbb{R}) \subset L^1(\mathbb{R}) \cap L^2(\mathbb{R})$.

If $f \in C_c(\mathbb{R})$ and $\operatorname{supp} f \subset (-\pi, \pi)$, then Parseval for periodic functions and (19) of §5.2 (with $F = f$ on $(-\pi, \pi)$, $\widehat{F}(n) = \frac{1}{2\pi}\widehat{f}(n)$) give

$$\frac{1}{2\pi}\int_{-\pi}^{\pi} |f(t)|^2\,dt = \sum_{n=-\infty}^{\infty} \Big|\frac{1}{2\pi}\widehat{f}(n)\Big|^2.$$

This also holds for $g(x) = f(x)e^{-i\alpha x}$ (with $\widehat{g}(n) = \widehat{f}(n+\alpha)$), so

$$\int_{\mathbb{R}} |f(x)|^2\,dx = \frac{1}{2\pi}\sum_{n=-\infty}^{\infty} |\widehat{f}(n+\alpha)|^2.$$

Integrating both sides over $\alpha \in [0, 1)$,

$$\int_{\mathbb{R}} |f(x)|^2\,dx = \frac{1}{2\pi}\int_0^1 \sum_{n=-\infty}^{\infty} |\widehat{f}(n+\alpha)|^2\,d\alpha = \frac{1}{2\pi}\int_{\mathbb{R}} |\widehat{f}(\alpha)|^2\,d\alpha.$$

In general, if $\operatorname{supp} f \subset (-\pi\lambda, \pi\lambda)$, put $g(x) = \sqrt{\lambda} f(\lambda x)$; then $\operatorname{supp} g \subset (-\pi, \pi)$ and $\widehat{g}(\alpha) = \frac{1}{\sqrt{\lambda}}\widehat{f}(\alpha/\lambda)$, so for every $f \in C_c(\mathbb{R})$

$$\int_{\mathbb{R}} |f(x)|^2\,dx = \int_{\mathbb{R}} |g(x)|^2 = \frac{1}{2\pi}\int_{\mathbb{R}} |\widehat{g}(\alpha)|^2\,d\alpha = \frac{1}{2\pi}\int_{\mathbb{R}} |\widehat{f}(\alpha)|^2\,d\alpha.$$

Let $\widehat{\mathbb{R}}$ denote the real line with $\frac{1}{2\pi}$ times Lebesgue measure. Then

$$\int_{\mathbb{R}} |f(x)|^2\,dx = \int_{\widehat{\mathbb{R}}} |\widehat{f}(\alpha)|^2\,d\alpha, \qquad f \in C_c(\mathbb{R}), \tag{19}$$

and the inversion formula reads

$$f(x) = \int_{\widehat{\mathbb{R}}} \widehat{f}(\alpha) e^{i\alpha x}\,d\alpha.$$

Regarding $C_c(\mathbb{R})$ as a subspace of $L^2(\mathbb{R})$, (19) gives an isometric linear map

$$f \mapsto \widehat{f} : C_c(\mathbb{R}) \to L^2(\widehat{\mathbb{R}}).$$

Since $C_c(\mathbb{R})$ is dense in $L^2(\mathbb{R})$, this map extends to an isometric linear map

$$\mathcal{F} : L^2(\mathbb{R}) \to L^2(\widehat{\mathbb{R}}).$$

Consistency on $L^1 \cap L^2$. For $f \in L^1(\mathbb{R}) \cap L^2(\mathbb{R})$, is $\widehat{f} = \mathcal{F}(f)$? By the proofs of Facts B.20–B.22, for $p \neq q \in [1, \infty)$ and $f \in L^p[a, b] \cap L^q[a, b]$ there is a simple $s : [a, b] \to \mathbb{C}$ with simultaneously

$$\|f - s\|_p < \epsilon, \qquad \|f - s\|_q < \epsilon.$$

Moreover, choosing in the proof of Fact B.21 a continuous g with

$$\mu(\{x : g(x) \neq s(x)\}) < \min\Big\{\Big(\frac{\epsilon}{2\|s\|_\infty}\Big)^p, \Big(\frac{\epsilon}{2\|s\|_\infty}\Big)^q\Big\}, \qquad |g| \leq \|s\|_\infty,$$

we get simultaneously

$$\|f - g\|_p < 2\epsilon, \qquad \|f - g\|_q < 2\epsilon. \tag{20}$$

Likewise (Fact B.22) every $f \in L^p(\mathbb{R}) \cap L^q(\mathbb{R})$ has $g \in C_c(\mathbb{R})$ with (20), so for $f \in L^1(\mathbb{R}) \cap L^2(\mathbb{R})$ there are $g_n \in C_c(\mathbb{R})$ with

$$\|f - g_n\|_1 \to 0, \qquad \|f - g_n\|_2 \to 0,$$

and then

$$\lim_{n\to\infty} \|\widehat{f} - \widehat{g_n}\|_\infty = \lim_{n\to\infty} \|\mathcal{F}(f) - \mathcal{F}(g_n)\|_2 = 0.$$

Since $\widehat{g_n} = \mathcal{F}(g_n)$, we conclude $\widehat{f} = \mathcal{F}(f)$ (Exercise 6.3.4).

Exercise 6.3.4

If a sequence $\langle h_n \rangle$ in $C_0(\mathbb{R})$ satisfies $\lim_n \|f - h_n\|_\infty = \lim_n \|g - h_n\|_2 = 0$, show that $f = g$ almost everywhere.

Solution [Supplement]. Since $\|g - h_n\|_2 \to 0$, by Fact B.18 a subsequence $h_{n_k} \to g$ almost everywhere. But $h_{n_k} \to f$ uniformly, in particular pointwise everywhere. Hence $f = g$ almost everywhere. □

Surjectivity. By the proof of Theorem 5.2.8, every $h \in C_c^2(\mathbb{R})$ is $\widehat{k}$ for a bounded integrable, hence L^2, $k = \frac{1}{2\pi}\widehat{h}(-\cdot)$, so $h = \mathcal{F}(k)$. The range is dense in $L^2(\widehat{\mathbb{R}})$ and closed (isometric image of a complete space), so $\mathcal{F} : L^2(\mathbb{R}) \to L^2(\widehat{\mathbb{R}})$ is a surjective isometry (a Hilbert space isomorphism, by polarization), the **Plancherel transform**. Summarizing:

Theorem 6.3.8 (Plancherel)

There is a unique Hilbert space isomorphism $\mathcal{F} : L^2(\mathbb{R}) \to L^2(\widehat{\mathbb{R}})$ with $\mathcal{F}(f) = \widehat{f}$ for every $f \in L^1(\mathbb{R}) \cap L^2(\mathbb{R})$.

(Uniqueness: $\mathcal{F}$ is determined on the dense subspace $C_c(\mathbb{R}) \subset L^1 \cap L^2$ and is continuous.)

If $f \in L^2(\mathbb{R})$, then $f_A = \chi_{[-A,A]} f \in L^1(\mathbb{R}) \cap L^2(\mathbb{R})$ and

$$\widehat{f_A}(\alpha) = \int_{-A}^{A} f(x) e^{-i\alpha x}\, dx.$$

Since $\lim_{A\to\infty} \|f - f_A\|_2 = 0$,

$$\lim_{A\to\infty} \|\mathcal{F}(f) - \widehat{f_A}\|_2 = 0,$$

the analogue of (18) for Fourier series. Finally, the statement that the Plancherel transform is a Hilbert space isomorphism reads

$$\int_{\mathbb{R}} f(x)\overline{g(x)}\, dx = \frac{1}{2\pi} \int_{\mathbb{R}} \mathcal{F}f(\alpha)\overline{\mathcal{F}g(\alpha)}\, d\alpha,$$

also called **Parseval's identity**.

Example 2

Applying Parseval to $f(x) = g(x) = e^{-|x|}$ (with $\widehat{f}(\alpha) = \frac{2}{1+\alpha^2}$),

$$\int_{-\infty}^{\infty} (e^{-|x|})^2\, dx = \frac{1}{2\pi} \int_{-\infty}^{\infty} \left[\frac{2}{1+\alpha^2}\right]^2 d\alpha,$$

so

$$\int_{-\infty}^{\infty} \frac{1}{(1+\alpha^2)^2}\, d\alpha = \pi \int_0^{\infty} e^{-2x}\, dx = \frac{\pi}{2}.$$

Exercise 6.3.5

Compute $\displaystyle\int_{-\infty}^{\infty} \frac{\sin a\alpha \, \sin b\alpha}{\alpha^2}\, d\alpha$ for $a, b > 0$.

Solution [Supplement]. By Example 1 of §5.2, $\widehat{\chi_{[-a,a]}}(\alpha) = \frac{2\sin a\alpha}{\alpha}$, real-valued. Parseval with $f = \chi_{[-a,a]}$, $g = \chi_{[-b,b]}$ (both in $L^1 \cap L^2$):

$$\int_{\mathbb{R}} \chi_{[-a,a]}\chi_{[-b,b]}\, dx = \frac{1}{2\pi} \int_{\mathbb{R}} \frac{2\sin a\alpha}{\alpha} \cdot \frac{2\sin b\alpha}{\alpha}\, d\alpha.$$

The left side is $2\min\{a,b\}$. Therefore

$$\int_{-\infty}^{\infty} \frac{\sin a\alpha \, \sin b\alpha}{\alpha^2} \, d\alpha = \pi \min\{a,b\}.$$

(For $a = b$ this gives the classical $\int_{\mathbb{R}} \frac{\sin^2 a\alpha}{\alpha^2} \, d\alpha = \pi a$.) □

Chapter 7. Complex Measures

Overview

Results from Chapters 1–3 are cited as *Fact B.k* (see the *Background* notes). This chapter allows measures with negative or complex values. Relative to a positive measure, every measure splits into an integral part and a singular part (Lebesgue–Radon–Nikodym), the core of measure theory; applied to Lebesgue measure on the Borel sets, this describes all Borel measures on $\mathbb{R}$.

7.1 Signed Measures and Complex Measures

Motivation. Signed and complex measures are differences of positive measures (Hahn, Jordan), so the positive theory still applies.

7.1.1 Definitions

A function $\lambda : \mathcal{S} \to [0, \infty]$ on a σ-algebra $\mathcal{S}$ of a set X satisfying

$$\{A_n \in \mathcal{S} : n = 1, 2, \dots\} \text{ pairwise disjoint } \implies \lambda\Big(\bigcup_{n=1}^{\infty} A_n\Big) = \sum_{n=1}^{\infty} \lambda(A_n) \tag{1}$$

with $\lambda(\emptyset) = 0$ was called a positive measure.

Definition (signed measure, complex measure)

A function $\lambda : \mathcal{S} \to \mathbb{R}^*$ satisfying (1) and $\lambda(\emptyset) = 0$ is a **signed measure**. A function $\lambda : \mathcal{S} \to \mathbb{C}$ satisfying (1) is a **complex measure**.

Convergence of $\sum \lambda(A_n)$ is assumed (to $\pm\infty$ allowed for signed measures). If $\lambda(E) = \infty$ and $\lambda(F) = -\infty$, then in

$$\lambda(E \cup F) = \lambda(E \setminus F) + \lambda(F \setminus E) + \lambda(E \cap F)$$

both $\pm\infty$ would appear on the right: *a signed measure cannot take both values* $\pm\infty$. Complex measures form a vector space.

Example 1

For a measurable function h on $\mathbb{R}$ define

$$\lambda(E) = \int_E h(x)\, dx, \qquad E \in \mathcal{M}. \tag{2}$$

If $h \geq 0$, λ is a positive measure (Fact B.7); if $h : \mathbb{R} \to \mathbb{R}^*$ is measurable with h_+ or h_- integrable, (2) is a signed measure; if $h : \mathbb{R} \to \mathbb{C}$ is integrable, a complex measure.

Given a complex measure $\lambda : \mathcal{S} \to \mathbb{C}$, let $\lambda_1(E), \lambda_2(E)$ be the real and imaginary parts of $\lambda(E)$. Then λ_1, λ_2 are signed measures and

$$\lambda = \lambda_1 + i\lambda_2. \tag{3}$$

So we first study signed measures, showing that a signed measure is a difference of positive measures — just as a real function is $f = f_+ - f_-$.

7.1.2 The Hahn decomposition

Definition (positive and negative sets)

Let λ be a signed measure on a σ-algebra $\mathcal{S}$ of X. A set $P \in \mathcal{S}$ is **positive** if

$$E \in \mathcal{S},\ E \subset P \implies \lambda(E) \geq 0,$$

and **negative** if

$$E \in \mathcal{S},\ E \subset P \implies \lambda(E) \leq 0.$$

E.g. in Example 1, $\{h \geq 0\}$ and $\{h < 0\}$ are positive and negative sets. Measurable subsets of positive sets are positive, and a union P of positive sets P_n is positive: for $E \subset P$, $E \in \mathcal{S}$, put

$$E_n = (E \cap P_n) \setminus (P_1 \cup \cdots \cup P_{n-1}), \qquad n = 1, 2, \ldots \quad (P_0 = \emptyset);$$

then $E_n \subset P_n$, so $\lambda(E_n) \geq 0$, and $\lambda(E) = \sum_n \lambda(E_n) \geq 0$.

Theorem 7.1.1 (Hahn decomposition)

For a signed measure λ on $(X, \mathcal{S})$, X splits into a positive set and a negative set, and two such splittings (P_1, Q_1), (P_2, Q_2) satisfy, for every $E \in \mathcal{S}$,

$$\lambda(E \cap P_1) = \lambda(E \cap P_2), \qquad \lambda(E \cap Q_1) = \lambda(E \cap Q_2).$$

Proof. First assume λ does not take the value $+\infty$. Define

$$\alpha = \sup\{\lambda(P) : P \text{ is a positive set}\};$$

$\emptyset$ is positive, so $\alpha \geq 0$. Choose positive sets $\langle P_n \rangle$ with $\alpha = \lim_n \lambda(P_n)$; their union P is positive. For each n, $P \setminus P_n \subset P$ is positive, so $\lambda(P \setminus P_n) \geq 0$ and

$$\lambda(P_n) \leq \lambda(P_n) + \lambda(P \setminus P_n) = \lambda(P) \leq \alpha.$$

Hence $\lambda(P) = \alpha < \infty$. Now put $Q = X \setminus P$. If a measurable $E \subset Q$ is positive, then $E \cup P$ is positive, so

$$\alpha \geq \lambda(E \cup P) = \lambda(E) + \lambda(P) = \lambda(E) + \alpha \geq \alpha,$$

and therefore $\lambda(E) = 0$. That is,

$$E \in \mathcal{S},\ E \subset Q,\ E \text{ positive} \implies \lambda(E) = 0. \tag{4}$$

It remains to show Q is negative. Let $E \subset Q$, $E \in \mathcal{S}$. If E is positive, $\lambda(E) = 0$ by (4). Otherwise let n_1 be the least natural number with some $F_1 \subset E$, $\lambda(F_1) < -\frac{1}{n_1}$. If $E \setminus F_1$ is positive, $\lambda(E \setminus F_1) = 0$ by (4) and $\lambda(E) = \lambda(F_1) < 0$. Otherwise repeat: if $E \setminus (F_1 \cup \cdots \cup F_{k-1})$ is positive, $\lambda(E) = \sum_{i<k} \lambda(F_i) < 0$; else take the least n_k and $F_k \in \mathcal{S}$ with

$$F_k \subset E \setminus (F_1 \cup \cdots \cup F_{k-1}), \qquad \lambda(F_k) < -\frac{1}{n_k}.$$

If the process never stops, put

$$F = E \setminus \Big[\bigsqcup_{k=1}^{\infty} F_k\Big].$$

Clearly $\langle n_k \rangle$ is nondecreasing.

If $\langle n_k \rangle$ is bounded, then $\lambda(\bigsqcup_k F_k) = -\infty$, and since $\lambda(F) < \infty$, $\lambda(E) = \lambda(\bigsqcup_k F_k) + \lambda(F) = -\infty$, and we are done.

If $\langle n_k \rangle$ is unbounded, we show F is positive. Suppose F contains $G \in \mathcal{S}$ with $\lambda(G) < 0$. Since $n_k \to \infty$, we can choose k with $\frac{1}{n_k - 1} < -\lambda(G)$. Then

$$G \subset F \subset E \setminus (F_1 \cup \cdots \cup F_{k-1}), \qquad \lambda(G) < -\frac{1}{n_k - 1},$$

contradicting the minimality in the choice of n_k. So F is positive, $\lambda(F) = 0$ by (4), and $\lambda(E) = \sum_{k=1}^{\infty} \lambda(F_k) < 0$. Hence Q is negative.

The case where λ does not take $-\infty$ is proved in the same way (apply the above to $-\lambda$).

For the second part, see Exercise 7.1.1. ■

Exercise 7.1.1

Prove the second part of the Hahn decomposition theorem.

Solution [Supplement]**.** For $E \in \mathcal{S}$, $E \cap (P_1 \setminus P_2) = E \cap P_1 \cap Q_2$ lies in the positive P_1 and the negative Q_2, so has measure 0; likewise $E \cap (P_2 \setminus P_1)$. Hence

$$\lambda(E \cap P_1) = \lambda(E \cap P_1 \cap P_2) + \lambda(E \cap P_1 \cap Q_2) = \lambda(E \cap P_1 \cap P_2) = \lambda(E \cap P_2).$$

The same argument with Q_1, Q_2 gives $\lambda(E \cap Q_1) = \lambda(E \cap Q_2)$. □

7.1.3 Jordan decomposition and total variation

Given a Hahn decomposition $X = P \sqcup Q$ for λ, define for $E \in \mathcal{S}$

$$\lambda_+(E) = \lambda(E \cap P), \qquad \lambda_-(E) = -\lambda(E \cap Q).$$

By Theorem 7.1.1 these do not depend on the decomposition. λ_+ and λ_- are positive measures and

$$\lambda(E) = \lambda_+(E) - \lambda_-(E), \qquad E \in \mathcal{S}.$$

Also define

$$|\lambda|(E) = \lambda_+(E) + \lambda_-(E);$$

$|\lambda|$ is a positive measure as well.

Exercise 7.1.2

If λ is defined by (2), show that for every $E \in \mathcal{M}$

$$\lambda_+(E) = \int_E h_+, \qquad \lambda_-(E) = \int_E h_-, \qquad |\lambda|(E) = \int_E |h|.$$

Solution [Supplement]**.** $P = \{h \geq 0\}$, $Q = \{h < 0\}$ is a Hahn decomposition: for $E \subset P$, $\lambda(E) = \int_E h \geq 0$, and for $E \subset Q$, $\lambda(E) \leq 0$. Since $h\chi_P = h_+$ and $h\chi_Q = -h_-$,

$$\lambda_+(E) = \int_{E \cap P} h = \int_E h_+, \qquad \lambda_-(E) = -\int_{E \cap Q} h = \int_E h_-,$$

and $|\lambda|(E) = \int_E (h_+ + h_-) = \int_E |h|$. □

Proposition 7.1.2

If λ is a signed measure on a σ-algebra $\mathcal{S}$ of X, then for every $E \in \mathcal{S}$

$$|\lambda|(E) = \sup\Big\{\sum_{n=1}^{\infty}|\lambda(E_n)| : \ E_n \in \mathcal{S},\ E = \bigsqcup_{n=1}^{\infty} E_n\Big\}. \tag{5}$$

Proof. Let α be the right side of (5). If (P, Q) is a Hahn decomposition, then

$$|\lambda|(E) = \lambda_+(E) + \lambda_-(E) = |\lambda(E \cap P)| + |\lambda(E \cap Q)| \le \alpha.$$

Conversely, if $E_n \in \mathcal{S}$ and $E = \bigsqcup_{n=1}^{\infty} E_n$, then

$$\sum_{n=1}^{\infty}|\lambda(E_n)| = \sum_{n=1}^{\infty}|\lambda_+(E_n) - \lambda_-(E_n)| \le \sum_{n=1}^{\infty}(\lambda_+(E_n) + \lambda_-(E_n)) = \sum_{n=1}^{\infty}|\lambda|(E_n) = |\lambda|(E),$$

so $\alpha \le |\lambda|(E)$. ∎

7.1.4 Total variation of a complex measure

From now on let λ be a complex measure on a σ-algebra $(X, \mathcal{S})$ and define $|\lambda|(E)$ for $E \in \mathcal{S}$ by (5). Clearly $|\lambda(E)| \le |\lambda|(E)$ for every $E \in \mathcal{S}$.

Proposition 7.1.3

If λ is a complex measure on a σ-algebra $\mathcal{S}$ of X, then $E \mapsto |\lambda|(E)$ is a positive measure.

Proof. For a disjoint sequence $\langle E_n\rangle$ in $\mathcal{S}$ with union E we must show

$$|\lambda|(E) = \sum_{n=1}^{\infty}|\lambda|(E_n).$$

If $E = \bigsqcup_{m=1}^{\infty} F_m$, then for all m, n

$$F_m = \bigsqcup_{n=1}^{\infty}(E_n \cap F_m), \qquad E_n = \bigsqcup_{m=1}^{\infty}(E_n \cap F_m).$$

Hence

$$\begin{aligned}\sum_{m=1}^{\infty}|\lambda(F_m)| &= \sum_{m=1}^{\infty}\Big|\sum_{n=1}^{\infty}\lambda(E_n \cap F_m)\Big| \le \sum_{m=1}^{\infty}\sum_{n=1}^{\infty}|\lambda(E_n \cap F_m)| \\ &= \sum_{n=1}^{\infty}\sum_{m=1}^{\infty}|\lambda(E_n \cap F_m)| \le \sum_{n=1}^{\infty}|\lambda|(E_n),\end{aligned}$$

and taking the supremum over partitions $\{F_m\}$ gives $|\lambda|(E) \le \sum_n |\lambda|(E_n)$. For the reverse, let $\epsilon > 0$ and choose for each E_n a partition $\{E_{n,m} : m \in \mathbb{N}\}$ with

$$|\lambda|(E_n) < \sum_{m=1}^{\infty}|\lambda(E_{n,m})| + \frac{\epsilon}{2^n}, \qquad n = 1, 2, \dots$$

Then $\{E_{n,m} : m, n = 1, 2, \dots\}$ is a partition of E, so

$$\sum_{n=1}^{\infty}|\lambda|(E_n) \le \sum_{m,n=1}^{\infty}|\lambda(E_{n,m})| + \epsilon \le |\lambda|(E) + \epsilon.$$

■

By (3), a complex measure λ is a linear combination of positive measures:

$$\lambda = (\lambda_{1+} - \lambda_{1-}) + i(\lambda_{2+} - \lambda_{2-}). \tag{6}$$

If $\{E_n\}$ is a partition of $E \in \mathcal{S}$, then

$$\sum_{n=1}^{\infty} |\lambda(E_n)| \le \sum_{n=1}^{\infty} \left(\lambda_{1+}(E_n)+\lambda_{1-}(E_n)+\lambda_{2+}(E_n)+\lambda_{2-}(E_n)\right) = \lambda_{1+}(E)+\lambda_{1-}(E)+\lambda_{2+}(E)+\lambda_{2-}(E),$$

so for every $E \in \mathcal{S}$

$$|\lambda|(E) \le \lambda_{1+}(E) + \lambda_{1-}(E) + \lambda_{2+}(E) + \lambda_{2-}(E).$$

Since λ is a complex measure, the values on the right are always finite. Hence $|\lambda|(E) < \infty$ for every $E \in \mathcal{S}$; in particular $|\lambda|(X) < \infty$, and we define

$$\|\lambda\| = |\lambda|(X).$$

Positive measures are in general not complex measures; **finite positive measures** ($\mu(X) < \infty$) are.

Exercise 7.1.3

For a σ-algebra $\mathcal{S}$ on X, show that the set of all complex measures on $\mathcal{S}$ is a vector space and that $\lambda \mapsto \|\lambda\|$ is a norm on it.

Solution [Supplement]**.** If λ, ρ are complex measures and $a \in \mathbb{C}$, then $(\lambda+\rho)(\emptyset) = 0$ and for disjoint A_n,

$$(\lambda+\rho)\Big(\bigsqcup A_n\Big) = \sum \lambda(A_n) + \sum \rho(A_n) = \sum (\lambda+\rho)(A_n)$$

(sum of convergent series); similarly for $a\lambda$. So they form a vector space. Norm: $\|\lambda\| \ge 0$. If $\|\lambda\| = 0$ then $|\lambda(E)| \le |\lambda|(E) \le |\lambda|(X) = 0$ for all E, so $\lambda = 0$. For any partition $\{E_n\}$ of X, $\sum |a\lambda(E_n)| = |a| \sum |\lambda(E_n)|$, so $|a\lambda| = |a||\lambda|$ and $\|a\lambda\| = |a|\|\lambda\|$. Finally $\sum |(\lambda+\rho)(E_n)| \le \sum |\lambda(E_n)| + \sum |\rho(E_n)| \le \|\lambda\| + \|\rho\|$; taking the supremum gives $\|\lambda+\rho\| \le \|\lambda\| + \|\rho\|$. □

7.2 The Radon–Nikodym Theorem and the Lebesgue Decomposition

Motivation. $\nu(E) = \int_E f\, d\mu$ (a density) holds exactly when ν ignores μ-null sets; this underlies densities, conditional expectation and the dual of L^p.

7.2.1 Absolute continuity and singularity

Definition

Let μ be a positive measure and λ a signed or complex measure on a σ-algebra $\mathcal{S}$ of X.

- λ is **absolutely continuous** with respect to μ, written $\lambda \ll \mu$, if

$$E \in \mathcal{S},\ \mu(E) = 0 \implies \lambda(E) = 0.$$

- λ is **singular** with respect to μ, written $\lambda \perp \mu$, if there are disjoint $A, B \in \mathcal{S}$ with

$$\lambda(E) = \lambda(A \cap E), \qquad \mu(E) = \mu(B \cap E), \qquad E \in \mathcal{S}. \tag{7}$$

(2) is absolutely continuous with respect to Lebesgue measure; the Dirac measure is singular.

Linear combinations of measures $\ll \mu$ are $\ll \mu$. For $\lambda_1, \lambda_2 \perp \mu$ take disjoint $A_i, B_i \in \mathcal{S}$ with, for $i = 1, 2$,

$$\lambda_i(E) = \lambda_i(A_i \cap E), \qquad \mu(E) = \mu(B_i \cap E), \qquad E \in \mathcal{S}.$$

Put $A = A_1 \cup A_2$, $B = B_1 \cap B_2$. Then A, B are disjoint and

$$(\lambda_1 + \lambda_2)(A \cap E) = (\lambda_1 + \lambda_2)(E), \qquad \mu(B \cap E) = \mu(E)$$

(indeed $\lambda_1(A \cap E) = \lambda_1(A_1 \cap A \cap E) = \lambda_1(A_1 \cap E)$, and $\mu(E) = \mu(B_1 \cap E) = \mu(B_2 \cap B_1 \cap E)$). So linear combinations of singular measures are singular.

If a complex measure λ is both absolutely continuous and singular with respect to μ, take disjoint A, B satisfying (7). For each $E \in \mathcal{S}$, $\mu(E \cap A) = \mu(E \cap A \cap B) = 0$, so $\lambda(E \cap A) = 0$ by $\lambda \ll \mu$. Hence $\lambda(E) = \lambda(E \cap A) = 0$, i.e. $\lambda = 0$.

Fix a positive measure μ. Part (8) of the next theorem, the **Lebesgue decomposition**, splits every complex measure into absolutely continuous and singular parts; part (9) gives the absolutely continuous part by a function, the **Radon–Nikodym derivative**. This theorem makes precise "measures are generalized functions".

Theorem 7.2.1 (Lebesgue–Radon–Nikodym)

Let μ be a σ-finite positive measure and λ a complex measure on a σ-algebra $\mathcal{S}$ of X. Then there are unique measures λ_a, λ_s with

$$\lambda = \lambda_a + \lambda_s, \qquad \lambda_a \ll \mu, \qquad \lambda_s \perp \mu. \tag{8}$$

Moreover there is a unique $h \in L^1(X, \mu)$ with

$$\lambda_a(E) = \int_E h \, d\mu, \qquad E \in \mathcal{S}. \tag{9}$$

If μ and λ are σ-finite positive measures, there are unique measures λ_a, λ_s satisfying (8) and a unique measurable $h : X \to [0, \infty]$ satisfying (9).

Proof. Let λ be complex; first uniqueness. Two decompositions give $\lambda_a - \lambda_a' = \lambda_s' - \lambda_s$, both $\ll \mu$ and $\perp \mu$, hence 0; $h \in L^1(\mu)$ is unique by Fact B.8(b).

Step 1: μ and λ finite positive measures. Then $\nu = \mu + \lambda$ is also a finite positive measure. For every $f \in L^2(\nu)$, by the Cauchy–Schwarz inequality,

$$\Big| \int_X f \, d\lambda \Big| \le \int_X |f| \, d\lambda \le \int_X |f| \, d\nu \le \Big(\int_X |f|^2 \, d\nu \Big)^{1/2} \nu(X)^{1/2},$$

and since $\nu(X) < \infty$, $f \mapsto \int_X f \, d\lambda$ is a bounded linear functional on $L^2(\nu)$. By Theorem 6.3.3 (Riesz) there is $g \in L^2(\nu)$ with

$$\int_X f \, d\lambda = \int_X f g \, d\nu, \qquad f \in L^2(\nu). \tag{10}$$

As $\nu(X) < \infty$, Fact B.29 gives $g \in L^1(\nu)$. Moreover, putting $f = \chi_E$,

$$\nu(E) > 0 \implies \frac{1}{\nu(E)} \int_E g \, d\nu = \frac{\lambda(E)}{\nu(E)} \in [0, 1],$$

so by Fact B.27, $0 \le g \le 1$ ν-almost everywhere. Hence we may assume $0 \le g \le 1$ while keeping (10). Put

$$A = \{x \in X : 0 \le g(x) < 1\}, \qquad B = \{x \in X : g(x) = 1\},$$

and for $E \in \mathcal{S}$

$$\lambda_a(E) = \lambda(E \cap A), \qquad \lambda_s(E) = \lambda(E \cap B).$$

Rewriting (10) (using $\nu = \mu + \lambda$),

$$\int_X (1-g) f \, d\lambda = \int_X fg \, d\mu, \qquad f \in L^2(\nu). \tag{11}$$

Putting $f = \chi_B$,

$$0 = \int_B (1-g)\, d\lambda = \int_B g \, d\mu = \mu(B),$$

so $\lambda_s \perp \mu$ (with the disjoint sets B and $A = X \setminus B$: $\lambda_s(E) = \lambda_s(B \cap E)$ and $\mu(E) = \mu(A \cap E)$). Next, putting $f = (1 + g + \cdots + g^n)\chi_E$ in (11),

$$\int_E (1 - g^{n+1})\, d\lambda = \int_E g(1 + g + \cdots + g^n)\, d\mu.$$

Let $n \to \infty$: as $(1 - g^{n+1})|_B = 0$ and $1 - g^{n+1} \nearrow 1$ on A, the left side tends to $\lambda(E \cap A) = \lambda_a(E)$, and with $h = \lim_n g(1 + g + \cdots + g^n)$ the right side to $\int_E h \, d\mu$ (monotone convergence). This gives (8), (9), $\lambda_a \ll \mu$, and $h \in L^1(\mu)$ as $\lambda_a(X) < \infty$.

Step 2: μ σ-finite, λ finite positive. Write $X = \bigsqcup_n X_n$ with $\mu(X_n) < \infty$ and choose positive numbers $\langle \alpha_n \rangle$ with $\sum_n \alpha_n \mu(X_n) < \infty$. For $E \in \mathcal{S}$ put

$$\mu_0(E) = \sum_{n=1}^{\infty} \alpha_n \mu(E \cap X_n); \tag{12}$$

μ_0 is clearly a finite positive measure. By Step 1 there are λ_a, λ_s with

$$\lambda = \lambda_a + \lambda_s, \qquad \lambda_a \ll \mu_0, \qquad \lambda_s \perp \mu_0.$$

If $\mu(E) = 0$ then $\mu_0(E) = 0$, so $\lambda_a \ll \mu_0$ gives $\lambda_a \ll \mu$. If $\mu_0(E \setminus B) = 0$, then $\mu((E \setminus B) \cap X_n) = 0$ for all n ($\alpha_n > 0$), so $\mu(E \setminus B) = 0$; thus $\lambda_s \perp \mu_0$ gives $\lambda_s \perp \mu$, proving (8). By Step 1 there is $h_0 \in L^1(\mu_0)$ with

$$\lambda_a(E) = \int_E h_0 \, d\mu_0, \qquad E \in \mathcal{S}.$$

Put

$$k = \sum_{n=1}^{\infty} \alpha_n \chi_{X_n}. \tag{13}$$

Then $\int_E d\mu_0 = \int_E k \, d\mu$, so

$$\int_X f \, d\mu_0 = \int_X fk \, d\mu$$

for simple f, hence for measurable $f \geq 0$. With $f = h_0 \chi_E$ and $h = h_0 k$ this gives (9), and $\int h \, d\mu = \int h_0 \, d\mu_0 < \infty$ ($h_0 \geq 0$).

This proves the theorem for σ-finite positive μ and finite positive λ; complex λ are combinations of four such by (6) (uniqueness as above). ∎

Exercise 7.2.1

Prove the Lebesgue–Radon–Nikodym theorem for σ-finite positive λ, and for signed λ that are differences of σ-finite positive measures. (Uniqueness cannot subtract possibly infinite measures.)

Solution [Supplement]. *Existence (λ σ-finite positive).* Write $X = \bigsqcup_m Y_m$, $\lambda(Y_m) < \infty$, and $\lambda^{(m)}(E) = \lambda(E \cap Y_m)$. The theorem gives $\lambda^{(m)} = \lambda_a^{(m)} + \lambda_s^{(m)}$, $d\lambda_a^{(m)} = h_m\,d\mu$, $h_m \geq 0$, $\lambda_s^{(m)}$ concentrated on A_m, $\mu(A_m) = 0$; we may take $A_m \subset Y_m$ and $h_m = h_m \chi_{Y_m}$. Put

$$\lambda_a = \sum_m \lambda_a^{(m)}, \quad \lambda_s = \sum_m \lambda_s^{(m)}, \quad h = \sum_m h_m, \quad A = \bigcup_m A_m.$$

Then $\lambda = \lambda_a + \lambda_s$, $\lambda_a(E) = \int_E h\,d\mu$ (monotone convergence), $\lambda_a \ll \mu$, $\mu(A) = 0$ and $\lambda_s(E) = \lambda_s(E \cap A)$, so $\lambda_s \perp \mu$.

Uniqueness. For another decomposition with h', A', $\mu(A') = 0$, let $N = A \cup A'$ ($\mu(N) = 0$). Singular parts vanish on N^c and absolutely continuous parts on N, so for every E

$$\lambda_s(E) = \lambda(E \cap N) = \lambda_s'(E), \qquad \lambda_a(E) = \lambda(E \setminus N) = \lambda_a'(E).$$

No subtraction was used. For h: λ_a is finite on each $Y_m \cap X_n$ ($\mu(X_n) < \infty$), so $\int_E (h - h')\,d\mu = 0$ for $E \subset Y_m \cap X_n$, and $h = h'$ a.e.

Signed case. For $\lambda = \rho_1 - \rho_2$ with σ-finite positive ρ_i (one finite), decompose each ρ_i and set $\lambda_a = \rho_{1,a} - \rho_{2,a}$, $\lambda_s = \rho_{1,s} - \rho_{2,s}$, $h = h_1 - h_2$; these are well defined, $\ll \mu$ and $\perp \mu$ respectively. Uniqueness: $\lambda_s(E) = \lambda(E \cap N)$ and $\lambda_a(E) = \lambda(E \setminus N)$ for *any* decomposition. □

When λ_a and μ satisfy (9) we write

$$d\lambda_a = h\,d\mu, \qquad h = \Big[\frac{d\lambda_a}{d\mu}\Big].$$

Exercise 7.2.2

If $d\lambda = g\,d\mu$, show that $\lambda(\{x : g(x) = 0\}) = 0$.

Solution [Supplement]. Let $Z = \{x : g(x) = 0\}$. Then $\lambda(Z) = \int_Z g\,d\mu = \int_Z 0\,d\mu = 0$. (More generally $\lambda(E) = 0$ and even $|\lambda|(E) = 0$ for every measurable $E \subset Z$.) □

7.2.2 Polar decomposition

The Lebesgue–Radon–Nikodym theorem has many applications in measure theory; we look at a few.

Theorem 7.2.2 (polar decomposition)

For every complex measure λ on a set X there is a unique $h \in L^1(X, |\lambda|)$ with

$$d\lambda = h\,d|\lambda|, \qquad |h| = 1.$$

Proof. Since $\lambda \ll |\lambda|$, there is a unique $h \in L^1(|\lambda|)$ with $d\lambda = h\,d|\lambda|$ (Theorem 7.2.1, the singular part being 0). If $|\lambda|(E) > 0$, then

$$\Big|\frac{1}{|\lambda|(E)} \int_E h\,d|\lambda|\Big| = \frac{|\lambda(E)|}{|\lambda|(E)} \leq 1,$$

so by Fact B.27, $|h| \leq 1$ almost everywhere. On the other hand, for $r > 0$ put $A_r = \{x \in X : |h(x)| < r\}$. If $A_r = \bigsqcup_{n=1}^{\infty} E_n$, then

$$\sum_{n=1}^{\infty} |\lambda(E_n)| = \sum_{n=1}^{\infty} \Big|\int_{E_n} h\,d|\lambda|\Big| \leq \sum_{n=1}^{\infty} r|\lambda|(E_n) = r|\lambda|(A_r),$$

so $|\lambda|(A_r) \leq r|\lambda|(A_r)$, and for $r < 1$ this forces $|\lambda|(A_r) = 0$. Hence $|h| \geq 1$ almost everywhere, and we may redefine h on a null set so that $|h| = 1$. ■

7.2.3 Integration with respect to a complex measure

The polar decomposition suggests how to define integration with respect to a complex measure. Given $d\lambda = h\, d|\lambda|$, for every $f \in L^1(X, |\lambda|)$ define

$$\int_X f\, d\lambda = \int_X fh\, d|\lambda|.$$

If we decompose λ into positive measures as in (6), then for every $f \in L^1(X, |\lambda|)$

$$\int_X f\, d\lambda = \Big(\int_X f\, d\lambda_{1+} - \int_X f\, d\lambda_{1-}\Big) + i\Big(\int_X f\, d\lambda_{2+} - \int_X f\, d\lambda_{2-}\Big). \tag{14}$$

For characteristic functions this is the definition of $d\lambda = h\, d|\lambda|$, hence it holds for simple functions. For $0 \le f \in L^1(X, |\lambda|)$ take simple $s_n \nearrow f$ (Fact B.5); as $f \in L^1(|\lambda|)$ and $f \in L^1(\lambda_{i\pm})$ ($\lambda_{i\pm} \le |\lambda|$), dominated convergence gives (14), hence (14) for all complex $f \in L^1(X, |\lambda|)$.

7.2.4 The dual of $L^p(\mu)$

Finally, another application of Radon–Nikodym: the dual of $L^p(\mu)$ is $L^q(\mu)$ ($1 \le p < \infty$); again σ-finiteness is needed. Recall from Example 2 of §6.1 that

$$g \mapsto \varphi_g : L^q(\mu) \to L^p(\mu)^*$$

is an isometry (using σ-finiteness for $p = 1$, $q = \infty$); we show it is onto: for $\varphi \in L^p(\mu)^*$ we find $g \in L^q(\mu)$ with $\varphi = \varphi_g$.

Finite μ. For $E \in \mathcal{S}$ put

$$\lambda(E) = \varphi(\chi_E).$$

Since $\chi_{E_1 \sqcup E_2} = \chi_{E_1} + \chi_{E_2}$, $\lambda(E_1 \sqcup E_2) = \lambda(E_1) + \lambda(E_2)$. If $E = \bigsqcup_{n=1}^{\infty} E_n$ and $A_n = E_1 \sqcup \cdots \sqcup E_n$, then

$$\lim_{n\to\infty} \|\chi_E - \chi_{A_n}\|_p = \lim_{n\to\infty} \left[\mu(E \setminus A_n)\right]^{1/p} = 0$$

(here $p < \infty$ and $\mu(E) < \infty$), and since $\varphi \in L^p(\mu)^*$,

$$\lambda(E) = \lim_{n\to\infty} \lambda(A_n) = \lim_{n\to\infty} \sum_{k=1}^{n} \lambda(E_k).$$

So λ is a complex measure. Clearly $\lambda \ll \mu$ (if $\mu(E) = 0$ then $\chi_E = 0$ in L^p), so taking the Radon–Nikodym derivative $g \in L^1(\mu)$,

$$\varphi(\chi_E) = \int_E g\, d\mu = \int_X \chi_E g\, d\mu = \varphi_g(\chi_E), \qquad E \in \mathcal{S}.$$

So $\varphi = \varphi_g$ on simple functions, dense in $L^\infty(\mu)$, where both are continuous ($\|f\|_p \le \mu(X)^{1/p}\|f\|_\infty$, $g \in L^1$); so $\varphi = \varphi_g$ on $L^\infty(\mu)$. It remains to show $g \in L^q(\mu)$ and $\varphi = \varphi_g$ on $L^p(\mu)$.

Exercise 7.2.3

For a finite positive measure μ, show that the vector space of simple functions is dense in $L^p(\mu)$ for $1 \le p \le \infty$. [Hint: treat $1 \le p < \infty$ and $p = \infty$ separately; Fact B.20.]

Solution [Supplement]. $p = \infty$. Take $|f| \le \|f\|_\infty = M$ everywhere. Cover $\{|z| \le M\}$ by disjoint Borel $B_1, \dots, B_N$ of diameter $< \epsilon$, pick $z_k \in B_k$, and let $s = \sum_k z_k \chi_{f^{-1}(B_k)}$; then $\|f - s\|_\infty \le \epsilon$ (no finiteness of μ needed). $1 \le p < \infty$. Take $f \ge 0$ and simple $0 \le s_n \nearrow f$ (Fact B.5). $|f - s_n|^p \le f^p \in L^1$ tends to 0, so $\|f - s_n\|_p \to 0$ by dominated convergence. (Finiteness of μ puts all simple functions, e.g. χ_X, in L^p.) □

$p = 1$, $q = \infty$. For every $E \in \mathcal{S}$,

$$\Big|\int_E g\,d\mu\Big| = |\varphi(\chi_E)| \le \|\varphi\|\|\chi_E\|_1 = \|\varphi\|\mu(E),$$

so by Fact B.27, $\|g\|_\infty \le \|\varphi\|$.

$1 < p < \infty$. The basic method is the same as for ℓ^p. For $n = 1, 2, \dots$ put $E_n = \{x \in X : |g(x)| \le n\}$ and $f_n = \chi_{E_n}|g|^{q-1}\cdot\operatorname{sgn} g$ (with $\operatorname{sgn} g = \overline{g}/|g|$ where $g \neq 0$, so that $f_n g = \chi_{E_n}|g|^q$). Then $f_n \in L^\infty(\mu)$ and

$$\int_{E_n}|g|^q = \int_X f_n g = \varphi_g(f_n) = \varphi(f_n) \le \|\varphi\|\|f_n\|_p = \|\varphi\|\Big(\int_{E_n}|g|^q\Big)^{1/p}.$$

Dividing both sides by the last factor and applying the monotone convergence theorem,

$$\|g\|_q = \lim_{n\to\infty}\Big(\int_X \chi_{E_n}|g|^q\Big)^{1/q} \le \|\varphi\|.$$

Now that $g \in L^q(\mu)$, φ_g is continuous on $L^p(\mu)$; since simple functions are dense in $L^p(\mu)$, $\varphi(f) = \varphi_g(f)$ for every $f \in L^p(\mu)$. So $L^p(\mu)^* = L^q(\mu)$ for finite μ and $1 \le p < \infty$.

σ-finite μ. As in the Lebesgue–Radon–Nikodym theorem, take μ_0 and $k \in L^1(\mu)$ from (12), (13), $d\mu_0 = k\,d\mu$. Then

$$K : f \mapsto k^{1/p} f : L^p(\mu_0) \to L^p(\mu)$$

is a surjective isometry, so $\varphi \circ K$ is a bounded linear functional on $L^p(\mu_0)$. By the finite case there is $g \in L^q(\mu_0)$ with

$$(\varphi \circ K)(f) = \int_X fg\,d\mu_0, \qquad f \in L^p(\mu_0).$$

Rewriting,

$$\varphi(Kf) = \int_X fgk\,d\mu = \int_X k^{1/p} f g k^{1/q}\,d\mu = \int_X (Kf)\cdot(gk^{1/q})\,d\mu$$

(with $1/\infty = 0$). Hence $gk^{1/q}$ is the function we were looking for.

Exercise 7.2.4

In the argument above, show that K is a surjective isometry from $L^p(\mu_0)$ onto $L^p(\mu)$, and that $gk^{1/q} \in L^q(\mu)$.

Solution [Supplement]. $k > 0$ everywhere ($\alpha_n > 0$). *Isometry:* $\int_X |k^{1/p} f|^p\,d\mu = \int_X |f|^p\,d\mu_0$. *Surjective:* for $u \in L^p(\mu)$, $f = k^{-1/p}u$ has $\int |f|^p\,d\mu_0 = \|u\|_p^p$ and $Kf = u$. And μ_0, μ have the same null sets. $gk^{1/q} \in L^q(\mu)$*:* for $1 < q < \infty$, $\int |gk^{1/q}|^q\,d\mu = \int |g|^q k\,d\mu = \int |g|^q\,d\mu_0 < \infty$. For $q = \infty$ ($p = 1$), $k^{1/q} = 1$ and $\|g\|_{L^\infty(\mu)} = \|g\|_{L^\infty(\mu_0)} < \infty$ because the null sets agree. □

Summarizing:

Theorem 7.2.3

For a σ-finite positive measure μ on a σ-algebra of X, $L^p(\mu)^* = L^q(\mu)$, where p, q are conjugate exponents and $1 \le p < \infty$.

Exercise 7.2.5

Let $X = \{a, b\}$ with $\mu(\{a\}) = 1$, $\mu(\{b\}) = \infty$. Is the dual of $L^1(\mu)$ equal to $L^\infty(\mu)$?

Solution [Supplement]. $f \in L^1(\mu)$ iff $|f(a)| + \infty \cdot |f(b)| < \infty$, i.e. $f(b) = 0$, so $L^1(\mu) \cong \mathbb{C}$ via $f \mapsto f(a)$ and $L^1(\mu)^* \cong \mathbb{C}$; but $\{b\}$ is not null, so $L^\infty(\mu) \cong \mathbb{C}^2$ (max norm). $g \mapsto \varphi_g$, $\varphi_g(f) = g(a)f(a)$, is onto $L^1(\mu)^*$ but not injective ($g = \chi_{\{b\}}$, $\|g\|_\infty = 1$, $\varphi_g = 0$): $L^1(\mu)^* \neq L^\infty(\mu)$, so σ-finiteness is needed in Theorem 7.2.3 for $p = 1$. □

Corollary 7.2.4

Let μ be a σ-finite positive measure on X and $1 \le p \le \infty$. If a measurable function g on X satisfies

$$h \in L^q(\mu) \implies \Big|\int_X gh\,d\mu\Big| \le M\|h\|_q \qquad (p, q \text{ conjugate}),$$

then $g \in L^p(\mu)$ and $\|g\|_p \le M$.

Proof. The hypothesis says $\varphi_g \in L^q(\mu)^*$ and $\|\varphi_g\| \le M$. So if $1 < p \le \infty$ (i.e. $1 \le q < \infty$), Theorem 7.2.3 applies and the isometry gives $\|g\|_p = \|\varphi_g\| \le M$. For $p = 1$, the hypothesis gives

$$\int_X |g|\,d\mu = \int_X g \cdot (\operatorname{sgn} g)\,d\mu \le M\|\operatorname{sgn} g\|_\infty \le M,$$

where $\operatorname{sgn} g = \overline{g}/|g|$ on $\{g \ne 0\}$ and 0 elsewhere. ∎

Corollary 7.2.5

For every $f \in L^1(\mathbb{R})$ and $g \in L^p(\mathbb{R})$ $(1 \le p \le \infty)$, $f * g \in L^p(\mathbb{R})$ and $\|f * g\|_p \le \|f\|_1\|g\|_p$.

Proof. For every $h \in L^q(\mathbb{R})$, writing $g_y(x) = g(x - y)$,

$$\begin{aligned}\Big|\int_{\mathbb{R}} (f * g)(x)h(x)\,dx\Big| &= \Big|\int_{\mathbb{R}}\int_{\mathbb{R}} f(y)g(x-y)\,dy\,h(x)\,dx\Big| \\ &\le \int_{\mathbb{R}} \Big|\int_{\mathbb{R}} g_y(x)h(x)\,dx\Big|\,|f(y)|\,dy \\ &\le \|g_y\|_p\|h\|_q\|f\|_1 = \|f\|_1\|g\|_p\|h\|_q\end{aligned}$$

(Fubini, Hölder, translation invariance). Corollary 7.2.4 gives the result. ∎

(To be careful, apply the computation first to $|f|, |g|, |h|$: Tonelli gives $\iint |f(y)g(x-y)h(x)|\,dy\,dx \le \|f\|_1\|g\|_p\|h\|_q < \infty$, which justifies Fubini and shows $(f * g)(x)$ is defined a.e.)

7.3 Borel Measures on the Real Line

Motivation. Finite Borel measures on $\mathbb{R}$ come from increasing functions; the discrete, singular continuous and absolutely continuous parts of the measure match those of the function, the last recovered by differentiation.

7.3.1 Regular Borel measures and their distribution functions

We find all complex measures on the Borel sets $\mathcal{B}$ of $\mathbb{R}$ (§4.1), the **complex Borel measures**. As they are combinations of positive measures, consider positive Borel measures finite on bounded intervals: **regular positive Borel measures**.

For a regular positive Borel measure λ on the real line define $\alpha_\lambda : \mathbb{R} \to \mathbb{R}$ by

$$\alpha_\lambda(x) = \begin{cases} \lambda((0, x]), & x > 0, \\ 0, & x = 0, \\ -\lambda((x, 0]), & x < 0. \end{cases}$$

Then α_λ is clearly nondecreasing. It is also right-continuous. First let $x > 0$ and take any decreasing sequence $\langle h_n \rangle$ in $(0, x)$ converging to 0. Then

$$\alpha_\lambda(x + h_n) - \alpha_\lambda(x) = \lambda((0, x + h_n]) - \lambda((0, x]) = \lambda((x, x + h_n]),$$

and since $\bigcap_{n=1}^{\infty}(x, x+h_n] = \emptyset$ and $\lambda((x, x+h_1]) < \infty$, continuity from above gives

$$\lim_{n\to\infty}\{\alpha_\lambda(x+h_n) - \alpha_\lambda(x)\} = 0.$$

Thus $\lim_{h\to 0^+}\alpha_\lambda(x+h) = \alpha_\lambda(x)$. The cases $x = 0$ and $x < 0$ are handled the same way, so α_λ is right-continuous.

Exercise 7.3.1

For a regular positive Borel measure λ on the real line, show that $\alpha_\lambda : \mathbb{R} \to \mathbb{R}$ is a nondecreasing right-continuous function.

Solution [Supplement]**.** In all cases $\alpha_\lambda(b) - \alpha_\lambda(a) = \lambda((a, b]) \geq 0$ for $a < b$ (e.g. for $a < 0 < b$, $\lambda((0, b]) + \lambda((a, 0])$), so α_λ is nondecreasing. For $h_n \searrow 0$, $\alpha_\lambda(x+h_n) - \alpha_\lambda(x) = \lambda((x, x+h_n]) \to 0$ by continuity from above ($\lambda((x, x+h_1]) < \infty$); by monotonicity α_λ is right-continuous. □

7.3.2 Riemann–Stieltjes integrals

For bounded $f : [a, b] \to \mathbb{R}$ the Riemann–Stieltjes integral $\int_a^b f\, d\alpha_\lambda$ is built like the Riemann integral (Fact B.13): for a partition $P = \{x_0, \ldots, x_n\}$, with U_P, L_P as in Fact B.13, the sums are

$$\begin{aligned} U_a^b(f, P, \alpha_\lambda) &= \sum_{i=1}^{n} M_i(\alpha_\lambda(x_i) - \alpha_\lambda(x_{i-1})) = \sum_{i=1}^{n} M_i\lambda((x_{i-1}, x_i]) \\ &= \int_{[a,b]} U_P\, d\lambda - f(a)\lambda(\{a\}) = \int_{(a,b]} U_P\, d\lambda, \end{aligned}$$

and similarly $L_a^b(f, P, \alpha_\lambda) = \int_{(a,b]} L_P\, d\lambda$. Hence for every partition $P \in P[a, b]$

$$L_a^b(f, P, \alpha_\lambda) \leq \int_{(a,b]} f\, d\lambda \leq U_a^b(f, P, \alpha_\lambda).$$

If $f : \mathbb{R} \to \mathbb{R}$ is continuous with support in a bounded interval (A, B), then f is Riemann–Stieltjes integrable on $[A, B]$ with respect to α_λ, so

$$\int_{\mathbb{R}} f\, d\lambda = \int_A^B f(x)\, d\alpha_\lambda(x).$$

So on $C_c(\mathbb{R})$, integration against a regular positive Borel measure is Riemann–Stieltjes integration. The latter struggles with discontinuous integrands; the measure integrals instead struggle with improper integrals.

7.3.3 Continuity of α_λ

Applying Lebesgue–Radon–Nikodym to Borel measures on $\mathbb{R}$, we first ask when α_λ is continuous or absolutely continuous. As $\bigcap_n(x - h_n, x] = \{x\}$ for $h_n \searrow 0$,

$$\lambda(\{x\}) = \lim_{n\to\infty}\lambda((x - h_n, x]) = \lim_{n\to\infty}\{\alpha_\lambda(x) - \alpha_\lambda(x - h_n)\}.$$

So α_λ is left-continuous at x iff $\lambda(\{x\}) = 0$. A measure for which every point has measure 0 is a **continuous measure**. Since α_λ is always right-continuous:

Proposition 7.3.1

For a regular positive Borel measure λ on the real line, α_λ is continuous at x if and only if $\lambda(\{x\}) = 0$.

The next theorem explains the shared term "absolutely continuous". Both measures must live on one σ-algebra; $m|_{\mathcal{B}}$ is also written m.

Theorem 7.3.2

For a regular positive Borel measure λ on the real line, the following are equivalent:

(a) $\lambda \ll m$;

(b) α_λ is absolutely continuous on every bounded closed interval.

Proof. (a)⇒(b). If $\lambda \ll m$, by the Radon–Nikodym theorem there is a Borel function $h : \mathbb{R} \to [0, \infty]$ with

$$\lambda(E) = \int_E h(x)\,dx, \qquad E \in \mathcal{B}.$$

Fix a bounded closed interval I; we show α_λ is absolutely continuous on I. Since $\int_I h = \lambda(I) < \infty$, $h \in L^1(I, m)$. By Fact B.15, for every $\epsilon > 0$ there is $\delta > 0$ with

$$A \in \mathcal{B},\ m(A) < \delta \implies \lambda(A) < \epsilon.$$

This δ works in the definition of absolute continuity of functions. If $\{(a_i, b_i) : i = 1, \dots, n\}$ are disjoint intervals in I with $\sum_{i=1}^n (b_i - a_i) < \delta$, then $m(\bigcup_{i=1}^n (a_i, b_i]) < \delta$, so

$$\sum_{i=1}^n |\alpha_\lambda(b_i) - \alpha_\lambda(a_i)| = \sum_{i=1}^n \lambda((a_i, b_i]) = \lambda\Big(\bigcup_{i=1}^n (a_i, b_i]\Big) < \epsilon.$$

(b)⇒(a). Let α_λ be absolutely continuous on each bounded closed I, with δ_I for ϵ. If $m(A) = 0$ then $m(A \cap I) = 0$, so there are disjoint open intervals (c_n, d_n) with

$$A \cap I \subset \bigcup_{n=1}^\infty (c_n, d_n), \qquad \sum_{n=1}^\infty (d_n - c_n) < \delta_I.$$

Writing (a_n, b_n) for the intersection of (c_n, d_n) with the interior $\operatorname{int} I$ of I,

$$A \cap \operatorname{int} I \subset \bigcup_{n=1}^\infty (a_n, b_n) \subset I, \qquad \sum_{n=1}^\infty (b_n - a_n) < \delta_I.$$

Hence $\sum_{n=1}^N [\alpha_\lambda(b_n) - \alpha_\lambda(a_n)] < \epsilon$ for every N, so $\sum_{n=1}^\infty [\alpha_\lambda(b_n) - \alpha_\lambda(a_n)] \le \epsilon$. Since α_λ is continuous (being absolutely continuous), $\alpha_\lambda(b_n) - \alpha_\lambda(a_n) = \lambda((a_n, b_n))$ (no point masses), and

$$\lambda(A \cap \operatorname{int} I) \le \lambda\Big(\bigcup_{n=1}^\infty (a_n, b_n)\Big) = \sum_{n=1}^\infty \lambda((a_n, b_n)) = \sum_{n=1}^\infty [\alpha_\lambda(b_n) - \alpha_\lambda(a_n)] \le \epsilon.$$

So $\lambda(A \cap \operatorname{int} I) = 0$ for every bounded open interval $\operatorname{int} I$, and therefore $\lambda(A) = 0$. ∎

7.3.4 Regularity

We now look for the condition on α_λ for $\lambda \perp m$. First some properties of regular positive Borel measures on the real line.

Theorem 7.3.3 (regularity)

A regular positive Borel measure λ on the real line satisfies, for every Borel set $E \in \mathcal{B}$,

$$\lambda(E) = \inf\{\lambda(U) : U \supset E,\ U \text{ open}\},$$
$$\lambda(E) = \sup\{\lambda(K) : K \subset E,\ K \text{ compact}\}.$$

Proof. Open sets satisfy the first identity and, as increasing unions of compact sets, the second. Let $\mathcal{C}$ be the Borel sets satisfying both; show $\mathcal{C}$ is a σ-algebra.

Countable unions. Take $\langle E_n\rangle$ in $\mathcal{C}$ and $E = \bigcup_n E_n$. First the case $\lambda(E) < \infty$: choose open $U_n \supset E_n$ with $\lambda(U_n) < \lambda(E_n) + \epsilon/2^n$ and let U be their union. Since $U \setminus E \subset \bigcup_n (U_n \setminus E_n)$,

$$\lambda(U) - \lambda(E) = \lambda(U \setminus E) \le \sum_{n=1}^{\infty} \lambda(U_n \setminus E_n) < \epsilon,$$

so E satisfies the first identity. If $\lambda(E) = \infty$ the first identity is trivial. For the second, for each j choose compact sets $\langle K_{ij} : i = 1, 2, \dots\rangle$ with

$$K_{ij} \subset E_j, \qquad \lim_{i\to\infty} \lambda(K_{ij}) = \lambda(E_j).$$

With $K_n = \bigcup_{i,j=1}^{n} K_{ij}$ we can show $\lambda(K_n) \to \lambda(E)$, so $E \in \mathcal{C}$ (Exercise 7.3.2).

Complements. For $E \in \mathcal{C}$ take open $U_n \supset E$ decreasing with $\lambda(U_n) \to \lambda(E)$; the compact $L_n = [-n, n] \setminus U_n \subset E^c$ have $\lambda(L_n) \to \lambda(E^c)$ (Exercise 7.3.2). For the first identity for E^c we first show

$$I \text{ a bounded closed interval},\ I \cap E \in \mathcal{C} \implies I \setminus E \in \mathcal{C}.$$

Choose an open $U \supset I \cap E$ with $\lambda(U) < \lambda(I \cap E) + \epsilon$. Then $K = I \setminus U$ is compact, $K \subset I \setminus E$, and since $\lambda(I) < \infty$,

$$\lambda(K) = \lambda(I) - \lambda(I \cap U) \ge \lambda(I) - \lambda(U) > \lambda(I) - \lambda(I \cap E) - \epsilon = \lambda(I \setminus E) - \epsilon,$$

so $I \setminus E$ satisfies the second identity. For the first, choose a bounded open interval $J \supset I$ with $\lambda(J) < \lambda(I) + \epsilon$, and a compact $K \subset I \cap E$ with $\lambda(K) > \lambda(I \cap E) - \epsilon$. Put $U = J \setminus K$; then U is open, $U \supset I \setminus E$, and

$$\lambda(U) = \lambda(J) - \lambda(K) < \lambda(I) + \epsilon - \lambda(I \cap E) + \epsilon = \lambda(I \setminus E) + 2\epsilon,$$

so $I \setminus E$ satisfies the first identity.

If $E \in \mathcal{C}$, then $E \cap [-n, n] \in \mathcal{C}$ for each n (one checks this as above), so $[-n, n] \setminus E \in \mathcal{C}$. Since $\mathcal{C}$ is already closed under countable unions, $E^c = \bigcup_n ([-n, n] \setminus E) \in \mathcal{C}$. ∎

Exercise 7.3.2

Show that every open set is an increasing union of compact sets, and in the proof of Theorem 7.3.3 that $\lambda(K_n) \to \lambda(E)$ and $\lambda(L_n) \to \lambda(E^c)$.

Solution [Supplement]. *Open sets.* For open $U \subset \mathbb{R}$ ($U \neq \mathbb{R}$) let $C_n = \{x \in U : |x| \le n,\ d(x, U^c) \ge \frac{1}{n}\}$; for $U = \mathbb{R}$ let $C_n = [-n, n]$. Each C_n is closed and bounded, hence compact, $C_n \subset C_{n+1}$, and every $x \in U$ has $d(x, U^c) > 0$, so $x \in C_n$ for large n.

$\lambda(K_n) \to \lambda(E)$. $K_n = \bigcup_{i,j\le n} K_{ij}$ is compact, $K_n \subset K_{n+1}$ and $K_n \subset E$, so $L = \lim_n \lambda(K_n) \le \lambda(E)$ exists. If some $\lambda(E_j) = \infty$, then $L \ge \lambda(K_{ij}) \to \infty$ as $i \to \infty$, so $L = \lambda(E) = \infty$. Otherwise fix N and i. For $n \ge \max(i, N)$, $K_n \supset \bigcup_{j\le N} K_{ij}$, and

$$\lambda\Big(\bigcup_{j\le N} E_j\Big) - \lambda\Big(\bigcup_{j\le N} K_{ij}\Big) \le \sum_{j\le N} \lambda(E_j \setminus K_{ij}) = \sum_{j\le N} \big(\lambda(E_j) - \lambda(K_{ij})\big) \xrightarrow[i\to\infty]{} 0.$$

Hence $L \geq \lambda(E_1 \cup \cdots \cup E_N)$ for every N, and continuity from below gives $L \geq \lambda(E)$.

$\lambda(L_n) \to \lambda(E^c)$. Suppose $\lambda(E) < \infty$. $L_n = [-n, n] \setminus U_n$ is compact, increasing (as U_n decreases), and $L_n \subset E^c$. Since $([-n, n] \cap E^c) \setminus L_n = [-n, n] \cap U_n \cap E^c \subset U_n \setminus E$,

$$0 \leq \lambda([-n, n] \cap E^c) - \lambda(L_n) \leq \lambda(U_n \setminus E) = \lambda(U_n) - \lambda(E) \to 0,$$

while $\lambda([-n, n] \cap E^c) \to \lambda(E^c)$ (continuity from below), so $\lambda(L_n) \to \lambda(E^c)$. (If $\lambda(E) = \infty$, apply this to $E \cap [-N, N]$, as at the end of the proof.) □

Corollary 7.3.4

If $\lambda \ll m$ and $\alpha'_\lambda = 0$ almost everywhere, then $\lambda = 0$.

Proof. By Theorem 7.3.2, α_λ is absolutely continuous, so Fact B.16 gives

$$\lambda((a, b)) = \alpha_\lambda(b) - \alpha_\lambda(a) = \int_a^b \alpha'_\lambda(x)\, dx = 0.$$

Hence $\lambda(U) = 0$ for every open U (a countable disjoint union of open intervals), and Theorem 7.3.3 gives $\lambda = 0$. ■

7.3.5 Singular measures

Theorem 7.3.5

For a regular positive Borel measure λ on the real line, the following are equivalent:

(a) $\lambda \perp m$;

(b) $\alpha'_\lambda = 0$ almost everywhere.

Proof. (b)⇒(a). Assume $\alpha'_\lambda = 0$ a.e. Write the Lebesgue decomposition $\lambda = \lambda_a + \lambda_s$; then $\alpha_\lambda = \alpha_{\lambda_a} + \alpha_{\lambda_s}$, both nondecreasing, so $0 \leq \alpha'_{\lambda_a} \leq \alpha'_\lambda = 0$ and hence $\alpha'_{\lambda_a} = 0$ almost everywhere. Since $\lambda_a \ll m$, Corollary 7.3.4 gives $\lambda_a = 0$, so $\lambda = \lambda_s$ is singular.

(a)⇒(b). Assume $\lambda \perp m$ and put

$$\nu(E) = \int_E \alpha'_\lambda\, dm, \qquad E \in \mathcal{B}. \tag{15}$$

Then obviously $\nu \ll m$. On the other hand, by Fact B.14,

$$\nu((a, b]) = \int_{(a,b]} \alpha'_\lambda\, dm \leq \alpha_\lambda(b) - \alpha_\lambda(a) = \lambda((a, b]).$$

Open sets are countable disjoint unions of intervals $(a, b]$, so $\nu(U) \leq \lambda(U)$ for open U, and Theorem 7.3.3 gives $\nu \leq \lambda$ on Borel sets, so $\nu \ll \lambda$. With $\lambda \perp m$, $\nu \perp m$ (Exercise 7.3.3), and $\nu \ll m$ forces $\nu = 0$. As $\alpha'_\lambda \geq 0$ a.e., $\alpha'_\lambda = 0$ a.e. ■

Exercise 7.3.3

Show that if $\nu \ll \lambda$ and $\lambda \perp m$, then $\nu \perp m$.

Solution [Supplement]. Choose disjoint A, B with $\lambda(E) = \lambda(E \cap A)$ and $m(E) = m(E \cap B)$ for all E. Then $\lambda(E \setminus A) = 0$, so $\nu(E \setminus A) = 0$ by $\nu \ll \lambda$, i.e. $\nu(E) = \nu(E \cap A)$. Together with $m(E) = m(E \cap B)$, the disjoint pair A, B shows $\nu \perp m$. □

7.3.6 The Radon–Nikodym derivative is α'_λ

For an arbitrary regular positive Borel measure λ with Lebesgue decomposition $\lambda = \lambda_a + \lambda_s$, define ν by (15). By Theorem 7.3.5,

$$\alpha'_\lambda = \alpha'_{\lambda_a} + \alpha'_{\lambda_s} = \alpha'_{\lambda_a}$$

almost everywhere, and by Fact B.16 (α_{λ_a} absolutely continuous by Theorem 7.3.2)

$$\nu((c,d)) = \int_{(c,d)} \alpha'_\lambda \, dm = \int_{(c,d)} \alpha'_{\lambda_a} \, dm = \alpha_{\lambda_a}(d) - \alpha_{\lambda_a}(c) = \lambda_a((c,d)).$$

Hence $\nu(U) = \lambda_a(U)$ for every open U, and since ν is also regular, $\nu = \lambda_a$ (Theorem 7.3.3). That is,

$$\lambda_a(E) = \int_E \alpha'_\lambda \, dm$$

for every Borel set $E \in \mathcal{B}$: $\alpha'_{\lambda_a} = \alpha'_\lambda$ is precisely the Radon–Nikodym derivative of λ_a with respect to m.

Corollary 7.3.6

If a regular positive Borel measure λ on the real line is absolutely continuous with respect to Lebesgue measure m, then $\left[\frac{d\lambda}{dm}\right] = \alpha'_\lambda$ almost everywhere.

7.3.7 Discrete, continuous singular, and absolutely continuous parts

For a regular positive Borel measure λ put

$$D = \{x \in \mathbb{R} : \lambda(\{x\}) > 0\},$$

and for $E \in \mathcal{B}$

$$\lambda_d(E) = \lambda(E \cap D), \qquad \lambda_c(E) = \lambda(E \setminus D).$$

For each n, $\lambda(D \cap [-n,n]) \le \lambda([-n,n]) < \infty$, so $D \cap [-n,n]$ is countable (for each k, at most $k\lambda([-n,n])$ points have mass $> 1/k$); hence D is countable, $D = \{x_n : n = 1,2,\dots\}$. Then for every $E \in \mathcal{B}$

$$\lambda_d(E) = \sum_{x_n \in E} \lambda(\{x_n\}) = \sum_{n=1}^{\infty} \lambda(\{x_n\}) \delta_{x_n}(E),$$

so λ_d is a sum of Dirac measures, and α_{λ_d} jumps at each $x \in D$; α_{λ_c} is continuous (Proposition 7.3.1), and if $\lambda_c \perp m$ its derivative is 0 a.e. (Theorem 7.3.5), as for the Cantor function.

Summary

Every regular positive Borel measure λ on the real line is the sum of three measures:

$$\lambda = \lambda_a + \lambda_s + \lambda_d,$$

an absolutely continuous λ_a, a continuous singular λ_s and a discrete λ_d, whose functions are absolutely continuous (α_a), continuous with derivative 0 a.e. (α_s), and a jump function (α_d). Then $\alpha = \alpha_a + \alpha_s + \alpha_d$ and

$$\lambda_a(E) = \int_E \alpha'_a \, dm = \int_E \alpha' \, dm, \qquad E \in \mathcal{B}.$$

Given a regular positive Borel measure λ, there is a corresponding nondecreasing right-continuous function α, and the two linear functionals on $C_c(\mathbb{R})$

$$f \mapsto \int_{\mathbb{R}} f \, d\lambda, \qquad f \mapsto \int_{\operatorname{supp} f} f(x) \, d\alpha(x)$$

coincide (Borel measure integral and Riemann–Stieltjes integral). Conversely, every nondecreasing function gives a regular positive Borel measure (next chapter).